

**DIVISION 15**

**MECHANICAL WORKS**

**DIVISION 15 - MECHANICAL**

**INDEX**

**TABLE OF CONTENTS**

**PART A – MECHANICAL WORKS**

|            |   |                                           |
|------------|---|-------------------------------------------|
| SECTION 1  | : | GENERAL REQUIREMENTS                      |
| SECTION 2  | : | INSULATION                                |
| SECTION 3  | : | AIR CONDITIONING EQUIPMENT                |
| SECTION 4  | : | DUCTWORK AND AIRSIDE EQUIPMENT            |
| SECTION 5  | : | PLUMBING WORKS                            |
| SECTION 6  | : | SOIL, WASTE, VENT AND STORM WATER SYSTEMS |
| SECTION 7  | : | FIRE FIGHTING SYSTEM                      |
| SECTION 8  | : | LIQUEFIED PETROLEUM GAS INSTALLATION      |
| SECTION 9  | : | IRRIGATION SYSTEM                         |
| SECTION 10 | : | SCHEDULE OF MANUFACTURERS                 |

**SECTION 1**  
**GENERAL REQUIREMENTS**

|      |                                               |    |
|------|-----------------------------------------------|----|
| 1.0  | GENERAL .....                                 | 5  |
| 2.0  | INTENT .....                                  | 5  |
| 3.0  | EXAMINATION OF SITE AND DRAWINGS .....        | 5  |
| 4.0  | PERMITS FEES AND INSPECTIONS. ....            | 6  |
| 5.0  | CONTRACT DRAWINGS.....                        | 6  |
| 6.0  | CONSTRUCTION DRAWINGS.....                    | 7  |
| 7.0  | SHOP DRAWINGS & MATERIAL SUBMITTAL .....      | 7  |
| 8.0  | RECORD DRAWINGS.....                          | 8  |
| 9.0  | TEMPORARY SERVICE. ....                       | 9  |
| 10.0 | COOPERATION .....                             | 9  |
| 11.0 | PROTECTION AND STORAGE .....                  | 9  |
| 12.0 | EXISTING WORK AND EQUIPMENT .....             | 10 |
| 13.0 | SUPERVISION .....                             | 10 |
| 14.0 | SCHEDULING .....                              | 10 |
| 15.0 | WORK TO CONTINUE .....                        | 10 |
| 16.0 | STANDARDS AND AUTHORITIES.....                | 11 |
| 17.0 | WORKMANSHIP .....                             | 11 |
| 18.0 | MATERIALS AND EQUIPMENT .....                 | 11 |
| 19.0 | CLEANING .....                                | 12 |
| 20.0 | ACCESSIBILITY .....                           | 12 |
| 21.0 | CUTTING AND PATCHING. ....                    | 12 |
| 22.0 | INSERTS, SLEEVES, ESCUTCHEONS AND CURBS.....  | 12 |
| 23.0 | ACCESS PANELS AND DOORS. ....                 | 14 |
| 24.0 | FLASHING.....                                 | 15 |
| 25.0 | GENERAL WELDING REQUIREMENTS.....             | 15 |
| 26.0 | TESTS FOR WELDER QUALIFICATION. ....          | 16 |
| 27.0 | TESTING OF WELDED PIPEWORK INSTALLATION. .... | 17 |
| 28.0 | PAINTING, IDENTIFICATION AND VALVE TAGS ..... | 18 |
| 29.0 | FLUSHING AND CLEANING. ....                   | 20 |
| 30.0 | INSPECTION, TESTING AND ADJUSTING. ....       | 20 |
| 31.0 | TEMPORARY USAGE .....                         | 22 |
| 32.0 | INSPECTION OF THE WORK. ....                  | 22 |
| 33.0 | OPERATING INSTRUCTIONS TO OWNER. ....         | 23 |
| 34.0 | COMMISSIONING OF SERVICES. ....               | 24 |
| 35.0 | SYSTEM ACCEPTANCE .....                       | 25 |
| 36.0 | CORRECTION AFTER COMPLETION.....              | 25 |
| 37.0 | GUARANTEES .....                              | 25 |
| 38.0 | MAINTENANCE .....                             | 26 |



## **1.0 GENERAL**

- 1.1 The Instructions to Tenderers, Form of Tender, Main Conditions of contract, Note on Pricing, the Preliminary Section and any Supplementary Conditions of Contract apply to and are part of this Division and all subsections.
- 1.2 Sections of this Services Specification which relate to other Divisions are not intended to supersede the specifications of these divisions, but to guide the Contractor in tendering and performance. In no event shall a quality less than that indicated here be acceptable.
- 1.3 The precedence of various Documents and Drawings shall be as determined in the General Contract Documents, subject to the limitation that any item shown or required in any of the documents or drawings shall be considered by the Contractor to be included in all Documents.
- 1.4 **Contractor scope will include:**
- Execution, Completion of the Works, Remedying of any defects during DNP and Maintenance (routine and preventive) during DNP. (DNP for all Works will be 1 year).
  - Secure & pass testing & commissioning or any other required tests as per the Specifications,
  - Provide relevant Warranty & Guarantees and Operational & Maintenances Manuals including training,
  - Handover all the check list, log of maintenance conducted by the Contractor during DNP and complete report of the maintenance to the Employer along with the Discharge letter.

## **2.0 INTENT**

- 2.1 Check design and accept responsibility for proper performance of all mechanical systems in accordance with relevant standards and codes and design parameters listed hereinafter.
- 2.2 Provide all items, articles, materials, operations, sundries, labour, supervision, guarantees, allowances for overhead and profit, etc., to achieve a fully functional and acceptable system.
- 2.3 Consider the specifications as an integral part of the work together with the drawings. Consider any item or subject omitted from one, but mentioned or reasonably implied on the other as properly and sufficiently indicated and provide the same under the work of this division.
- 2.4 The Contractor is responsible for developing his own take off of materials and is to make this available as requested to the Consultant.

## **3.0 EXAMINATION OF SITE AND DRAWINGS**

- 3.1 Visit the site of the proposed works and obtain all information as to existing conditions and limitations and all proposed works on adjacent sites and in adjacent areas which might affect the works on this site, whether by Private Individuals or by Government Authorities or others.
- 3.2 Examine the documents including the Specifications and Drawings of all other Divisions before bidding and again before commencing any portion of the works.
- 3.3 Neither the Owner nor the Consultant will be responsible for any claim for extra work or expense resulting from the failure of the Contractor to be fully aware of Site Conditions, Drawing implications, etc.

#### **4.0 PERMITS FEES AND INSPECTIONS.**

- 4.1 Arrange for inspection of all work by the Authorities having jurisdiction over the works. This is to occur on an on-going basis throughout construction to avoid delays at the termination of the project. On completion of the work present to the Consultant, for the Owner, final unconditional approval certificates of the Inspecting Authorities.
- 4.2 Pay fees/charges as levied by statutory authorities for inspections, approvals, services connections, rerouting of existing services, etc. at no additional cost to contract.
- 4.3 Before commencing any work on site, submit the required number of copies to the relevant Authorities (after coordination with present building Contractor) for checking and approval. Follow the progress of such drawings to permit the timely approval of them by the Authorities. Comply with any changes requested by the Authorities, but notify the Consultant immediately of any such change for proper processing of these requirements. The Contractor, working on a regular basis with the Authorities is expected to know which changes are likely at the time he bids the job, and no extra cost will be allowed for such changes.

#### **5.0 CONTRACT DRAWINGS**

- 5.1 The drawings for services works are performance drawings, diagrammatic, and intended to convey the scope of work and indicate general arrangement and approximate locations of apparatus, fixtures, pipe and duct runs, etc. The drawings do not intend to indicate architectural or structural details, nor do they show any fabrication or installation details.
- 5.2 Do not scale drawings. Obtain accurate dimensions to structure and architectural items from drawings of those trades. Confirm by site measurement. Verify location and elevation of all services (Water, Electrical, Telephone, Sanitary, Storm Drainage, Gas, etc..) before proceeding with the work.
- 5.3 Make at no extra cost, any changes or additions to materials, and/ or equipment necessary to accommodate structural conditions (pipes or ducts around beams, columns, etc.)
- 5.4 Alter, at no additional cost, the location of materials and/ or equipment as directed, provided that the changes are made before installation and do not necessitate additional material.
- 5.5 Install all ceiling mounted components (Diffusers, grilles, detectors, light fixtures, emergency lights, fire detectors, loudspeakers, camera points, etc.) in accordance with the reflected ceiling drawings which are to be prepared by the Contractor and coordinated with all trades. These must be submitted for approval and be approved before any work commences on site.
- 5.6 Leave space clear and install all work to accommodate future materials and/or equipment as indicated and/or supplied by another division of work of the contract. Install all pipe runs, conduit runs, cable trays, etc., to maintain maximum headroom and clearances, and to conserve space in shafts and ceiling spaces and under floors, and to provide adequate space for service and maintenance.
- 5.7 Confirm on the site the exact location of outlets and fixtures. Confirm also location of outlets and fixtures provided by any other division of work under the contract.

## **6.0 CONSTRUCTION DRAWINGS.**

- 6.1 Prepare drawings in conjunction with all trades concerned, showing sleeves and openings for all passages through structure and all insert sizes and locations.
- 6.2 Prepare composite construction drawings in colour fully dimensioned, of piping and equipment in tunnels, shafts, mechanical equipment rooms and areas, and all other critical locations to avoid a conflict of trades. Composite drawing shall include all MEP services in coordination with the structure and false ceiling heights given in the Architectural drawings.

Base equipment drawings upon shop drawings and include but do not necessarily limit to, all details pertaining to access, cleanouts, tappings, sleeves, electrical connections, drains, location and elevation of pipes, ducts, conduits, etc., obtained from consultation with, and agreement of, all trades involved.

The location of final fix items such as supply / return air diffusers and sprinkler heads shown in the contract drawings are indicative only. Contractor shall co-ordinate with the reflected ceiling plans and Interior design drawings and agree the actual location in relation to other items such as light fittings, smoke detectors, speakers etc. Final location should be based on the approved reflected ceiling plans prepared by Contractor and obtain approval from the Architect / Engineer.

- 6.3 Prepare drawings of equipment bases, pump pits, anchors, inertia slabs, floor and roof curbs, wall openings, trenches, pertaining to mechanical work.
- 6.4 Prepare all drawings to scale as agreed with the Consultant. Forward these drawings, approved by all trades concerned to the Consultant for his records. Provide transparencies and provide printed copies in a number as specified elsewhere in the Contract but not less than four sets.
- 6.5 Bind one complete set of construction drawings showing "as built" conditions in each operating and maintenance instruction manual. The extent of these drawings will be indicated to the Contractor in advance by the Consultant.

## **7.0 SHOP DRAWINGS & MATERIAL SUBMITTAL**

- 7.1 Submit shop drawings and samples for materials and equipment as listed in this and in each subsequent section.

Provide transparencies wherever possible and provide printed copies in a number as specified elsewhere in the contract documents. The Consultant may retain the sample of each item at his discretion until the completion of the contract.

- 7.2 Present a schedule of shop drawings and material submittal within one month after award of the contract, indicating the anticipated date when the drawings will be submitted for review. Assume full responsibility for timely submission of all drawings. Allow a minimum of three weeks for the Consultant's review. At time of submission indicate order deadlines and cost implications, etc., otherwise the Consultant will advise the Owner that cost and completion date will be unaffected.
- 7.3 Drawings production and presentation is a Contractual matter and any delay on the Contractor's behalf in making these submissions will be considered a Contractual delay and may be subject to Contractual penalties and other remedies as determined by the Consultant.
- 7.4 The Consultant will only consider shop drawings bearing the stamp of the Contractor and all Subcontractors involved. Check for all pertinent information such as physical dimensions, make, performance, electrical characteristics and indicate the intended use and location before submitting these drawings. Use reference symbols or enumeration to correspond to the design drawings.

- 7.5 Assume responsibility for accuracy of equipment dimensions related to space available, accessibility for maintenance and service, compliance with inspection authorities' codes. Ensure that shop drawings indicate the shipping and working weights of all equipment.
- 7.6 The submission of samples will be subject to the same procedure as that of shop drawings. One set of such samples shall be required to be brought to site and kept there after approval till substantial completion.
- 7.7 The Consultant will mark the drawings "resubmit specified item", "rejected", "no exception taken", or "make correction noted". In the last case all revisions will be clearly marked on the returned print and corrected prints may be issued for manufacture and construction.
- Make the revisions shown on the "make corrections as noted" prints onto the tracings as soon as practicable and forward copies to the Consultant for his records.
- 7.8 The Consultant is not responsible for any delays caused by the inadequacy of the Contractor's drawings or his failure to obtain initial or subsequent approval. Any time taken by the Contractor to obtain approval after the originally scheduled date will be considered as a delay to the contract caused by the Contractor.
- 7.9 When drawings are marked "resubmit specified item", the Contractor is to resubmit the item as originally specified or as may be determined to be equivalent by the Consultant. The Consultant is the sole arbiter of whether any item is satisfactory or equivalent.
- 7.10 When drawings are marked "rejected" a complete resubmission of the particular drawing is necessary, subject to the same conditions as outlined above.
- 7.11 The Consultant's review shall not relieve the Contractor from responsibility for deviations from the Contract documents, unless he has, in writing, called the Consultant's attention to such deviations at the time of submission of drawings. The Consultant's review shall be construed to apply to, and only to, general arrangements and shall not relieve the Contractor from the entire responsibility. Any approval by the Consultant shall be on the understanding that any item submitted shall be ordered with options and modifications to fully meet the specification. Any fabrication, erection, setting out or other work done in advance of receipt of stamped drawings shall be done entirely at the Contractor's risk and cost.
- 7.12 Furnish prints of the reviewed details to all other parties who may require them for proper coordination of their work, and furnish all information necessary for the work as a whole.
- 7.13 Obtain Manufacturers' installation directions to aid in the proper execution of the work. Submit two copies of such directions to the Consultant prior to installation, for use in inspecting the work.
- 7.14 Bind one complete set of checked shop drawings into each operating and maintenance manual.

## **8.0 RECORD DRAWINGS**

- 8.1 As the job progresses, mark on one set of prints to accurately indicate the status of installed work. Have the white prints available for inspection at the site at all times, and present them for scrutiny at all job meetings. Transfer all information onto the set of transparencies. Obtain the Contractor's approval before forwarding these to the Consultant. Submit two sets of white prints to the Consultant for checking and approval. Upon approval, submit one set of transparencies and five sets of white prints to the consultant for onward transmission to the client.

- 8.2 Show on the record transparencies the installed inverts of all services entering and leaving the building and the property. Dimension underground services at key points of every run-in relation to structure and building. Record all elevations for underground services in relation to floor level of the building and give reference datums to Municipal benchmarks.
- 8.3 Indicate exact location of all services left for future work. Show and dimension all work embedded in the structure.

#### **9.0 TEMPORARY SERVICE**

- 9.1 Do not use any of the permanent service facilities during construction, unless specific written approval is obtained from the Consultant and the Owner or where specifically allowed elsewhere in the Contract Documents.
- 9.2 The use of permanent facilities for temporary services will not affect, in any way, the commencement day of the period in which the correction after completion is the Contractor's responsibility. Such period will commence only when the overall project is completed and certificates of acceptance are issued.
- 9.3 In the event that mains power is not available or is not of sufficient capacity during construction, supply, install and operate diesel generator(s) of the correct capacity (which must be approved by the Consultant.)

Any generator must have full safety features and must be maintained regularly to ensure site power at all times.

#### **10.0 COOPERATION**

- 10.1 Confer with all trades installing equipment which may affect the work of this division, and arrange equipment in proper relation with that equipment installed under all Divisions of the Contract.
- 10.2 Furnish all items to be built in by others, in time, complete with all pertinent information, commensurate with the progress of the work.
- 10.3 Store materials neatly and out of the way and clean up all refuse caused by the work daily.

#### **11.0 PROTECTION AND STORAGE**

- 11.1 Protect the buildings and structures from damage due to carrying out of this work.
- 11.2 Protect all mechanical & electrical works from damage. Keep all equipment dry and clean at all times.
- 11.3 Cover all openings in equipment and materials. Cover all temporary openings in ducts and pipes with polyethylene sheets or caps until final connection is made.

The quality of such cover must be determined with due regard to how long it may be until final connection.

- 11.4 Be responsible for and make good any damages caused directly or indirectly to any walls, floors, ceilings, woodwork, brickwork, finishes, services, roads, gardens, etc.

## **12.0 EXISTING WORK AND EQUIPMENT**

Before this Contractor undertakes work in any area he must prepare a list of deficiencies in that area which affect his works, or which could possibly be construed as being caused by himself if not noted. In the event that such deficiency list is not prepared, then he shall be deemed responsible for such deficiencies. Any list shall be brought to the attention of the Consultant forthwith.

## **13.0 SUPERVISION**

13.1 The Subcontractor will maintain at site, as necessary for the performance of the Contract, qualified personnel and supporting staff, with proven experience in erecting, testing, and adjusting projects of comparable nature and complexity.

13.2 Before commencing work the Contractor will submit details of the proposed Engineers and Supervisors, including copies of their Certificates. If in the Consultant's opinion the proposed Engineers or Supervisors are not adequately qualified or are otherwise unacceptable, the onus is on the Contractor to submit alternates until such approval is given.

13.3 Where the Contractor's or Subcontractor's staff is, or becomes during the Contract deficient in the choice of adequate or acceptable personnel, the Contractor is to remedy the situation by appropriate measures.

13.4 Approval of the Contractor's or Subcontractor's Engineers or Staff shall in no way prevent the withdrawal of that approval at any time during the Contract should the Consultant so desire. In the event of such disapproval then the Contractor will be required to rectify the position as stated above within 14 days.

13.5 In the event of any negligent or severely detrimental behavior the Consultant has the right to order the removal from site of any Engineer, Supervisor, or worker on a "forthwith" basis.

The Contractor will remedy the situation on a similar forthwith basis and in any event, will not be allowed to proceed with any area of work which as a result is left without proper supervision or technical backup until such time as the Consultant is again satisfied by the measures taken.

## **14.0 SCHEDULING**

14.1 In the event that a completion date has been set before tendering it will be the obligation of the Contractor to provide a schedule adequate for all trades to meet the completion date.

14.2 If no completion date is specified it will be the obligation of the Contractor to provide to the Consultant a proposed schedule of work including mechanical trades, within 21 days of award of Contract. This schedule will be either approved by, or negotiated with, the Consultant.

## **15.0 WORK TO CONTINUE**

15.1 Unless specifically instructed otherwise by the Consultant the works are to continue to completion in the most expeditious manner.

## **16.0 STANDARDS AND AUTHORITIES.**

- 16.1 All equipment and installations shall be in accordance with local Codes and Regulations. In general, all equipment and installation shall also be in accordance with:
- Uniform plumbing code of USA.
  - National Plumbing Code
  - Institute of Plumbing (UK)
  - ASTM
  - ISO
  - AWWA
  - ASME
  - UL Laboratories
- 16.2 Obtain approval of shop drawings from Authorities having jurisdiction including but not limited to Drainage Department, Municipality, Ministry of Water and Electricity, Fire Brigade, Department of Health, and local telephone authority. All costs towards such approvals to be borne by the contractor.

## **17.0 WORKMANSHIP**

- 17.1 Install equipment, ductwork, conduit and piping in a workmanlike manner to present a neat appearance and to function properly to the satisfaction of the Consultant. Install ducts and pipes parallel and perpendicular to the building planes. Install all piping and ductwork concealed in chases, behind furring, or above ceiling, except in unfinished areas. Install all exposed systems neatly and group to present a neat appearance.
- 17.2 Install all gauges, thermometers, etc., to permit easy observance.
- 17.3 Install all equipment and apparatus which requires maintenance, adjustment, or eventual replacement with due allowance for this.
- 17.4 Install control valves to guarantee proper sensing. Shield elements from direct radiation and avoid placing them behind obstructions.
- 17.5 Install all panels and boards, etc., to permit easy operation.
- 17.6 Include in the work all requirements of Manufacturers as shown on their drawings.
- 17.7 Replace all work unsatisfactory to the Consultant without extra cost and to the standard required by the Consultant. This applies to any item which is found to be defective in service during the maintenance period, or extended maintenance period as appropriate.
- 17.8 Where height dimensions are given, the installation shall be reasonably expected to be within 1/8" (3mm) of the design dimensions. Where room center or proportion dimensions are not shown these shall be reasonably expected to be located on site to within 1/2" (13mm) of actual position.
- 17.9 Items displaying a horizontal dimension or edge must be plumbed with a level, and must be to the Consultant's satisfaction. Any apparent discrepancy between "level" electrical items and adjacent items MUST be reported immediately to the Consultant.

## **18.0 MATERIALS AND EQUIPMENT**

- 18.1 All materials used on the project shall be brand new and of high quality. Obtain approval of all manufacturers from the Consultant and the Owner.

18.2 "New" is defined as newly manufactured, "state of the arts," tested and proven item of equipment. Items which have been held in stock for any extended period of time by either the Manufacturer or the Supplier will be rejected.

18.3 All materials used on this project shall bear the third-party quality assurance stamp like British Standard kitemark, UL, FM, NFPA and/or other quality assurance authority as stated herein elsewhere.

## **19.0 CLEANING**

19.1 Each day as the work proceeds and on completion, clean up and remove from the premises all rubbish, surplus material, equipment, machinery, tools, scaffolds, and other items used in the performance of the work. Clean out dirt and debris and leave the buildings broom clean with no stains and in a condition acceptable to the Consultant.

19.2 Where electrical items form part of the visible finish in the rooms, protect from over-painting, etc. and give all items a final cleaning before handing over of the project.

## **20.0 ACCESSIBILITY**

20.1 Each item of equipment shall be located so as to be accessible for maintenance or repair without removing adjacent structures, equipment, piping, ducts, or other materials. For the large Air Handling units, the Contractor shall ensure that these can be built up on site from components which can be taken into the area. Any openings, framing or special lifting equipment which may be required shall be included by the General Contractor.

20.2 Any item of equipment needing maintenance shall be located so as to be accessible for maintenance or repair without removing adjacent structures, equipment, piping, ducts, or other materials.

20.3 Cleanouts shall be located to permit rodding of all drain lines. These shall be located wherever possible external to occupied areas, and to minimize spillage problems during rodding.

## **21.0 CUTTING AND PATCHING.**

21.1 Inform all other Divisions in time concerning required openings. Obtain the approval of the Consultant before doing any cutting.

21.2 Obtain the approval of the Consultant before doing any cutting. Supporting members of any floor, wall or the building structure shall only be cut and in such a manner as approved by the Consultant. All reinstatement work must be done to the same standard as the original work.

## **22.0 INSERTS, SLEEVES, ESCUTCHEONS AND CURBS.**

22.1 Under this section supply necessary sleeves and other inserts to other trades.

22.2 Use only factory made, threaded or toggle type inserts as required for supports and anchors, properly sized for the load to be carried. Place inserts only in portions of the main structure and not in any finishing material.

22.3 Use factory made expansion shields where inserts cannot be placed, but only where approved by the Consultant and for light weights.

22.4 Do not use powder activated tools except with the written permission of the Consultant.



- 22.5 Supply and locate all inserts, holes, anchor bolts, and sleeves in good time when walls, floors, and roof are erected.
- 22.6 Pass insulation unbroken where pipe or duct is insulated. Size sleeves to provide adequate clearance all around.
- 22.7 Use the following materials for pipe sleeves:
- a) Through all interior walls above grade use uPVC Class D pipes, machine cut, flush with finished structure. For non metallic pipe crossing in fire wall use steel pipe (medium).
  - b) Through all exterior walls above grade use uPVC Class D pipes, machine cut, flush with finished structure inside and to suit flashing on outside.
  - c) Through all exterior walls below grade and all other waterproof walls use uPVC Class D sleeves, machine cut. Check flashing specification for further details.
  - d) Through all waterproof floors, janitor's closets, mechanical rooms, kitchens, roofs, use uPVC Class 150 sleeves, machine cut. Extend sleeves 100 mm above finished floor upwards and cut flush with underside of floor. Refer to flashing detail through waterproof floors.
  - e) Provide 100 mm high, 100 mm wide watertight concrete curbs with 20 mm chamfered edges around all pipes passing through waterproof floors except where furred in. Read Concrete specification before proceeding with curbs. All gaps to be filled up with waterproof type sealant.
  - f) Where uPVC pipes pass through fire rated walls/partitions, provide fire rated sleeves.
- 22.8 Pack all sleeves between the insulated pipe and the sleeve or where un-insulated between the pipe and the sleeve with loose fibreglass insulation. Seal the annular space as follows:
- a) For all horizontal sleeves in exposed areas, use a seal equal or better fire rated than the wall to be sealed.
  - b) For all vertical sleeves through roofs, janitor's closets, equipment rooms, use permanently resilient silicone based sealing compound, non- inflammable and waterproof.
- Ensure that the seal is compatible with floor and ceiling finishes. Check the room finishing schedules for further details and clarify if necessary with the Consultant.
- c) Cover exposed floor and wall pipe sleeves in finished areas with satin finish chrome or nickel plated solid brass or with satin finished stainless steel escutcheons with non-ferrous set screws. Split cast plates of the screw locking type may not be used. Do not use stamped steel friction type split plates
- Through roofs, provide curbs and sleeves as shown on drawings and to suit flashing requirements.
- 22.9 Use the following sleeving for ducts:
- For rectangular duct openings through walls and floors a removable hardwood box-out shall be provided of the required size. Soft wood or plywood will not be acceptable. Hardwood shall be further protected with anti-termite paint / solvent.
- Through fire walls, build fire dampers into wall, or make detailed fixing in accordance with DW 144 / SMACNA Standards.

Through floors where ducts are not furred in or enclosed in a ductshaft, provide 100 mm high and 100 mm wide watertight concrete curbs, with 25 mm chamfered edges all around. Extend sleeves where used flush to top of curb. Read Concrete Specification before proceeding. Through floors where duct is enclosed in a duct shaft or furred in, provide the watertight curbs at the extreme top and bottom only.

22.10\* after ducts are installed, pack the opening and seal the packing as follows:

- a) Use fibreglass insulation for packing
- b) Through all vertical walls seal the fibreglass packing using permanently resilient silicone based sealant.

22.11 Brace duct sleeves and box-outs to retain their position and shape during the pouring of concrete and other work.

22.12 Provide bracing for each duct at every passage through structure to prevent sagging.

22.13 Cover exposed duct sleeves and openings in exposed areas. Use 100 mm galvanized steel escutcheons in the form of a duct collar. Over curbs extend the collar 30 mm down the side of the curb, similar to counter flashing. Fix collar in place with Cadmium plated screws.

## **23.0 ACCESS PANELS AND DOORS.**

23.1 Install all concealed mechanical equipment requiring adjustment or maintenance in locations easily accessible through access panels or doors. Install systems and components to result in a minimum number of access panels. Indicate access panels on as-built drawings.

23.2 Provide the respective Division of work with panels, doors or frames, complete with all pertinent information for installation.

Ensure that access doors are installed in a manner to match the building grids where applicable

23.3 Prepare detail drawings showing location and type of all access doors in coordination with other trades before proceeding with installation and hand these to the Contractor to obtain approval.

23.4 Size all access doors to provide adequate access and commensurate with the type of structure and Architectural finish. Should it be necessary for persons to enter, provide a minimum opening of 600 x 450 mm.

23.5 Ensure proper fire rating of access doors in fire separations

23.6 Lay-in type ceiling tiles, if properly marked may serve as access panels.

23.7 Provide panels in glazed tile walls of 2.6mm thick 304 alloy stainless steel, no. 4 finish, with recessed frame secured with stainless steel, countersunk, flush-headed screws.

23.8 Provide panels in plaster surfaces with dish shaped door and welded metal lath, ready to take plaster. Provide a plastic grommet for door key access.

- 23.9 Provide other access doors of welded 2.6 mm thick steel, flush type with concealed hinges, lock and anchor straps, complete with factory prime coat. Submit details to Consultant for approval.

#### **24.0 FLASHING**

- 24.1 Flash all mechanical and electrical parts passing through or built into a roof, an outside wall, or a waterproof floor.
- 24.2 Provide uPVC sleeves with puddle flanges for services crossing water proofed walls in basement.
- 24.3 Provide stainless steel puddle flanged connections where pipework penetrates walls of the water storage tank.
- 24.4 All flashing shall suit roof angle and shall extend a minimum of 400 mm on all sides. Leave the flashing as directed by the Contractor for him to build into the roofing system to render a completely watertight connection.
- 24.5 Provide counter flashing on all stacks, ducts, and pipes passing through roofs to fit over flashing or curb.
- 24.6 Flash pipes through waterproof floors as per approved detail.
- 24.7 Provide pipes and sleeves passing through outside walls with lead or copper flashings and as directed by the Consultant. All visual aspects of such sleeves to be approved by the Consultant.
- 24.8 Pay special attention to the waterproofing conditions of basements and walls and floors that may exist. Cooperate at all times with the waterproofing trade and do not cut or destroy any waterproofing seal without the consent of the waterproofing trade.

Provide piping sleeves passing through waterproof walls with asphalt roofing felt wrapped around so as to leave a 25 x 50 mm recess on both sides of the wall. These recesses and the space between pipe and sleeve will be caulked by the General Contract, Moisture protection, or Waterproofing, Division.

#### **25.0 GENERAL WELDING REQUIREMENTS.**

- 25.1 All welding shall be generally in accordance with ANSI B31. This will be modified where appropriate for other materials and may be relaxed or varied by order of the Consultant, provided that the Contractor has made a reasonably comprehensive request for an alternate.
- 25.2 Tack welds shall be performed by fully qualified welders and all tack welds shall be of a length equal to twice the pipe thickness and shall fully penetrate the pipe walls.
- 25.3 Where welding is carried out in the proximity of inflammable materials special precautions shall be taken to prevent risk of fire or other damage to the building fabric.
- 25.4 Where oxyacetylene cutting equipment or any welding plant is being used by an operative for any of the works defined in the contract documents, then fire extinguishers shall be supplied and carried as part of the equipment. The operators of cutting and welding equipment shall be trained in the use of the fire extinguishers which they carry and all extinguishers shall be fully charged and ready for use. In all cases, extinguishers shall be positioned immediately adjacent to the position where cutting and welding is being carried out and shall be readily accessible for use in the event of an emergency.

- 25.5 All accommodation, benches, tools, welding plant, acetylene, oxygen or electricity, filler rods and electrodes, which are necessary for installations where welding is required shall be provided as part of this contract.
- 25.6 Welding shall not be done at a temperature of 5 degrees Centigrade or below unless the parent metal is preheated by torches or other approved means until it is warm to the hand (about 27 degrees C) for about 150mm either side of the joint. No welding shall be done below minus 18 degrees C. After preheating, the heated portion including the welded joints shall be covered with muffs or suitable insulation materials to allow cooling free from draughts. Any open ends of the pipe or assembly shall be effectively sealed to prevent heat flow by convection.
- 25.7 All welded pipe assemblies shall be constructed so that individual welded joints do not affect each other. The distance between the centres of adjacent welds shall be not less than twice the bore diameter of the pipe.
- 25.8 No welded joints shall be left partially completed. Any joints tacked in position must be promptly finished within the working day. The Consultant will reject all work not done in accordance with this instruction.
- 25.9 Where work is rejected, pipes must be machine cut at least 150 mm either side of rejected welds and proper weld preparation must be used on the shortened sector. Where shorter fill-in sections are required because of such rejection and reworking, then new full-sized lengths must be supplied by the Contractor.
- 25.10 Where pipes with longitudinal seams are specified, pipe seams shall be arranged such that adjacent seams are opposed 45 deg. from each side of top dead centre and branches shall be made only with weldable fittings
- 25.11 All filler metals which are coated shall be protected from excessive moisture changes. Filler materials or flexes which show any sign of deterioration shall not be used. If requested by the Consultant samples of filler rods to be used shall be submitted to him for approval before any work is done on site.

These may be submitted, at the Contractor's cost, to an independent testing laboratory for verification.

## **26.0 TESTS FOR WELDER QUALIFICATION.**

- 26.1 The purpose of the welder's qualification tests is to determine the ability of the welders to make sound and acceptable welds. Before any site welding on the contract is allowed, each proposed welder shall carry out the tests required in the presence of the Consultant, or the Employer's Insurance Inspector.
- 26.2 Any weld test specimens which have been suitably marked and approved shall be kept on site by a responsible person, so that they can be produced at any time, at the request of the Consultant.
- 26.3 All accommodation, benches, tools, welding plant, acetylene, oxygen, electricity, test pieces, filler rods, electrodes, facilities for cutting and grinding, polishing, bending and examining, which are necessary for welder's qualification tests shall be provided by the Contractor. In the absence of any items for inspecting the welds, the Consultant may submit the finished samples to an independent laboratory for testing at the Contractor's expense.
- 26.4 Under no circumstances shall a welder be employed on the contract, either on or off the site for welding operations other than those for which that welder is qualified.
- 26.5 Copies and records of all test reports shall be promptly given to and kept by the Consultant.

- 26.6 Even welders holding a certificate for welding are to be tested at this job site in full accordance with the specifications.

26.7 Tests for Qualification on Steel Pipes

26.7.1 The Contractor shall be responsible for the quality of welding and brazing and shall provide the testing of the welders employed. Certificates of such proficiency test, together with stamped samples provided for Engineer's review before welding work is commenced. The welder's proficiency standard to be equal to Lloyd's Standard Class 2, certified by an authorized testing laboratory paid for the contractor.

26.7.2 Qualification tests will be made for butt and branch welds in pipes. One test piece per welder will be required for the qualification of welding of pipes exceeding 100mm nominal diameter.

26.7.3 In addition, test pieces will be required for each position of a pipe. These positions will be horizontal and rotated while welding, horizontal and fixed while welding, vertical and fixed while welding for each range of nominal diameters specified.

**27.0 TESTING OF WELDED PIPEWORK INSTALLATION.**

- 27.1 Testing of welded pipework installation shall be by either destructive or non-destructive test method.

- 27.2 Each butt-welded test piece to be subjected to two root bend tests and two nick break tests of the welding.

Each branch welded test piece will be subjected to two root bend tests and two fillet break tests of the welding.

Each test piece weld to be radiographically examined.

- 27.3 The Contractor shall assign an identification symbol to identify each welder and this will be used to stamp the work done by that welder.

- 27.4 The Contractor will compile a set of safety instructions concerning welding procedures for the welders and after review by the Engineer, these will be issued to the welding operatives and foremen.

- 27.5 The Contractor shall allow for two percent of all the welds to be radiographically examined.

Determination of piping sections to be tested will be designated by the Engineer.

The timing of the radiographic examination of a weld or welds designated to be so examined will be confirmed in writing by the Engineer during the installation and commissioning.

Radiographic examination of welds will be carried out by a specialist employed by the Contractor subject to review by the Engineer. The specialist so employed will be responsible for the proper indexing, interpretation and reporting of the results of the radiographic tests.

- 27.6 Any weld, whether radiographically tested or not which are considered by the Engineer to be faulty, either in whole or in part, or which have failed under hydraulic or other tests so designated by the Engineer, will be removed and replaced by the Contractor.

- 27.7 The faulty part of a weld may be removed by flame gouging, chipping or grinding. Otherwise, the Engineer may instruct that the portion of the works containing the fault weld be cut out and replaced by the welding in of a new section of pipe or member.
- 27.8 Should any of the welder's work (as defined by the Engineer) be found to be faulty during radiographic tests, then that welder shall be removed from site and all other welds undertaken by that welder shall be checked by radiographic means and replaced as necessary at the expense of the contractor.
- 27.9 The extra radiography involved shall not be part of the 2% inspection previously stipulated.

## **28.0 PAINTING, IDENTIFICATION AND VALVE TAGS**

- 28.1 All painting of plant, equipment, storage vessels, and the like, and all surfaces to be painted in the areas where such items are installed, including generator area, condenser areas, etc., shall be carried out by an approved specialist painting contractor.
- 28.2 Painting shall be applied to all exposed pipework, ductwork and associated components, valves, fittings, etc., equipment, supports of any kind, insulation, plain mild steel, copper, or cast-iron surfaces. Where movement is required between adjacent surfaces, request clarification from the Consultant.
- 28.3 The following items do not require site painting:
- (a) Insulation having any of the following finishes:
- (1) Aluminium foil.
  - (2) Hot Dip Galvanized angles / channels
  - (3) Polyisobutylene sheeting/ bitumen coating.
  - (4) Vinyl-glass/ resin-aluminium foil laminate.
- (b) Equipment or plant or supports or frames delivered to site with any of the following finishes:
- (1) Painted finish, other than primed only, provided that the finish is not damaged in any way.
  - (2) Stainless steel or plastic-coated steel. (3) Stoved enamel.
- 28.4 All surfaces to be painted shall be prepared by thoroughly cleaning and removing all rust, grease, oil, dirt and surface corrosion, using wire brush, emery paper and/ or degreasing medium as required. The paint shall be applied in accordance with the maker's instructions and the type of paint to be used shall be in accordance with the following:
- Ferrous surface, one coat of Calcium Plumbate metal primer plus wash primers as necessary, followed by 2 undercoats and one finishing coat.  
Non-Ferrous surface, one coat of zinc chromate primer plus wash primers as necessary, followed by 2 undercoats, and one finishing coat.
- All paint shall be manufactured by approved manufacturer and type of paint shall be selected by the manufacturer based on the location i.e. Indoor / Outdoor.
- 28.5 For factory applied finishes, repainting or refinishing of any surfaces damaged during shipping, erection or construction shall be done using only factory supplied materials.

- 28.6 After insulation or final painting is completed, identify each piped and ducted service. Locate identification bands, stickers and arrows:
- (a) Behind each access door.
  - (b) At each change of direction on all joining pipes and ducts.
  - (c) At no more than 10 meters apart in straight runs of exposed pipes and ducts, but on both sides of sleeves.
  - (d) At no more than 10 meters apart in straight runs of pipes and ducts behind removable enclosures such as lay-in ceiling but on both sides of sleeves.
  - (e) Above each floor or platform for pipes in shaft and vertical exposed pipes, preferably 1.5 meters above floor or platform level.
- 28.7 Identification Bands, stickers and arrows shall be made of self-adhesive plastic long lasting type. The stickers when used outdoor shall be weatherproof type suitable for the temperature in UAE.
- 28.8 Identification of ductwork shall be carried out as per DW 144 standards coloured triangle with the type of service. Additionally, the AHU / FAN reference no shall be identified in the ducts.
- 28.9 All insulated or un-insulated pipework shall be identified as detailed above by means of PVC tape identification bands in accordance with British Code to BS 1710.
- In addition, the name of the service and flow direction arrow shall be provided.
- 28.10 The access doors and fire dampers in duct shall be identified by a suitable sticker.
- 28.11 All identifying band colours and finishing colour of services shall be agreed with the Consultant prior to application.
- 28.12 All equipment shall have a nameplate secured to the item giving the following information:
- Equipment reference number (as indicated on the record drawings.) Make / Model No.
- System.
- Room/ Area served. Duty/ output information.
- Power fed from:
- The name plate shall be 100 x 100 mm approximately, of white plastic 3 mm thick with the above information engraved in black lettering and the plate shall be secured by screws, bolts, clips, etc., as appropriate to the item concerned.
- This plate is in addition to any name plate supplied by the manufacturer of the item giving detailed specification information for the equipment.
- 28.13 All main valves in plant room, roof, and risers shall be numbered system wise, each with a unique number. The number shall be engraved in a tag 30mm die and suspended with a stainless-steel chain from the valve handle.
- The tag shall be of aluminium material for outdoors and plastic indoor.
- The piping schematic diagrams shall have the valve numbers with valve chart and the same shall be enclosed as a part of operating instruction and shall be framed with glass and displayed in the relevant plant rooms.

## **29.0 FLUSHING AND CLEANING.**

- 29.1 Flush all fluid-carrying systems after completion with a stream of water or appropriate fluid at the highest obtainable pressure and velocity. Discharge the flushing fluid through all strainers and out through line sized valves with hose ends. Clean all strainers. Repeat flushing operation to the approval of the Consultant until the water is running clear, colourless and odourless and the conductivity is no more than 30 ms/cm above the mains water.
- 29.2 If the water pressure normally available on site is, in the Consultant's opinion, inadequate to perform proper flushing, then the Contractor shall provide at his own expense, pumps, tanks, etc., to give pressure and volume flows to the Consultant's satisfaction.
- 29.3 For layer systems, the flushing and cleaning shall be carried out in sections of every three floors.
- 29.4 During flushing and cleaning all cooling coils, heat exchangers shall be bypassed in order to avoid debris / dirt going to coils.

## **30.0 INSPECTION, TESTING AND ADJUSTING.**

- 30.1 All the works provided as part of this Contract shall be inspected and commissioned in accordance with all relevant Specifications, and Codes of Practice and to the entire satisfaction of the Consultant.
- 30.2 The Contractor shall employ the services of a specialist company approved by the Engineer regularly engaged in providing a testing balancing and commissioning service and who has been in continuous business for not less than 7 years employing fully trained staff having not less than 3 years dedicated experience.

A senior experienced Commissioning Engineer with minimum dedicated experience of 7 years shall be responsible for supervising and directing the activities for the testing and commissioning team.

- 30.3 Carry out all tests specified. Carry out all tests required by Authorities having jurisdiction. Test equipment to the requirements of, and where necessary, in the presence of the Manufacturer.
- 30.4 Provide all equipment, labour, instruments, loading devices, incidentals, and pay for all fuel, power and sundries required to carry out the tests.
- 30.5 All installations shall be inspected and tested in sections as the work proceeds and on completion as composite systems and it shall be noted that the Consultant or any of the other relevant Authorities may require inspecting or test any equipment during manufacture at the Manufacturer's works. All necessary arrangements shall be made as part of this Contract.
- This will generally not apply to specified items unless specifically noted in these documents, but may be necessary for alternate equipment, should this be considered at all.
- 30.6 All tests shall be arranged in cooperation with the Consultant and his Engineer and all other concerned parties and shall be subject to at least five (5) days' notice in writing of the time, location and nature of the test to be performed. No test shall be considered valid unless the Consultant is present.
- 30.7 All necessary skilled and unskilled labour shall be provided for attendance during the tests (including pre- and post- test activities,) and the test media shall be provided and subsequently disposed of except where specifically stated otherwise.



- 30.8 The testing and adjusting is the contractual responsibility of the Contractor but actual performance of the tests is expected to be the sole responsibility of the Subcontractor.
- 30.9 Have all testing and balancing performed by only persons who are thoroughly versed in this type of testing and balancing and with proven ability. Submit names, complete with experience records, and references for approval of the Consultant.
- 30.10 Any defects occurring at any time during the test duration shall be made good and a complete re-test shall be carried out, all at no cost to the Contract.
- 30.11 Where failure occurs during a test, inspection or commissioning procedure which results in damage to the building fabric and/ or any services not provided as part of this Contract, or requires subsequent builder's work to be carried out, carry out all such repair work to the entire satisfaction of the Consultant at no cost to the Owner or the Consultant.
- 30.12 All test points shall be provided which are necessary to carry out the specified tests and commissioning procedures including facilities for measuring or monitoring temperature, pressure, pressure drop, volume flow, in-duct sound power or sound pressure, humidity, or other relevant conditions in both air side and water side systems. Such points shall be fitted with removable plugs, flanges, or other appropriate and approved devices.
- 30.13 Prepare test report forms for each test to be performed and submit these to the Consultant at least two weeks prior to the commencement of any tests.
- 30.14 Test only after the system installation has been completed and the system has been put into continuous operation. Perform the testing, adjusting, and balancing when outside conditions are commensurate with the design conditions for the given system. Add dummy loads to the system if outside and inside conditions are less severe than the specified points.
- 30.15 Prepare a complete list of instruments for each test containing for each instrument:
- Name of instrument Manufacturer.
  - Scale and full-scale accuracy.
  - Date of last calibration test.
  - Name of last calibrating company.
- All instruments and consumables, such as recording paper, necessary for conducting the tests shall be provided, including but not limited to the following:
- Tube type velometer.
  - Sloping tube manometer.
  - Pitot tubes of various lengths.
  - Mercury in glass thermometers.
  - Weekly recording thermometers & digital thermometers.
  - Weekly recording relative humidity meter.
  - Anemometer for diffusers, with collector.
  - Ammeter, voltmeter, wattmeter.
  - Power factor meter.
  - Insulation tester.
  - Earth loop impedance tester.
  - Tachometer.
- 30.16 Duplicate signed test certificates shall be provided after each test which will be countersigned by the Consultant who witnesses the test. The test certificate shall give the following particulars: -
- Apparatus or section under test.
  - Maker's number (if any).

- Nature, duration and conditions of test.
- Result of test.

No test shall be valid until the test certificate is provided.

- 30.17 Duplicate copies of test certificates carried out at Manufacturer's works shall be forwarded to the Consultant for approval prior to despatch of the article to site.
- 30.18 No section of the works shall be insulated or in any other way concealed prior to testing and inspection and subsequent concealment where applicable shall only take place following written authority from the Consultant.
- 30.19 All necessary facilities, measuring and recording instruments including test pumps and gauges for inspection, testing and commissioning requirements shall be provided and shall be checked or calibrated as necessary before use.

The Consultant reserves the right to call for a demonstration of the accuracy of any instruments provided.

- 30.20 All representatives present during inspection, testing and commissioning shall be fully conversant with the system concerned and the method of system and instrument operation.

Manufacturer's or Specialist Subcontractors' representatives shall attend where specifically indicated elsewhere in the specification or where necessary to ensure full service and cooperation is available to the Consultant to enable the works to be tested and commissioned in accordance with the requirements of the specification.

- 30.21 All necessary precautions shall be taken to safeguard against damage during inspection, testing or commissioning. Any damage so caused shall be made good at no cost to the contract.
- 30.22 All tests shall last for the minimum time period stated or longer if necessary to ensure all sections have been fully examined as required by the test.

One set of test instruments shall be made available during the warranty period to check the system whenever required.

### **31.0 TEMPORARY USAGE**

- 31.1 The Consultant reserves the right to use any piece of electrical and mechanical equipment, device, or material installed under this section for such reasonable lengths of time as may be required during construction (wild air) and to make a complete and thorough test of same before the final completion and acceptance of the work.

Such tests shall not be construed as acceptance of any part of the work, and it is agreed and understood that no claim for damage will be made for any injury or breakage to any part or parts of the above due to this use, where caused by weakness, inaccuracy of parts, or by defective materials or workmanship.

### **32.0 INSPECTION OF THE WORK.**

- 32.1 The representatives of the Owner and the Consultant will make periodic visits to the site during construction to ascertain that the work is being executed in reasonable conformity with all plans and specifications, but will not execute quality control at all times. Each Subcontractor must maintain the quality control as intended in the contract documents.

32.2 Correct all deficiencies immediately as noted during field inspections.

32.3 Request in writing that a final inspection of all services.

Do not issue this written request until:

All deficiencies noted during job inspections have been corrected.

All systems have been balanced and tested and are ready for operation. All balancing reports have been submitted and reviewed.

All instruction manuals have been submitted and reviewed. The cleaning up is finished in all respects.

All spare parts and replacement parts specified have been provided and receipt of the same acknowledged.

### **33.0 OPERATING INSTRUCTIONS TO OWNER.**

33.1 Instruct the Owner's staff how to make minor adjustments, carry out necessary maintenance and how to operate each system.

For new equipment, provide the Consultant with three copies of complete operating and maintenance instructions for equipment at the time of delivery of the equipment.

Equipment will be rejected unless accompanied by instructions. Such documents must be received at least one month prior to the completion date of the relative section of the works.

Instructions shall be bound in a suitable loose-leaf booklet or binder, and shall include:

- a) Prints of the following drawings:
  - 1) List of all equipment installed.
  - 2) General layout.
  - 3) Wiring diagram of control panels.
  - 4) Non-dimensional layout, showing location of all electrical devices (Limit Switches etc.).
- b) Operation instructions include start-up and shutdown procedure.
- c) Maintenance instructions, including preventative maintenance instructions for components of the equipment.
- d) Lubricating instruction and recommended cycle of lubrication.
- e) Complete parts list, showing manufacturer's name and catalogue number.
- f) Complete list of recommended spare parts for 2-year operation with price list for each item of equipment.

The contractor is to prepare a draft for discussion with the Consultant prior to finalizing the documents.

Portable tools and spare parts shall be correctly labelled and handed over to the Consultant.

- 33.2 The section dealing with complete systems shall be subdivided into each service with a ready means of reference and detailed index. The function and manner of operation of each system shall be clearly described together with illustrations and line diagrams in schematic form showing the location and function of control valves, items of equipment and spaces or areas which are served by these items. The colour coding and identification systems employed shall be explained.

#### **34.0 COMMISSIONING OF SERVICES.**

The Commissioning of HVAC systems shall be carried out by one of the specialized commissioning agencies approved by the Engineer. The agency shall be appointed by the MEP Contractor at an early stage of the project.

- 34.1 All piping systems shall be flushed, chemically cleaned and then filled with treated water, or appropriate fluid, vented as necessary, and brought to operating conditions and the flows then regulated to the design values.
- 34.2 All water circuits shall be balanced by means of the regulating valves provided and flow rates shall be determined on a temperature and pressure drop basis. Flow through pumps shall be measured by relating the pressure drop across the pump to manufacturers test curves. A copy of the test curve indicating the final operating point shall be forwarded to the Consultant.
- 34.3 All refrigerant systems shall be vacuumed, purged and charged according to A.S.H.R.A.E. standard methods. All safety controls and interlocks will be tested as part of the commissioning procedures.
- 34.4 Flow through coils shall be adjusted to give the design temperature difference. In general water systems shall be balanced in accordance with the guidelines specified in NEBB Manual. Upon completion of balancing and testing operations for heating and cooling systems, temperature measurements shall be taken in all rooms and the readings tabulated in schedule form together with hourly ambient temperature readings taken over the measuring period. Two copies of the schedule shall be forwarded to the Consultant.
- 34.5 All ventilating systems shall be commissioned in accordance with the procedures recommended in the AABC / CIBSE Manual. Duplicate schedules of commissioning results shall be forwarded to the Consultant and shall detail the recorded air volume and percentage deviation from design air volume, for each air input and extract terminal. Wet and dry bulb temperature measurements shall be taken in all rooms served by air supply systems and the results indicated on the schedule of commissioning results.
- These shall be recorded together with external ambient wet and dry bulb temperatures measured at hourly intervals over the measuring period.
- 34.6 Sound pressure levels (dBA) shall be measured in all rooms containing supply or extract terminals, all plant rooms, all rooms immediately adjacent to plant rooms and all rooms located above or below plant rooms. The Consultant may at his discretion request a spectrum band analysis of sound pressure in any locations.
- 34.7 All automatic controls shall be commissioned by the controls supplier.
- 34.8 Fault conditions shall be simulated and all alarms and safety devices shall function correctly. Such proving tests shall be carried out in the presence of the Consultant and certificates shall be provided specifically detailing all check procedures which have been carried out.

34.9 Following proving tests all installations shall be left in working order ready for handover.

34.10 All electrical systems shall be tested and commissioned in accordance with IEE regulations and local authority codes.

### **35.0 SYSTEM ACCEPTANCE.**

35.1 The ultimate condition for system acceptance is that the Owner and Consultant have inspected the system and found it to be acceptable, and indicated this in writing.

Issuance of the final payment certificate does not necessarily indicate system acceptance, neither does release of final payment holdback in whole or in part. The Consultant's acceptance may be contingent on any or all of the following if applicable:

35.1.1 Submit original copies of letters from manufacturers of all systems indicating that their technical representatives have inspected and tested the respective systems and are satisfied with the methods of installation, connections and operation.

35.1.2 Submit "as built" drawings and operation and maintenance manuals.

35.1.3 Train owner's maintenance staff.

35.1.4 Satisfactory completion of 12 months guarantee period.

### **36.0 CORRECTION AFTER COMPLETION.**

36.1 Remedy all work in accordance with the General Conditions of Contract for a period of one year from the date of acceptance of the completed work, or if so noted, for longer periods for parts of the work.

36.2 Attend immediately to any and all the defects occurring during the period defined above and repair in a manner to prevent recurrence. This contractor is responsible for all work required by other trades necessary to repair the works of this section, or necessary to repair damage caused by the failure of any part of this section.

36.3 Instruct all Suppliers and Manufacturers that guarantees on equipment will commence when the completed work is accepted and not from the date the equipment is put into operation. In the event that this condition is omitted by the supplier, or if subsequent cost to the Owner is involved, this Contractor shall be liable to the full extent of the guarantee as indicated in the specification.

### **37.0 GUARANTEES**

37.1. The Contractor will guarantee all material and workmanship for at least one year after preliminary take over by the Owner.

37.2. All warranties from equipment suppliers will be vested in the Owner, regardless of whether the Contractor who supplied the equipment is still associated with the project or not.

37.3. Guarantees will be full guarantees and will include all overhead, profit, incidental charges and sundries.

37.4. Where damage is caused to any other item by any failure of the item guaranteed, then the guarantee shall also include the costs incurred in rectifying that damage.

### **38.0 MAINTENANCE**

- 38.1. Maintenance is defined as the Contractual Liability to maintain the equipment in working condition, PLUS the regular checks and servicing of equipment during the maintenance period to keep the equipment in best working order.
- 38.2. Regular maintenance shall be as necessary, but in any event not less frequently than monthly.
- 38.3. In the event that the Owner has his own staff, the Contractor is still to check monthly and advise on any problems and is still to assume responsibility.

End of Section

**SECTION 2**  
**INSULATION**

## SECTION 2

### INSULATION INDEX

|     |                                |    |
|-----|--------------------------------|----|
| 1.0 | INSULATION GENERAL .....       | 29 |
| 2.0 | CONCEALED COLD AIR DUCTS ..... | 30 |
| 3.0 | EXPOSED COLD AIR DUCTS.....    | 31 |
| 4.0 | ACOUSTIC INSULATION.....       | 31 |
| 5.0 | APPARATUS CASINGS. ....        | 31 |
| 6.0 | PIPE INSULATION .....          | 32 |
| 7.0 | LAGGING ADHESIVE .....         | 33 |
| 8.0 | ALUMINIUM CLADDING .....       | 33 |
| 9.0 | SAMPLES .....                  | 33 |



## 1.0 INSULATION GENERAL

- 1.1 The whole of the insulation work shall be carried out by an approved specialist insulation Contractor. All allowances shall be included for arranging a specialist subcontract accordingly and for informing the specialist Sub-contractor of all conditions relating to the Contract and for coordinating his works with the remainder of the works.
- 1.2 All allowances shall be included for informing the specialist Sub-contractor of all details of the building structure, programme arrangements, and other relevant details at the time of tender and for all necessary visits to site by the Sub-contractor or his workers.
- 1.3 All insulation shall be in accordance with British Standard Specifications and Codes of Practice in all respects. Insulation meeting A.S.T.M. Standards will be considered if properly submitted.
- 1.4 In addition to complying with the relevant standards, all insulating material shall be free from Asbestos.
- 1.5 Insulating materials shall be acceptable only if they are equal to or better than the grades or classes of fire resistance as follows:

B.S. 476, Part 4, (Noncombustible grade.)

B.S. 476 Part 5 Class P, (Not easily ignitable.)

B.S. 476, Part 6, (Fire Propagation Index a maximum of 12.6.) B.S. 476, Part 7, Class 1, (Surface spread of flame.)

B.S. 476 Part 9, (Production of emitted smoke shall not give more than 35% obscuration of the light beam.)

All insulation finishes and coverings shall be classified as Class 1 (one) surface spread when tested in accordance with B.S. 476, Part 7.

- 1.6 All adhesives, mastics, coatings, sealers and primers shall be classified as Class 1 surface spread when tested in accordance with B.S. 476, part 7.

They shall not in any way attack the insulation or the surface to which the insulation is being applied and shall be suitable for the working temperatures.

- 1.7 Insulating materials shall have thermal conductivity values not more than those listed as follows:

| MATERIAL<br>W/M/DEG.C | TYPE      | THERMAL<br>CONDUCTIVITY |
|-----------------------|-----------|-------------------------|
| Mineral wool          | Sectional | 0.04                    |
| Mineral wool          | Slabs     | 0.04                    |
| Fibre Glass           | All       | 0.034                   |
| Closed Cell           | All       | 0.038                   |
| Polyurethane          | Sectional | 0.025                   |
| Styrofoam             | Rigid     | 0.026                   |

All conductivity figures are rated at an average temperature of 20 deg C.

- 1.8 The closed cell elastomeric foam insulation shall have following characteristics in addition to clause 1.5:

- a. Water vapour permeability - 0.09 ugm/Nh, BS 4370/2 (= 7000 to DIN 52615)

- b. Density - 55 to 70 kg / m<sup>3</sup> or 4Lb / ft<sup>3</sup> to 5Lb / ft<sup>3</sup>
- c. Closed cell content - > 90%.
- d. Corrosion risk assessment - Din 1988 / 7 Certified.
- e. Noise reduction - Din 4109 up to 32 dB (A).
- f. Thermal conductivity of 0.037 w/mk at 10°C and 0.038 w/mk at 20°C to BS 874 PART 2 – 86 (DIN 52613, 52612).

- 1.9 All material delivered to site shall be new and fully dried out and so maintained throughout the progress of the works.

All insulating materials shall be stored in storage sheds, and in accordance with the Manufacturer's recommendations.

- 1.10 In order to ensure that the insulation applied is in all respects in accordance with the specification, sections shall, as required by the Consultant, be cut from the finished insulation.

The Contractor is to allow in his price for the removal and replacement of two sections of each type of insulation.

If, however, defects are revealed, further sections shall be cut out for inspection, and all cutout sections shall be replaced at no cost to the Contract.

If further defects are revealed then the Consultant shall have the right, when in his opinion it is necessary, to issue instructions for any part or the whole of the insulation to be removed and replaced. The replacement with new insulation shall be to the satisfaction of the Consultant and the cutting out and replacing shall be at no cost to the Contract.

- 1.11 Particular attention shall be paid to the finished appearance of all thermal insulation which must present a neat and symmetrical appearance running true in line with pipe layouts, etc.

- 1.12 Any rough, irregular and badly finished surfaces shall be stripped down and re-insulated to the Consultant's satisfaction.

- 1.13 In certain cases the type of insulation specified will require a painted finish, in addition to identifying bands and any signs which are to be applied.

- 1.14 All systems are to have been tested and approved by the Consultant prior to installation of insulation.

- 1.15 Manufacturer shall have production process control in accordance with ISO 9002 / UNI EN 29002 or any other equivalent standard.

- 1.16 In case of heavy ducts and pipes wherein the supports are fixed directly on ducts / Pipes the supports shall be also insulated with a thermal break / barrier.

## **2.0 CONCEALED COLD AIR DUCTS**

- 2.1 Insulate all rectangular supply, return and extract ductwork with 25 mm thick, 24 Kg/M<sup>3</sup> density aluminium foil faced fibreglass duct insulation, finished with a 20x20 woven glass fibre cloth covering adhered between two coats of fungicidal protective, fire resistant lagging adhesive. Fasten the insulation with adhesive on 200 to 250 mm centers. Butt all joints tightly and seal all breaks and joints by adhering a 75 mm Aluminium foil vapor barrier tape or sheet with a fire-retardant adhesive.

- 2.2 Insulate flexible connections and connections to diffusers with 25 mm thick, 24 Kg/M<sup>3</sup> density reinforced aluminium foil faced, flame resistant flexible fibreglass insulation. Overlap onto adjacent insulation and seal with adhesive duct tape to give good closure.

### **3.0 EXPOSED COLD AIR DUCTS**

- 3.1. For ducts exposed inside conditioned spaces, insulate as described above for concealed air ducts but using aluminium foil faced fibreglass boards with density 48 kg/m<sup>3</sup>.

Then apply a 20x20 woven glass fibre cloth cover adhered between two coats of fungicidal protective, fire resistant lagging adhesive.

- 3.2 Where exposed cold air ductwork runs through occupied or public areas, it shall be clad with sheet Aluminium, 0.9mm or thicker.

- 3.3. For ducts exposed in non-air-conditioned areas, insulate using the method described for concealed ducts, but using insulation with a minimum thickness of 50 mm, 48 Kg/M<sup>3</sup> density fibreglass insulation. If necessary due to market availability, this may be installed in two layers, but with each layer properly finished.

Then apply a 20x20 woven glass fibre cloth cover adhered between two coats of fungicidal protective, fire resistant lagging adhesive.

Then cover with sheet Aluminium, 0.9 mm or thicker.

All ductwork exposed externally to the building is to be clad with aluminium.

Where ducts penetrate the building shell, the duct shall be flashed and waterproofed before any insulation is applied.

### **4.0 ACOUSTIC INSULATION**

- 4.1 All low-pressure ductwork shall be lined with acoustic insulation up to the first take off point, or 3 meters from the fan/ unit outlet.

- 4.2 Duct acoustic lining shall be 12 mm thick, fibreglass, 32 Kg/M<sup>3</sup> density. Air stream surface shall be faced with a black coated mat bonded to a black fibreglass substrate.

- 4.3 Liner shall be tested to UL181 or approved equal for air velocity rating of 25m/s and air erosion rating of 63.5m/s.

- 4.4 Liner will be attached by a fire-resistant adhesive such as Foster 81-99, or equivalent. In addition, a galvanized steel mesh (size 10) held in place with self-adhesive fasteners shall be used to prevent erosion of the liner.

- 4.5 All abutting edges must be caulked and at the extremities of lining shall have a sheet metal nosing of at least 40 mm length.

### **5.0 APPARATUS CASINGS.**

- 5.1 Insulate apparatus casings which are not provided with insulation as follows:

Apply generally 50 mm of rigid polyurethane insulation, 24 Kg/M<sup>3</sup> density. On the underside of coil sections, apply 100 mm thick polyurethane. All joints are to be sealed in cold adhesive compound.

- 5.2 Ensure that any access panels are insulated to the same standard and are openable.

- 5.3 For small units, cover the entire outside assembly with waterproof self-adhesive glass fibre tape. This must be at least 75 mm minimum width and must overlap at least 50% on each application, to achieve watertight conditions.

- 5.4 For larger units, provide a weather resistant enclosure outside the insulation, subject to Consultant's approval.
- 5.5 Ensure that all equipment meets these requirements, regardless of manufacturer's standard practice, and add insulation as required to the above standard.
- 5.6 Where insulation, reinsulating or additional insulation is required to the casing of externally mounted air conditioning units, the insulation shall be covered with sheet Aluminium, of 0.9 mm or thicker.
- 5.7 Insulation to drain pans is specified as part of the A.H.U. specification. Regardless of the Manufacturer's standard supply, the contractor shall reinsulate on site if necessary so that the specification is complied with in full. This standard of insulation is also required for fan coils.

## **6.0 PIPE INSULATION**

- 6.1. Pipe covering and insulation shall be manufactured by a recognized manufacturer, and shall be neatly installed to the following specifications. The general intent is to provide a sealed insulation which will not permit sweating of the pipes and which will not retain moisture to the detriment of its insulating capability.
- 6.2. Pipes shall be thermally insulated with rigid sections of glass fibre insulation with density of 65 Kg/M3, having a thermal conductivity factor of 0.03 W/m.K at 10 deg. C with self-applied aluminium craft paper.
- 6.3. The insulation, in sizes indicated shall be applied over clean, dry surfaces. Adjoining sections of insulation should be butted firmly together with the longitudinal seam of the jacket located on the bottom half of the pipe.
- 6.4. Pipes should be banded with at least three Aluminium bands per section. One at the center and one covering the edge of each circumferential strip.
- 6.5. Insulate and finish valves and fittings in the same manner and same thickness as piping in which such items are installed. Molded, factory shaped sectional pipe covering, factory- or job-fabricated may be used subject to satisfactory visual checking by the Consultant.
- 6.6. Direct contact between pipe and hanger shall be avoided. Hangers shall pass outside of the sheet metal protection saddle, which shall cover a section of high density insulation, of sufficient length to support the pipe without crushing the insulation. The vapor barrier shall be lapped over the saddle and securely cemented to it. Minimum thickness of metal saddle is 1.5 mm.
- 6.7. Internal chilled water pipe shall be covered by 50 mm of rigid insulation plus a water proof 20x20 woven glass fibre cloth cover painted with two coats of fungicidal protective, fire retardant lagging adhesive.  
  
External chilled water pipe shall be covered by 75 mm of rigid insulation, plus a waterproof 20x20 woven glass fibre cloth cover, painted with two coats of sealant.  
  
Where any insulated piped service is run external to the building, including any insulated pipework run on roof O.T.S, car park, etc., the piped services shall be clad using an outer covering of sheet Aluminium, of 0.9 mm. or thicker.
- 6.8. Similar protection is to be applied to pipes running in trenches, and in plant rooms.
- 6.9. Refrigerant piping shall be insulated with 25 mm of flexible cellular foam insulation. Exposed refrigeration pipework shall be clad with Aluminium sheet as specified for external piped services, or as specified for piped services exposed to damage.

- 6.10 Where exposed, insulated pipework runs through occupied or public areas, the exposed pipework shall be clad with sheet Aluminium, of 0.9 mm. or thicker.
- 6.11 All condensate drain pipework shall be insulated with 25mm thick fibreglass insulation and finished as indicated in clause 6.7.

## **7.0 LAGGING ADHESIVE**

- 7.1 Adhesive shall be a flexible, fire resistive, fungicidal resistant compound suitable for vapor sealing insulated ducts and pipes.
- 7.2 Adhesive shall be suitable for indoor and outdoor use and in high humidity environments.
- 7.3 Water vapor permeance shall not exceed 0.05 perms at 0.030 inch dry film thickness when tested in compliance with ASTM F 1249.
- 7.4 When tested for surface burning characteristics in compliance with ASTM E84, flame spread shall be 10 and smoke developed 15.
- 7.5 Adhesive shall be UL classified and shall meet or exceed the requirements of NFPA 90A and 90 B 25/50.

## **8.0 ALUMINIUM CLADDING**

- 8.1 Aluminium cladding shall be provided to ducts with neat 90-degree joints and finish. Material shall be with stone finish.
- 8.2 In pipework the longitudinal joints shall be cleated or held in position with stainless steel bands and clips. Elbows and other fittings shall be made with mitered section and riveted. All valves strainers, flexible connection etc. shall have easy removable boxes with stainless steel aluminium toggle clips.
- 8.3 In ductwork the joints shall be riveted or banded with stainless steel band and clips to avoid sagging. Boxes to be made around duct traverse joints.
- 8.4 All gaps shall be to a minimum and shall be sealed with silicon mastic suitable for outdoor.

## **9.0 SAMPLES**

- 9.1 Submit samples of all types of insulation for approval. Submit pipe insulation samples 300 mm long x 25 mm inside x 125 mm outside diameter, and flat samples 300 mm x 300 mm x 25 mm, minimum size.
- 9.2 Submit with samples, the full technical brochures.
- 9.3 Where the specification calls for additional treatments, such as wrapping and waterproofing, submit also a completed sample of similar size.
- 9.4 All samples after approval are to be retained on site to be a reference for future work.

**-End of Section-**

### **SECTION 3**

#### **AIR CONDITIONING EQUIPMENT**

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## INDEX

|      |                                                      |    |
|------|------------------------------------------------------|----|
| 1.0  | SCOPE OF WORKS.....                                  | 3  |
| 2.0  | DESIGN PARAMETERS .....                              | 3  |
| 3.0  | FRESH AIR HANDLING UNIT WITH RECOVERY SECTIONS ..... | 5  |
| 4.0  | REFRIGERATION MACHINES.....                          | 8  |
| 4.0  | CHILLED WATER PIPING SYSTEM .....                    | 8  |
| 7.0  | FANS CENTRIFUGAL.....                                | 15 |
| 8.0  | FANS, AXIAL.....                                     | 16 |
| 9.0  | PARTICULAR FAN.....                                  | 17 |
| 10.0 | ELECTRIC MOTORS .....                                | 17 |
| 11.0 | MOTOR STARTERS, CONTROLS SWITCHGEAR.....             | 19 |
| 12.0 | PANEL BOARDS AND MOTOR CONTROL CENTRE. ....          | 19 |
| 13.0 | ELECTROSTATIC AIR CLEANER .....                      | 20 |
| 14.0 | CLOSE CONTROL UNITS .....                            | 22 |
| 15.0 | SMOKE CONTROL STATION.....                           | 25 |
| 16.0 | CARBON MONOXIDE DETECTION SYSTEM.....                | 25 |
| 17.0 | SUBMISSIONS.....                                     | 26 |

## 1.0 SCOPE OF WORKS.

1.1 The works shall include supply, installation, testing, commissioning, balancing and adjustment and setting to work of the following systems:

- Central and local air-conditioning of all areas using Ducted & Decorative chilled water Units as shown on the drawing.
- Ventilation
- The entire air-conditioning system shall be monitored by DDC based BMS System (if any) & MEP
- Contractor to provide provision accordingly.
- Smoke control as per the design and DCD regulation and NFPA code

1.2 Contractor is to assume full responsibility for the correct functioning of the system and to carry any liability or other insurance or guarantee as may be necessary to protect all parties in this regard

1.3 Contractor shall be responsible for coordination with other trades and services and shall provide all materials, labour and supervision, equipment, tools, appliances, services, etc. for carrying out the following items of work:

- Supply and install brand new air conditioning and ventilation plant and equipment
- Insulated ductwork complete with air terminals, dampers, supports, etc.
- Insulated refrigerant pipework complete with valves, fittings, supports, etc.
- Controls and control panels.
- Motor control centers and starter panels complete with cabling from the panels to the respective equipment. Power supply up to the panel shall be the responsibility of the electrical subcontractor.
- Power supply to fan coil units, air handling units, fans etc. Electrical power supply terminating in an isolating switch within 2 meters of the equipment shall be the responsibility of the electrical subcontractor.
- Insulated drain piping from each air handling unit/fan coil unit to the nearest drain point. The drainage subcontractor shall be responsible for provision of a floor drain in each mechanical room.
- Sound attenuation requirements for the ducts shall be studied and supplied and installed by the HVAC contractor up on Engineer's requirements and approval

1.4 The contractor shall also include all overheads at office and site, labour, sundries, plant, appliances and consumable both for the works and for balancing, testing and commissioning, and during the maintenance period.

## 2.0 DESIGN PARAMETERS

Design, in general, shall be based on ASHRAE recommendation with the following particular requirements:



## 2.1 DESIGN TEMPERATURE CONDITIONS

|        |             |   |                                 |
|--------|-------------|---|---------------------------------|
| Summer | a) External | - | 46 deg. C DB 30 deg.C WB.       |
| Summer | b) Internal | - | 24 deg. C DB± 1 deg.C 50± 5% RH |

## 2.2 VENTILATION REQUIREMENTS As

shown on ventilation plans.

## 2.3 NOISE LEVELS

|   |             |       |
|---|-------------|-------|
| - | Offices     | NC35  |
| - | Prayer room | NC 35 |
| - | Bedrooms    | NC 30 |
| - | Corridors   | NC 40 |
| - | Retail area | NC 35 |
| - | Living      | NC 30 |

Noise level shall be achieved at high speed selection of FCU / maximum airflow.

## 2.4 Air Distribution

Duct Velocity, supply : 6.5 (m/s)  
(Pressure drop to be limited to a maximum of 0.6 Pa/m)

Duct velocity, return : 6 (m/s)  
Duct velocity, Kitchen exhaust : 7.5 (m/s)

## 2.5 AIR FILTRATION

| <u>Area</u> | <u>Filter Efficiency %</u> |                      |
|-------------|----------------------------|----------------------|
|             | Filter bed<br>No.1*        | Filter bed<br>No. 2* |
| A.H.U.'s    | 25                         | 85                   |

(\*) Based on ASHRAE Standard 52-76

## 2.6 PRESSURIZATION PHILOSOPHY

Fresh air requirements to each zone shall be generally determined by the level of dirty extract to that same zone, or to a value as required for the levels of occupancy in each zone.

Areas requiring no dirty extract shall be positively pressurized or maintained neutral as recommended by ASHRAE.

Areas requiring dirty extract shall be pressurized delta negative.

The whole building is positively pressurized to avoid any hot air infiltration.

## 2.7 FIRE PHILOSOPHY AND SMOKE MANAGEMENT

Smoke fire motorized damper shall be provided for all ventilation ducts crossing main corridors, and shall receive a signal from Fire Alarm system.

Also, fire dampers shall be positioned in all air ductwork crossing fire walls/slabs. They shall be triggered from a fusible link connection sited in the damper.

## 2.8 AIR INTAKE SYSTEM

Fresh air intake to each air handling unit shall incorporate a sand trap louver and/or in line filter within a sheet metal plenum chamber.

Air intake louvers shall, in general, be located as far away as possible from extract louvers and drainage vent stacks to avoid short circuiting of dirty air.

## 3.0 FRESH AIR HANDLING UNIT WITH RECOVERY SECTIONS

### 3.1 General

Supply and install as indicated in the schedule of equipment, packaged modular Air Handling Units of Double Decker Type, (as shown on drawings) each capable of the duty as mentioned in the schedule of equipment. The space available for the unit shall be confirmed with the civil and architectural proposals of the main contract and sizes of units shall be selected to fit into the spaces available. Where necessary the units may be built on site, subject to acceptance of the finished units for warranty purposes by the original supplier and his local agent.

Unless otherwise stated, the air handling units shall be of sectionalized construction with pre-filter, bag filter, chilled water-cooling coil and supply air fan. For Fresh Air Recovery AHU shall be additional with Heat Recovery wheel, necessary empty sections with access doors, liquid coupled heat exchanger, return air fan. As indicated in the schedules, the unit configuration should suit the ducting arrangement as shown on the drawings viz. double deck construction.

### 3.2 Quality Assurance

The equipment manufacturer shall strictly adhere to following standards & specification:

- (1) ISO 9001 certificate required.
- (2) Air Handling Unit manufacturer shall strictly adhere to CEN standards (Committee European de Normalization).
  - (a) Casing strength to CLASS 2 A
  - (b) Casing air leakage minus 400 Pa to CLASS A.
  - (c) Casing air leakage plus 700 Pa to CLASS A

|     |                       |           |
|-----|-----------------------|-----------|
| (d) | Thermal transmittance | CLASS T4. |
| (e) | Thermal bridge factor | CLASS TB3 |

(3) AHU panel insulation shall be fiber glass

### 3.3 Unit Construction

The unit casings shall be of double skinned panels. Casing shall be assembled with self supporting modular panel elements with an integrated base frame made of zincated steel and sections along upper sides of the units.

Sheet metal thickness shall be not less than 1.5 MM for the inner skin and 0.8 MM for the outer skin and shall be made from zincated steel sheets. The outside of the outer skin shall be plastic coated to a thickness of 0.18 mm for additional protection. Inside and outside of panel walls shall be completely smooth. The overall heat thermal transmittance (Heat Transfer Coefficient) shall not exceed  $k=0.59 \text{ W/m}^2\text{k}$ .

All casing panels shall be insulated glass fiber layer of 50mm thick. incombustible with class of inflammability:

A1 in accordance to DIN 4102 O in accordance

to ISO 1182.2

For units with polyurethane injected foam insulation, sheet metal thickness shall be 1.0mm for the inner and outer skins.

The base frame of the units shall be made from sendzimir galvanized sheet metal for size with largest dimensions upto 2500 MM, and hot dip galvanized U-profile for larger units. Service doors shall be provided with special gasket and locking device. Access doors 400MM and larger shall be provided with hinges and handles for external and internal opening.

Access doors with width smaller than 400MM shall be removable type fitted with quick release flap turn lock with spring pressure arrangement. Sealant between panels shall be an anti fungicide sealant material.

All air handling units installed outdoors shall be fitted with aluminum weather canopy.

The AHU manufacturer shall guarantee that no condensation shall take place on the exterior of panels. In the event that any condensation problems appear after installation, the contractor shall undertake all remedial measures to rectify and to the satisfaction of the consultants. Any stacked or double height coils shall have separate drain pans to reduce carry over.

Acoustical insulation through the panel :

| HZ MID FREQUENCY OCTAVE BAND |     |     |      |      |      |      |
|------------------------------|-----|-----|------|------|------|------|
| 125                          | 250 | 500 | 1000 | 2000 | 4000 | 8000 |
| 14                           | 25  | 28  | 27   | 29   | 31   | 34   |

### 3.4 Vibration Isolation

The Air Handling Unit shall have internal vibration isolation system by mounting fan, motor and drive assembly on spring isolators. The fan discharge shall be connected to the air handling unit casing through canvas connection to prevent vibration transfer. In addition to the above the entire unit shall be mounted on additional vibration isolators.

#### Filter Section.

- 3.5 Filter cells shall be of standard sizes and shall be obtained from reputed European manufacturer. The filters shall be sealed against the filter frame using a permanently elastic gasket to a standard compatible with the filter efficiency.

Pressure drop tapings shall be integrated into the frame to allow a manometer or filter monitor to be fitted. Filter materials shall be flame-retardant, incombustible, non-odorous and offer no sustenance to vermin. Each filter section has to be provided with manometer.

The filter material shall be pleated to provide a large effective area. Filter section should be provided with an inspection door.

The contractor shall supply one set of all filters per AHU as spares for replacement after testing and commissioning and prior to handing over of the installation.

#### PRE FILTER:

The Prefilter material shall be synthetic, pleated to provide a large effective area and shall be supported by a wire mesh and frame. The filter cells shall be suitable for side withdrawal on the inspection side. The filter class shall be EU-3 or EU-4 as per Eurovent 4/5 or G3 as per EN 778.

#### BAG FILTER:

Bag filters material shall be glassfibre, of standard readily available sizes. The filters shall be clamped against the frame using a cam locking bar. The filter class should be EU7 as per Eurovent 4/5 or G7 as per EN 778.

#### Cooling Coil Section

The coil shall be tested to 21 bar and designed to operate at 16 bar NP.

The coil shall be mounted on guide rails for side withdrawal. Header connection through the unit casing shall be sealed with plastic grommet.

- 3.7 Cooling coils shall be fabricated from nominal 3/8" (10mm) size copper tubing expanded into fins to give a mechanical bond. Return bends shall be die formed. Headers shall be heavy section seamless copper tubing. All joints shall be silver brazed. Fittings shall include plugged vent and drain taps for each section. The fins shall be Aluminium.

- 3.8 The coil shall be designed to operate at 200 psi and 300 degrees Fahrenheit (149 degrees Centigrade.) The performance shall be not less than that listed in the schedules.

The coil shall be assembled in a galvanized steel housing which is drilled to meet the adjacent sections of the A.H.U.

- 3.9 Coils shall have a minimum of 3 rows for recirculation service and 8 rows for fresh air service.

3.10 Rotary Heat Exchanger (Thermal Heat Wheel)

The wheel shall be made of layers of corrugated and intervening flat aluminium foil of uniform width to ensure smooth surface. This wheel material is bonded together to form a rigid transfer medium forming a multitude of narrow channels thus ensuring a laminar flow. The rotary heat exchanger shall be of hygroscopic type for the transfer of sensible and latent heat. The whole rotor shall be chemically treated by controlled corrosion process to produce hygroscopic layer on aluminium foil. The chemically attached layer should have high affinity to moisture. Pre-coated aluminium foil should not be used.

Rotor shall be mounted in a sturdy casing with an access door and sealed against casing by means of wear resistant, adjustable seal. The casing shall be equipped with purging sector. The wheel shall be cleanable by spraying its surface with compressed air, low temperature steam, hot water or by vacuum cleaning without affecting its latent heat recovery properties. The casing shall be equipped with adjustable brush seals which minimize carry-over to max. 0.5-0.2%. The wheel shall be belt driven along its perimeter. A constant speed drive motor shall be used. The motors shall be mounted on a self adjusting base to provide correct belt tension.

The rotary wheel should confirm to ASHRAE 84-78, EN 29001 and BS 7550.

For one piece rotor, hub and spoke shall be in aluminium. On sectorized rotor, hub made of steel, painted with anti-corrosion paint, galvanized sheet steel spokes.

The direction of airflow through the rotor should be reversing continuously to give excellent self cleaning effect. In order to prevent fouling and clogging of the rotor, a standard EU-4 filter (synthetic washable) should be in position and the rotor should always be rotating when the fans in the system are running. A double glass inspection window should be provided in this section.

The Heat Recovery Wheel shall be constant speed. The HRW shall be provided with speed detector.

## 6.0 DX SPLIT UNITS (SMALL CAPACITIES – UPTO 5 TONS NET)

The split system shall consist of a ceiling suspended indoor unit (evaporator unit) an outdoor unit (condensing unit) and a wall mounted remote control switch, all factory assembled and fully charged with refrigerant R410A. All units shall be start tested and checked at the factory. Units shall be designed for continuous operation at outdoor ambient temperature of 52 deg. C. Coils and fins shall have heresite or equal approved anti-corrosion coating.

### 6.1 Evaporator Unit

The indoor evaporator unit shall be of the slim and compact design with a decorative cabinet. Units shall be suitable for flush-to-wall installation to conceal pipe connections.

Where indicated on the drawings and schedules, evaporator units shall be of the horizontal concealed type with return air plenum suitable for ducted insulation. Unit casing shall be insulated with at least 12mm thick insulation of density 24 kg/m<sup>3</sup>.

Evaporator coils shall be seamless copper tubing mechanically bonded to aluminum fins and shall be factory leak tested at a pressure of 28 Kg/cm<sup>2</sup>. A drain pan made out of sheet steel shall extend throughout the length of the unit. The drain pan shall be coated with epoxy resin enamel and insulated to prevent condensation.

Evaporator fans shall be direct driven, forward curved, centrifugal type made of galvanized steel to deliver an accurate air flow at a low noise level. Fan driving components shall be mounted on rubber pad isolators to reduce noise and vibration.

A single phase squirrel cage motor shall be used to drive the evaporator fan. A fan speed control shall be provided to facilitate speed adjustment as required.

Each unit shall be provided with a permanent washable filter and swing flow louvers to eliminate uncomfortable hot spots.

### 6.2 Condensing Unit

Casing of the unit shall be of sheet steel suitably treated to be rust resistant for a long life.

Units shall have high efficiency, hermetic, line starting compressor. Each compressor shall be equipped with an internal thermostat, over current relay, a low pressure switch and a crankcase heater. Condenser coil shall be of seamless copper tubes mechanically bonded to aluminum plate fins with protection louvers. Each coil shall be factory leak tested at a pressure of 33 Kg/cm<sup>2</sup>. Coils and fins shall have here site or equal approved anti-corrosive coating.

Condenser fans shall be direct driven by a weatherproof squirrel cage induction motor. Fans shall be of the propeller type.

## HEAT EXCHANGER:

### GENERAL

The installation of plate heat exchangers will provide total separation between the district cooling suppliers' chilled water system and client chilled water system. Total separation ensures that client cannot be held responsible for any maintenance issues which may arise in the district cooling providers system and vice versa.

Based upon an outline cooling demand as per the design, it is estimated that four chilled water plate heat exchangers will be supplied and installed by Empower over the entire site. Currently Empower are reviewing the requirements of the project and will provide further instructions.

Empowers chilled water temperatures on the primary side will operate at flow and return water temperatures of 4.5°C 5°C (+1°C, allowable tolerance) and 13.5°C respectively, whilst the secondary side will operate at flow and return water temperatures of 5.5°C (+1°C, allowable tolerance) and 14.5°C respectively.

Empower shall supply and install the Plate Heat Exchanger, Primary Side Piping and Primary Side instrumentation and control.

Any part of the machines or system failing to meet the contract requirements shall be adjusted, repaired or replaced.

### **Pipe work, Pumps and Water Treatment**

Variable speed primary pumping will be used to distribute Chilled Water to the roof level and through the building within vertical risers, ceiling voids and car park areas.

Secondary side chilled water will be conveyed around client on a variable flow basis through a thermally insulated two pipe, flow and return system of steel pipework to FAHU's. AHUs and FCUs.

Variable speed pumps will be provided per heat exchanger and will discharge into a common header to provide system flexibility as per figure 06.B. The provision of standby pumps will provide system redundancy during planned and preventative maintenance. Should two pumps be non-operational, the cooling capacity of the ETS will reduce to 80%. Should three pumps be non-operational, the cooling capacity of the ETS will reduce to 40%.

The pumps will be selected with N+1 redundancy such that the failure or maintenance of any single pump will not impact on the development CHW distribution system.

The type of pumps proposed will be investigated further during the next stage, but currently horizontal split case pumps are under consideration as the configuration will allow for maintenance of the impeller without the need to modify the pipework connections which is problematic on large diameter pipe connections. Alternatively long coupled end suction pumps may be considered after discussions with specialist manufacturers which the PQS may calculate as cheaper.

All CHW secondary circuits will operate using the principal of variable speed control based on constant differential pressure. This will be achieved by using variable frequency drives for pump motors with the speed of chilled water pumps controlled via the BMS, under the dictates of differential pressure control and the associated pump VFD's.

The differential pressure transducers, located at 2/3 hydraulic distance of each pumped circuit, detect a reduction or increase in chilled water demand due to the modulation of terminal unit control valves; When the 2-port chilled water control valves associated with secondary low loss header drives towards the closed position, due to a reduction in building load, this has the effect of increasing the circuit resistance. This is sensed by the differential pressure transmitter. Consequently the control system slows down the chilled water pump speed accordingly maintaining the system differential pressure at the designated set point and matching the CHW flow rate to the building demand. Similarly as the load increases the control valves drive to

the open position, decreasing the pressure sensed at the transmitter, causing the pump to ramp up to maintain the required pressure set point.

An automatic by-pass valve will be installed to the index run's farthest point to ensure secondary the minimum flow of the CHW pumps can be maintained during periods of low loads.

Where multiple pressure sensors are utilised in large or split-main CHW networks the pump speed is adjusted to ensure the required minimum flow is set by the sensor with the largest demand to ensure all circuits receive sufficient CHW at all times.

The pressurization unit will be equipped with a microprocessor and fascia mounted display panel including volt free contacts for control, switching, monitoring and the raising of alarms at the BMS for: power failure, low water level, pump running, pump tripped, high/low pressure cut outs, excessive pump operation, current operating pressure and an audible alarm and muting facility will be incorporated.

#### Pipe Work Expansion :

Expansion of pipework within a large central cooling system such as this development must be considered. Generally the final design and selection of all thrust/expansion protection is the responsibility of the installing contractor based upon final material selection and routing of the network.

#### Dosing and Flushing

The closed chilled water system shall incorporate facilities to enable flushing, cleaning and ongoing maintenance so as to prevent bacteriological contamination (in accordance with the latest BSRIA Guidance) and shall include biocide treatment of the incoming/fill water. This will be achieved by the provision of strainers and flushing bypasses in the design with the specification prepared to require the contractor to undertake pre-commissioning flushing, chemical cleaning and regular bacteriological water quality checks throughout the installation, filling and commissioning phases.

A manual chemical feeder dosing pot of sufficient capacity shall be fitted to the distribution system to allow chemical treatment to be administered in the system and for water to be drawn off for analysis.

Prior to handover, the water in the system shall be analyzed and an appropriate water treatment added to maintain the efficiency of the system and prevent corrosion etc.

It is not intended that the building secondary chilled water pumps be used for flushing of the chilled water system in order to minimise the risk of damage to the pump impellers. Capped off line size connections and isolation valves shall be provided for the installation of temporary flushing pumps as part of the pre-commissioning cleaning process.

#### Chilled Water Hydraulic Balancing and Control

In any Hydronic system it is necessary to provide suitable valving arrangements to ensure the design can be balanced during the commissioning stage, such that each terminal unit will receive the required flow rate under peak conditions and be controllable under part loads.

In the past, when constant pumping was common place, this was achieved via the use of Double Regulating Valves (DRVs) to equalise the resistance of all sections of the circuit, and 3 Port Control valves to control the terminal unit flow with the excess simply by-passed when the load is less than the design duty. This system, whilst robust, is inefficient as the system pumps constantly circulate the full design flow rate under all conditions



throughout the year regardless of actual system demand and will bypass CHW resulting in Low Delta T syndrome,

As described a variable volume pumping scheme will modulate the flow rate through the system to match the development building load reducing pumping energy consumption. This, however, creates challenges ensuring stable operation at part loads as the action of the modulating control valves cause pressure fluctuations in the pipelines resulting in unpredictable flows patterns in the system and can itself result in "Low Delta T Syndrome" where by excessive flow passes through control valves resulting in the return water temperature reducing below the design condition. This is a significant issue to the District Cooling Provider carrying financial penalties under the utility service contract.

Controlling the pressure fluctuations in the system is key to efficient and controllable operation and is achieved by introducing Differential Pressure Control Valves (DPCVs) into the design. These valves include a sprung valve plug whose position moves automatically to ensure that the pressure after the valve set is constant allowing for accurate control valve operation.

As an alternative to a traditionally balanced system (manual setting of Double Regulating Valves) k w e c propose the use of Pressure Independent Control Valves (PICVs)

Essentially a PICV is a combination valve consisting of close coupled Motorised Valve and Differential Pressure Control Valve at each terminal unit. This allows for the omission of manual balancing valves and differential pressure control valves and will substantially reduce the time required to commission the system on site as the PICVs are factory set at the required flow and then the branch flow is simply verified by the commissioning agent to ensure the correct terminal flow rate is achieved.

## FAN COIL UNITS:

### GENERAL

- 1.1 This section covers the supply, delivery, installation, testing and commissioning of the fan coil units (FCU) and the associated accessories.
- 1.2 The Fan Coil Unit (FCU) shall be factory fabricated, tested and of efficient design and construction. The FCU shall be blow-through type and operate with low noise level. The FCU shall be supplied and installed with reference to the tender drawings and details as described in the Schedule of Equipment Data. Each FCU shall consist of but not limited to the following components:
  - i. Casing made of rigid galvanized steel;
  - ii. Fan assembly consist of fan wheel, 3-speed fan motor;
  - iii. Coil section with cooling coil of copper tubes and aluminum fins;
  - iv. Return plenum with filter housing and filter media;
  - v. Insulated dual pitched sloping drain pans;
  - vi. Vibration-isolating bases, vibration isolator and other accessories necessary for operation.
- 1.3 FCU shall be IEC60335 safety certified.
- 1.4 FCU manufacturer shall be ISO 9002 certified.
- 1.5 FCU shall be controlled by 3-way valves(on/off)
- 1.6 Off coil condition: 54.5 deg F- 55 deg F db,  
53.5 deg F-54.5 deg F wb.
- 1.7 On coil conditions: 76/63 deg F db/wb respectively

## **FCU CASING**

- 2.1 The casing shall be fabricated with rigid galvanized steel. All components shall be easily accessible for the inspection and servicing of internal components such as fan assembly and cooling coil.
- 2.2 The FCU shall be insulated with 108 kg/m<sup>3</sup> high density non-flammable 6mm Polyurethane Foam on the fan board and the coil section. In case the return plenum of the FCU is ducted, and then it shall also be internally insulated thermally with the same material.
- 2.3 Exposed units shall be with a decorative cabinet.
- 2.4 Ducted concealed units up to 1400 cfm nominal air flow, shall have full casing enclosing the complete fan/motor assembly and the coil assembly, with the chilled water coil header and U-bends completely concealed from sight, within the unit casing.
- 2.5 Ducted Concealed units shall be provided with a rear return plenum / panel filter combination.

## **3 FAN / MOTOR ASSEMBLY**

- 3.1 The fan wheel shall be constructed of metal and shall be both statically and dynamically balanced to ensure comfortable sound levels.
- 3.2 The fan shall be centrifugal, forward curved, multi-blade, double width, double inlet, and shall be directly driven by the motors via the shaft.
- 3.3 The fan housing shall be made of galvanized sheet steel.
- 3.4 Motors shall have an internal inherent thermal temperature cutout, be permanent split-capacitor, three speed, tap wound, induction type for maximum efficiency.
- 3.5 Motors shall have a permanently lubricated sealed for life sleeve bearings and all direction, vibration isolating mountings to ensure vibration free operation and minimum noise.
- 3.6 All motors shall be factory run tested in the assembled FCU prior to shipment.
- 3.7 Motors shall be rated for operation at (230V-50Hz-1ph) and fan performance shall be given at High speed.
- 3.8 The motor capacitor shall be totally enclosed in a metal shield, and attached to the motor.
- 3.9 The shaft shall be supported by the permanently lubricated sealed for life bearings.

## **4 COIL ASSEMBLY**

- 4.1 The FCU manufacturer shall fabricate the cooling coil suitable for chilled water application.
- 4.2 The chilled water cooling coil shall be suitable for a 2 pipe system and shall be constructed of 3/8 inch OD seamless copper tubes, mechanically bonded to Wavy 3B aluminum fin collars.
- 4.3 A manual air vent with drain line to the drain pan shall be facilitated, to avoid any water drips when venting.
- 4.4 All coils shall be tested at 1.5 times the working pressure.
- 4.5 The working pressure for the units shall be at least 13 bar, however subject to the operating head some units will be identified in the equipment schedules to operate at 21 bar. The responsibility to identify the units and building levels operating at that pressure shall be up to the installing contractor.

## **5 DRAIN PANS**

- 5.1 Drain pans shall be 25mm depth, manufactured in 0.8mm thick galvanized steel with epoxy resin coating.
- 5.2 To ensure maximum leak protection, the drain pan shall be a one-piece stamping, free from any seams or joints.
- 5.3 The drain pan shall be insulated with 5mm thick 27kg/m<sup>3</sup> density PE foam
- 5.4 The drain pan shall be levelable independently of the fan coil unit and shall have a 3/4 inch male pipe thread (JIS Standard: B 0203-1966) connection.
- 5.5 An auxiliary condensate drain pan shall be placed under the chilled water valve and accessories.

## **6 FILTER**

A 25mm, Thick air filter, permanent and cleanable type, made of aluminum and arranged in a solid frame. Arrangement shall be made for passing all fresh and recirculated air through the filters before passing through the coils, or otherwise separate filters on the fresh side shall be provided and disposal. Synthetic temporary filter shall be supplied with all FCU for commissioning purposes.

## **CHILLED WATER PUMPS:**

- 1- Supply and install as shown on the drawings identical pumps centrifugal split coupled vertical inline type, having duties as shown on the drawings (one as stand by), each pump shall be with a flexible coupled T.E.F.C. motor having Class "F" insulation. The pump & the motor assembly shall be mounted on an inertia base with spring isolators under each (detailed calculation of vibration study should be submitted to the Engineer for approval) and shall have two pressure gauges – one at suction side and one at discharge side along with all standard fittings as shown on the drawings.
- 2- The pump shall be constructed from class 30 C.I. casing, bronze, impeller, S.S. shaft, with mechanical seal. The speed of the pumps shall not exceed 1500 R.P.M.
- 3- The pump head to be calculated by the contractor according to the approved shop drawings to consultant approval.
- 4- The pumps shall be of horizontal split casing or end suction type according to the drawings.
- 5- Pump to be selected to meet the specified duties with an efficiency of 70-75.
- 6- Each pump shall be with mechanical seal and shall provide with flexible connectors, isolating valve, before and after each pump. End suction diffuser with permanent stainless steel strainer and multipurpose valve and end suction diffuser shall be provided by the same manufacturer as the pump. All pumps shall be factory tested prior to shipment.
- 7- The Flexible connectors shall be constructed of peroxide cured EPDM covers and liners with Kevlar type cord fractioning. Any alternative material shall have equal or superior characteristics. Solid steel rings shall be provided to prevent pull out of the raised rubber flanges.
- 8- The flexible connectors shall be installed on the equipment side of the isolation valves

#### **CHILLED WATER PRESSURIZATION UNIT:**

1. The Chilled water pressurization and make up unit shall be provided, as indicated on the drawings and wherever required.
2. The unit shall be supplied as a packaged set, complete with:
  - a. Two Pumps (one duty and one stand by) with controls.
  - b. Expansion vessel including (pressure gauge pressure relief valve ,support legs, flanged or screwed inlet connection)
  - c. Air Separator
  - d. Water tank with water supply & drain/over flow piping.
  - e. High pressure / low pressure switches
  - f. Pressurization unit shall be of the members type, compressing a welded mild steel cylindrical
3. Vessel with internal rubber diaphragm to separate the water from the pressuring gas.
4. The pressure vessel shall be manufactured in accordance with B.S 4814, suitable for A maximum working pressure of 16 bars.
5. The diaphragm shall be of the removable rubber bag type, and the internal surface of the vessel shall be treated with an epoxy/ polyester coating. Nitrogen shall be used as pressuring gas.
- 6 Size of vessel to be determined finally by the contractor according to the actual calculated System volume temperature variations and working pressure to consultant approval

#### **CHILLED WATER CHEMICAL TREATMENT:**

### **PART – 1 CHEMICAL PROGRAMS & PERFORMANCE – CHILLED WATER SYSTEMS**

- (i) All chemical products supplied to the client must be manufactured by ISO 9002 or ISO 14001 registered companies.
- (ii) All chemical products must be in containers sealed by the manufacturer.
- (iii) Chemical programmes must be submitted to the client for approval before acceptance.
- (iv) Flushing & chemical cleaning to be accomplish by the temporary pump provided by the contractor.
- (v) Methodology for the draining of plain flushing & chemical cleaning water.(should be as per DM regulation)  
It is Contractor responsibility to circulate the chilled water system after the final chemical treatment as per Manufacturer recommendation till the final handover to client. The chemicals used for cleaning & water treatment

(Temporary & permanent) must be compatible with the installed system materials.(like Chillers,Fcu ,Pumps, valve)

## 1.1 CHILLED WATER

The objective of the water treatment programme for the chilled water system is freedom from corrosion, deposition and microbiological activity along with plant integrity and operational efficiency. Environmental considerations are also paramount and preference will be given to companies pursuing operational excellence.

- 1.1.1 As this is a closed system protection will primarily be required against corrosion. Secondary consideration will be given to deposit control and microbiological activity.
- 1.1.2 Corrosion in the chilled water system shall be controlled at 2 mpy maximum.
- 1.1.3 Allowance should be made in the programme for an occasional (3 monthly) dose of biocide.
- 1.1.4 Sufficient chemicals should supply for a one year period based on initial system fill plus 5% losses per month.

## PART – 2 PROCEDURES FOR CHEMICAL TREATMENT OF CHILLED WATER SYSTEMS

### Static and dynamic flushing:-

The purpose of this pre-flushing is to remove inert suspended matter such as iron fines, jointing compound, hemp, welding slag, art, grit, etc

Following the flushing all the system strainers are to be removed by the contractor, cleaning and refitted prior to the chemical cleaning phase.

### Chemical cleaning phase:

The purpose for chemical cleaning is to remove oxide deposit from the metal surfaces. the system will be dosed with the specified amount of cleaning chemical agent as per system volume.

After the dosing of the chemical it is to be left circulating continuously for a period of not less than 24 hours and not more than 48 hours prior to draining and flushing.

Chemical will be circulated in he system until two successive samples indicate that no further increase in iron level is occurring prior to starting to flush, a reference sample should be collected by the contractor from the system in the sample bottle. At this point, draining and flushing will be commenced by the contractor and will continue until the dissolved iron level is less than 1 pats per million (ppm) and the traces of the chemical and the sediment removed.

The system should be flushed until it is running clear, colorless & odorless, the iron level is less than 1ppm & the conductivity is no more than 500 micro siemens/cm above the mains water.

Upon completion of flushing, the contractor should than remove, clean & reinstall the strainers.

### Passivation:-

When the system has been satisfactory cleaned it will be passivated by using appropriate chemical by maintaining higher phosphate residual level (20-30 ppm) for 3 days.

This step above ensures that a uniform mono molecular film is formed on the metal surface.

### Biological control:

After the passivation period of 5 days, micro biocide chemical is manually dosed @ 50 ppm on the system volume & circulated for 24 hours.

After 24 hours circulation period the main blow down valve is opened and the system water is drained till water runs clear & the chemical residuals are brought to normal operating levels.

#### **Regular Treatment:-**

After passivation & biological control treatment the automatic dosing system is switched on & regular treatment is started by dosing the chemical as per specialist recommendation.

After the final commissioning specialist shall provide follow up visits to site during the initial period of stabilization & subsequently after stabilization the service visit will be once in a month basis during the maintenance period

### **PART-3 CONTROL EQUIPMENT**

Control equipment is a necessary support for the chemical water treatment programme. Where control equipment is supplied, appropriate manuals must also be supplied.

#### **4.2 CHILLED WATER**

In order to apply appropriate levels of Balticare CB55 (OR EQUAL) and Biocide ITA a pot doser shall be used.

- 3.1.1 The pot doser shall be manufactured with welded end caps, air cock and filling funnel. Non return valves shall be included at its base, inlet, outlet and drain connections. 22mm bronze globe valves shall be used.
- 3.1.2 The volume should be 13.5 liters.
- 3.1.3 As an alternative, a high pressure dosing pump may be used. If so it should be fitted with low level protection.
- 3.1.4 A suitably sized corrosion rack shall be supplied which has space for 2 corrosion coupons.
- 3.1.5 Conductivity controller & control module.

#### **4.2 Automatic chemical dosing system:**

The automatic dosing system(for each circuit) comprises of the following:

Scale & corrosion inhibitor Dosing pump having adjustable stroke & speed with injection valve, foot valve & tubing, Dosing tank, Low level switch, Control module, conductivity controller ,corrosion racks with MS &

Cu

coupons, nitrate test kits & accessories..

- The scale corrosion inhibitor dosing pump doses the chemical in the common suction header of the chilled water system.
- The dosing pump operation is controlled through the control module.
- There are two modes operation manual & automatic mode.
- The manual mode is selected to dose the chemical if the system losses are more than the design losses.
- In automatic mode the dosing pump operates based on signal received from the conductivity controller. & from impulse water meter (for cooling tower), the system also comprises of a low level-switch for dry run protection of dosing pump,.

#### **4.2 Manual chemical dosing system:**

- A manual dosing pot of 13.5 liter capacity shall be provided with funnel,NRV, control valve & accessories(for each circuit).

### **CHILLED WATER PIPE & INSULATION:**

### Chilled Water Pipe Work (CHWP)

It should be of Electric resistance welded pipe (ERW) Schedule 40.

### Insulation

All exposed chilled water pipe work shall be insulated with closed cell elastomeric foam insulation with aluminum foil jacket and shall be protected by aluminum cladding for all pipe run on the roof / shaft & other unconditioned areas.

#### A) Material

The material shall be Closed Cell Elastomeric Foam Insulation preformed sections with the following properties.

1. Thermal conductivity (  $\lambda$  ) BS 874 Part 2-86 (DIN 52613, 52612)

$-20^{\circ}\text{C} \leq 0.034 \text{ W / mK}$

$0^{\circ}\text{C} \leq 0.036 \text{ W / mK}$

$20^{\circ}\text{C} \leq 0.038 \text{ W / mK}$

Thermal Conductivity values shall be governed by a Third Party Supervisory Control Agreement which shall be available for attestation of conformity purpose.

2. Water vapor diffusion resistance (  $\mu$  ) DIN 53122 Part 2, DIN 52615.

$\mu \geq 7000$ .

Water Vapor Diffusion Resistant (  $\mu$  ) values shall be governed by a Third Party Supervisory Control Agreement which shall be available for attestation of conformity purpose.

3. UL Listed & FM Approved

4. Water vapor permeability -  $0.09 \mu\text{gm} / \text{Nh.}$ , BS 4370 / 2.

5. Density 55 to 70 kg/m<sup>3</sup> or 3lb/ft<sup>3</sup> to 4lb/ft<sup>3</sup>.

6. Closed cell content - > 90%.

7. Fire rating:-

BS 476 Part 6 1989 (Fire Propagation): Class O rated

BS 476 Part 7 1987 (Surface spread of flame): Class 1 rated

(UNI 845 - UNI 9174).

Reaction to Fire Certification shall be available for all commercial insulation thickness up to 50mm

8. Corrosion risk assessment DIN 1988 / 7 certified," Material should be nitrite free and can contain no more than 0.2% of ammonia. Insulating material for stainless steel must contain no more than 0.05% of water soluble chloride ions “.

Corrosion risk assessment values shall be governed by a Third Party Supervisory Control Agreement which shall be available for attestation of conformity purpose.

9. Noise reduction DIN 4109 up to 32 dB (A).

#### **Important note:-**

- a) Manufacturer should be certified in accordance with ISO 9002 / UNI EN 29002 which includes production control procedures and systems as a pre – requisite.

- b) All tests should be conducted by independent, accredited institutes and the supervisory (third party) control agreements should be available for attestation of conformity purposes.

At all support points for the pipes, the Contractor shall provide manufacturer's recommended pipe support or approved equal material subject to Engineer's approval.

The insulation ladders shall be trained by the manufacturer's local agents before commencement of works.

The manufacturer's local agents shall visit the site periodically to inspect the insulation work.

The insulation thickness shall be as follows:-

In conditioned areas : Up to 4"  $\varnothing$  pipes - 19mm thickness

5"  $\varnothing$  and above - 25mm thickness

In unconditioned areas: Up to 5"  $\varnothing$  pipes - 25mm thickness

Above 5"  $\varnothing$  to 10"  $\varnothing$  pipes - 32mm thickness

Above 10"  $\varnothing$  - 50mm thickness

Aluminum cladding shall be applied to all insulated chilled water pipes and ducts running exposed on roof, shafts & other

Unconditioned areas. Aluminum cladding shall be of 22 gauge, it shall be used as protection against weather and

Mechanical damage .Fabricated 18 gauge aluminum covers shall be used for valves and fittings. The covers shall be in two sections hinged together and held in place by suitable stainless steel/aluminum clips.

## **VALVES AND ACCESSORIES**

### **Isolating Valves**

All isolating valves up to 50 mm shall comply to BS 5154 Series- B, wedge type, bronze construction, screwed bonnet, non-rising stem, screwed ends to BS21 or capillary to BS 864 Part 2 to match pipe work and enamelled aluminium hand wheel similar to Crane D159 or equal and approved. Alternatively, high quality ball valves are also acceptable such as Crane D171, D171c or equal and approved with PTFE seats and stem seals.

Isolating valves 65mm and above shall be of cast iron construction non-rising stem flanged ends to BS 4504 with cast iron hand wheel clearly marked to show direction of closing similar to Crane FM63 or equal and approved.

### **Globe Valves**

All globe valves up to 50mm shall be bronze valves to BS 5354 Series – B. metal to metal seat screwed ends to BS21 and enamelled aluminium hand wheel.

Globe valves 65 mm and above shall be cast iron construction flanged ends to BS 4504

### **Double Regulating Valves**

Double Regulating valves up to 50mm shall be to BS 7350 section 3.1, copper alloy.

Ends shall be threaded to BS 21 or flanged to match pipe work as indicated

Material shall be Bronze to BS 5154-Series B; oblique or Y pattern; inside screw rising stem; manufacturer's standard trim material. Provide drain plug facility.

Double regulating valves 65mm and over shall be cast iron, to BS 7350 section 3.1

Ends shall be Flanged to BS 4504, Cast iron to BS 5152, Oblique or Y pattern, copper alloy or nickel alloy valve face; rising stem outside screw; manufacturer's standard materials. Provide drain plug facility.

### **Flow Measurement Devices**

Up to 50 mm, Bronze, to BS 7350, section 3.2 – type 3 of fixed orifice either integral with or as a fixed orifice fitting close coupled to a double regulating globe valve.

Ends shall be threaded to BS 21 or flanged to BS 4504 to match pipework as indicated.

Material shall be bronze to BS 5154 series B and close coupled fixed orifice fitting to BS 7350 table 6 and Independent means for positive isolation on pressure tapping or adapter.

Flow measurement devices, 65mm and over shall be cast iron, to BS 7350 section 3.2

Ends shall be flanged to BS 4504, Cast iron to BS 5152 and close coupled fixed orifice fitting to BS 7350, table 6. Provide drain plug facility and independent means for positive isolation on pressure tapping or adapter.

### **Drain Valves**

Drain valves shall be bronze with hose union, blank cap and chain.

### **Test and Gauge Cocks**

Test and gauge cocks shall be bronze polish finish.

### **Check Valves**

Check valves to BS 5154 up to 50mm shall be bronze construction swing type screwed ends.

Check valves to BS 5154 65mm and above shall be cast iron construction swing type flanged ends.

### **Strainers**

Strainers up to 50mm shall be bronze construction Y type strainers screwed ends. Brass basket, screen perforations 0.7-0.9mm dia. with 250% FA.  
Strainers 65mm and above shall be cast iron construction Y type strainers flanged ends. Brass basket. 1.5-1.8mm screen perforation 200% FA  
Provide plugged connection for drain vent and differential pressure monitoring threaded to BS 21 .  
Screens shall be manufactured from perforated stainless steel sheet.

### **Expansion Joints**

Expansion joints shall be axial bellows types, to BS 4504 stainless steel multi ply construction with stainless steel inner sleeve with flanged ends.

### **Flexible Connections**

Flexible connections shall be stainless steel corrugated metal hose with stainless steel wire braid for absorbing vibration.  
Flexible connectors manufactured with multiple layers of nylon tire cord fabric and neoprene of high quality.

### **Test Plugs, Valve Controlled**

Provide self valve controlled test plugs for measurement of temperature and pressure, complete with captive cap for sealing when not in use. Ensure test plugs are suitable for system operating temperature and pressure. Provide one thermometer and pressure gauge for each range of conditions, for use with test plugs.

### **Flanges**

All flanges 65 mm and above shall be forged steel 68 kg with welding neck.

### **Thermometers**

Thermometers shall be provided on the supply and return of each zone, boiler, chiller, condenser, air handling unit, calorifier and anywhere else required or shown on drawings. Thermometer pockets shall be provided on the supply and return of each riser and sub-circuit. Thermometers shall be straight or angle pattern red mercury filled 250 mm scale 0oC - 80oC, for chilled water. The casing shall be V shape design moulded of GE valox polyester, black textured finished with glass protection against rattles by spring action.

The accuracy shall be 1% of the scale range.

### **Pressure Gauges**

Pressure gauges shall be provided on the inlet and outlet of each boiler, chiller, condenser pump and AHU's heating and cooling coils.

Gauges shall have a brass case with flanged neck. The movement shall be mounted on an independent frame and shall be free from strain caused by transit, or when fixing. The tube shall be of specially treated bronze and shall have a removable backplate to facilitate inspection or adjustment. Gauges diameter of dial shall not be less than 150 mm dia. Pressure range shall not exceed system working pressure more than 1.5 times.

### **Air Release Tanks**

The air release tanks shall be connected where shown on the drawing with flanged connections for easy removal. For type and fittings, see detail.



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**7.01 FANS CENTRIFUGAL.**

- 7.02 Furnish, install and connect up complete, all supply, return, exhaust, transfer and exhaust fans. All fans shall be of the non-overloading centrifugal type.
- 7.03 Fans shall have direction of rotation, discharge direction indicated on the casing. Fan arrangement shall be selected to suit the space conditions.
- 7.04 Fan housing shall be rigidly built and braced. Where fan scroll is 480 mm or more in width, provide access door with frame and gasket. All access doors shall be so fabricated that the inner surface is flush with the inside of the scroll. Raised frame doors of the pan type shall be provided on all fans where insulation is required. The doors shall be secured to the frame by hand-grip latches and shall be provided with lift handles. Bolted doors are not acceptable,
- 7.05 Fans shall not produce excessive noise as compared to units of like size and power when used in conjunction with the specified vibration isolation. Fan manufacturers shall furnish for approval for each fan, certified sound power ratings with an octave band analysis and also the volume- horsepower-pressure characteristic curves from shut-off to free delivery. Wheels shall have ample strength and shall be statically and dynamically balanced to avoid vibration and shall have blades designed to ensure quiet efficient operation.
- 7.06 Fans, unless otherwise indicated, shall be belted to respective motors by "V" belt drives. Sheaves shall be cast steel. Drives requiring 2 grooves or more shall have variable pitch motor sheaves and companion type fan sheaves. Belts shall have a rated capacity of not less than 150% of the motor horsepower. All "V" belt drives for integral horsepower motors shall have not less than two belts and shall have sufficient belt capacity to drive fans with one belt broken. Sheave ratio shall be selected so that top fan speed is not less than 105% and not more than 110% of rated speed. Drives for fractional horsepower motors may have one belt rated at 200% of motor horsepower. If to balance an air system, a fan other than the one provided should be required, necessary changes shall be made in the "V" belt drive by the Contractor at his expense. Supply and install belt guards of perforated metal, or as approved, for all sheaves and belts. Belt guards shall be grommetted at fan and motor shafts for ease in taking tachometer readings.
- 7.07 Each fan motor shall be sized to drive it's respective fan when fan is operating at a speed (due to pulley adjustment ) of 10% in excess of that required to meet fan performance, and when the fan requires the maximum power at this speed.

- 7.08 All fans shall carry the AMCA, or similar approved Certification Association performance and construction seal.
- 7.09 All fans shall be factory coated with one coat of primer and one coat of machine enamel. The interior of all fans and all wheels shall be painted as per manufacturer's standard.
- 7.010 All fans discharging directly up through the roof shall be equipped with a 38 mm casing drain which shall be piped to the nearest floor drain.
- 7.011 All fans scheduled to be arranged as belted vent sets shall have overhung wheels and motors mounted on a common base with the bearings. All such belted vent sets shall be furnished with weatherproof motor housings where located outdoors.
- 7.012 All double width, double inlet fans shall be provided with galvanized steel inlet guard screens.
- 8.00 FANS, AXIAL.**
- 8.01 Axial flow fans shall be capable of giving the design flow when tested to B.S. 8410.
- 8.02 Each fan shall be supplied with a suitably rated contactor/ starter/ isolator of approved pattern.
- 8.03 Fan casings shall be constructed of mild steel plates with angle stiffeners, with the casing hot dip galvanized after manufacture. The inlets and outlets of the axial flow fans shall be flanged for connection to the system. A drain plug shall be fitted and the casing designed to permit removal of the impeller.
- 8.04 An access door is to be provided on the casings of all fans. Casings shall cover both impeller and motor so that fans can be removed without disturbing adjacent ductwork or other components of the system. A flameproof external terminal box shall be fitted on the casing.
- 8.05 All lubrication points are to be extended to the outside of the casing and in a position that will permit access in relation to the adjacent plant, services or building structure.
- 8.06 Impellers shall be die cast in Aluminium alloy and X-rayed during manufacture. The impellers shall be capable of running continuously at 20% in excess of the rated speed. The impeller shall be keyed and locked onto the shaft, which shall be statically and dynamically balanced and tested at overspeed before dispatch from the Manufacturer's works.
- 8.07 Flexible connections shall be supplied and installed at both inlet and outlet of each fan.
- 8.08 The fan bearings shall be of the sleeve type wherever possible. The bearings are to be truly aligned and rigidly mounted on to the casing.

- 8.09 Graphs of the performance curve of each fan are to be forwarded to the Consultant together with a test report on the sound level at three diameters in a free field condition prior to the order being confirmed.
- 8.010 Blade angle shall be adjustable over at least a 30 degree range, with markings at the root to indicate the blade angle.
- 8.011 Axial flow fans are to be driven by electric motors of commercially silent pattern carrying a Makers guarantee in this respect. The motors shall be totally enclosed, fan cooled.
- 8.012 All motors are to be positioned to permit effective ventilation of the motor and all component parts of the fan and motor are to be suitable to withstand the temperature conditions expected in the fan.
- 8.013 Electrical connections to the motors shall be in a totally enclosed terminal box secured to the exterior of the casing. Wiring within the axial flow fans shall be suitable for the conditions within the casings.
- 8.014 A suitable steel support is to be provided for each fan, and the frame is to be fabricated from rolled steel channel with adequate cross members for bolting the fan into position. The frame shall be of welded construction with anti-vibration mountings.
- 8.015 The anti-vibration mountings are to be rubber in shear pattern and of a type that are bolted to both the fan and the support.

#### **10.0 PARTICULAR FAN**

##### **10.1 Inline Duct Mounted**

- 10.1.1 Unit casing shall be manufactured in epoxy coated aluminium alloy and shall house twin fan assemblies, each comprising double inlet forward curved centrifugal impellers running in an individual scroll. Impellers may be either direct driven or belt driven depending upon the fan duty. Motors shall be manufactured to BS 5000, TEFV type with sealed for life bearings. Motor insulation shall be Class 'F'.
- 10.1.2 Fans shall be fitted with air flow sensors and shall discharge into a common plenum chamber through a linked shutter system. Motors and flow sensors shall be prewired to fitted isolator accessible from outside.
- 10.1.3 Units shall be supplied for either roof mounting or plant room use as indicated on the drawings. Units shall be supplied with auto changeover panels with duty/stand-by selector switch, run/fail indication lamps, cyclic relay for duty sharing. Starter/Auto changeover panels shall be incorporated within electrical power supply panels as indicated on electrical schematic drawings.

#### **11.0 ELECTRIC MOTORS**

- 11.1 Unless otherwise specified in the equipment specifications, all motors supplied and installed as a part of mechanical works shall conform to the following requirements:

- 11.2 Motors shall be of high efficiency (I.E 3), and shall be constructed and type tested in compliance with relevant sections of BS. 4999 and BS. 5000. These shall be of totally enclosed fan cooled (TEFC) type to IP54 either with foot-mounting or flange-mounting suitable for its application. Unless otherwise specified, motors shall be designed for continuous running duty type S1 for operation during 3 years without maintenance. They shall be of such size as to drive adequately their associated equipment under the range of operating conditions, without overloading.

Windings shall be insulated with Class F insulation material to BS. 2757, and temperature rise of windings shall be class B (I.E. maximum of 80 deg.C. as measured by resistance method).

- 11.3 Electric motors up to and including 0.37 Kw shall be suitable for operation at 240 volts single phase. Motors rated above 0.37 Kw shall be of the 3 phase squirrel cage induction type with drip proof housing. Motors located directly in air streams shall be totally enclosed, fan cooled. All motors and equipment shall be tropicalized.
- 11.4 All motors shall be fitted with a power factor correction capacitors rated at 85% of the motor magnetizing KVA. Providing a power factor of not less than 0.93. This connection shall be in accordance with the B.S. Code of Practice. The capacitors shall be directly connected to the motor terminals. This connection shall be made by the equipment manufacturer. The capacitors shall be complete with discharge resistors.
- 11.5 All the motors shall be provided with a substantial earthing terminal tapped into the frame or clamping device located on the cable end box side of the motor. When air inlets are protected by a screen, such a screen shall be of corrosion resistant material.
- 11.6 Terminal connections shall be constructed in such a way that direct contact between screws, bolts or nuts and the conductor is avoided.
- 11.7 The terminals shall be alphabetically or numerically marked in such a way that the cyclic phase sequence in time of the supply corresponds to the sequence of terminal markings, the direction of rotation is clockwise when facing the driving end.
- 11.8 The terminal box shall be of sturdy construction with ample space for connecting cables. An earthing bolt of adequate size shall be provided inside the terminal box for connecting to the cable earthing. This will be in addition to the earthing bolt provided in the motor enclosure. The terminal box shall be adequately weather proofed with a suitable non sticking reusable non-hygroscopic gasket material. These terminal boxes also shall be provided with suitable cable sealing end boxes or cable glands.
- 11.9 The rating plate or plates shall be made of a corrosion resistant metallic material, fixed to a non-removable part of the frame, and shall be marked with the appropriate items in accordance with IEC 34-1, whereby the values given shall be those actually measured.
- 11.10 In addition to these items the following information shall be indicated:
- a) Type of enclosure in accordance with IEC 34-5 (IP classification).
  - b) Locked rotor torque on % of rated torque  
Locked rotor current in % of rated torque

- c) Allowable running-up time in seconds (ART) bearing type, size, fit, re-greasing period and type of grease.
- d) Net weight.
- e) Purchase order number.

11.11 All motor shafts shall be provided with a key way unless the motor is of the flange mounted type in which case they shall be complete with adjustable rails.

11.12 The frame shall be painted in accordance with manufacturer's standards. However, the painting shall at least consist of one coat of primer and two coats of oil-resistant finishing paint.

## 12.0 MOTOR STARTERS, CONTROLS SWITCHGEAR

All starters and other control components shall be supplied and installed under this section but the contractor shall ensure that all components supplied shall comply with the requirements of the relevant sections of the electric specifications.

## 13.0 PANEL BOARDS AND MOTOR CONTROL CENTRE.

Construction of all panel board, motor control centres, control panels etc. shall comply with the relevant sections of the electrical specifications.

### 13.1 Air handling unit starter panels:

Each air handling unit shall be supplied with a starter cubicle including the following:

- a) A door interlocked isolator handle.
- b) A three position HAND/OFF/AUTO control switch.
- c) A "motor run" lamp (green).
- d) A "motor trip" lamp (red).
- e) Motor speed selector switch
- f) Motor starter assembly.
- g) Control fuse.
- h) Filter pressure drop (filter clog indication)
- j) Inter-connection with fire alarm system and extract fans.
- k) Power fuses with carriers and bases.
- l) Moisture elimination heating coil.

- m) Heating element contactors and controls.
- n) Heating element 'energized' lamp (Green)
- o) Heating element 'tripped' lamp (Red) Note

:

|                       |   |                                                                         |
|-----------------------|---|-------------------------------------------------------------------------|
| Control switch "Off"  | - | Motor shall not run.                                                    |
| Control switch "Test" | - | Motor shall run except for local lock-off stop buttons. Control switch  |
| "Auto"                | - | Motor shall start and stop under dictation from remote control circuit. |

Starter panels, where installed outdoor, shall be of the weatherproof type in IP 65 enclosure.

### 13.2 Fan Coil Unit Control Enclosure

Control components of each fan coil unit like contactors, transformers, controllers, etc. shall be housed within an enclosure located besides the unit within the ceiling void.

## 14.0 ELECTROSTATIC AIR CLEANER

### 14.1 Electrostatic Air Cleaner

#### 14.1.1 Equipment Description:

The Electrostatic preceptor shall be supplied as a one-piece factory assembly of individual modules selected to treat kitchen exhaust air. The unit shall be ETL or UL labeled per tests and compliance with UL 867, 3<sup>rd</sup> Edition standards. The units shall also be in complete accordance with ASHREA standard 52-76 for air cleaners test and rating of Efficiency, Resistance & Dust holding capacity. And DOP for oil mist.

#### 14.1.2 Equipment Specifications:

The unit shall be designed and constructed and supplied as follows by a manufacturer specializing in the research, design and manufacture of products specified in this section with a minimum of five years of documented experience, who issues complete catalog data on the total product.

#### 14.1.3 Unit Housing:

Housing shall be 1.0/1.5 mm thickness welded galvanized sheet construction to protect against rust and corrosion. Each section shall include single door access, located one side of the unit. The access door shall be mounted on steel hinges and secured with adjustable, gasket sealed lever latches allowing for component access and removal. All doors shall be gasketed to prevent air and water leakage. Doors to charged high voltage components shall be equipped with electrical interlocks, for interconnection into the primary power supply, to prevent access when the components are charged. All supplied with positive pressure cell contact plungers for power disconnection when doors are open or cell is not fitted in the right direction of air flow.

#### 14.1.4 Base construction:

The modules are to be mounted on a channel for rail or floor mounting or ceiling suspension. Lifting lugs shall be incorporated in the base channel to allow for rigging.

#### 14.15 Finish:

The external casing finish shall be a durable industrial grade semi gloss baked- on epoxy ester, not less than 3 mil minimum thickness.

#### 14.1.6 Electrostatic Precipitator Modules:

The electronic air cleaner shall be the two-stage dual voltage plate type. rated at not less than 95% efficiency per the { D.O.P. Test Method for oil mist} {ASHRAE test standards for dry particulate}. The collection cells shall be in a single-pass arrangement to provide for maximum collection efficiency.

#### 14.1.7 Air Inlet and Outlet Collars:

Air inlet and outlet collars will contain {nominal 20mm metal mesh}, air baffle, accessible from the outside of cabinet through hinged access door. Doors will be gasketed to prevent air leakage from cabinet.

#### 14.1.8 Ionizing-Collecting Cells:

Ionizing-Collecting cell shall be of one-piece construction 14" inches deep in direction of airflow. All support framing, end plates and ionizer ground electrodes shall be 0.090 inch thick aluminum. Both repelling and collector plates shall be 0.032 inch thick aluminum, 9.41 inch deep in direction of airflow and rigidly retained in place with tubular spacers and tie rods. Spacing between plates shall be no less than 0.276 inch. Ionizing electrodes shall be minimum 7 numbers of 24 gauge stainless steel spiked design, for prevention of corroding or breaking & more effective air cleaning, rigidly supported both vertically and laterally (ionization wires are not acceptable). Ionizing electrode to ground electrode spacing shall be 1inch and ionizer ground electrodes will be 3.74 inch in direction of airflow. Face area of each cell shall be 0.15 sq.m and the effective collecting area 5.92 sq.m. High voltage support insulators shall be of self-glazing Cordierite standoff ceramic with all surfaces, including center hole, glazed to enhance dielectric strength and retard tracking, not less than 14 numbers per cell, each with min 3.74 inch in diameter and 1.58 inch in height, to reduce arc tracking on insulators for better efficiency, longer cell life, easy cleaning even the tacky contaminants and to prevent cracks and eliminate imperfections. Flexible wiring connected to the insulators shall be insulated with form fitting nitirle boots sized to the specific wire size to insulate and maintain collection efficiency. Insulators shall be mounted out of the airstream, to reduce contaminant buildup. All high voltage electrical connections within each tier of cells, shall be between cells and automatically made when cells are installed. All high voltage electrical connections between cell tiers and between power packs and cells shall be located on the access door end of the cabinet and manually connected for ease of service. High Voltage wire connections to the cells shall be shielded by means of a removable insulator boot.

#### 14.1.9 Power Supply

Power supplies shall be pulse width modulated & 100% solid state, High frequency with built in short circuit and arc protection, providing a dual high voltage output of (+)12 KVDC for the ionizer and (+) 6.0 KVDC for the collector. A regulated output of up to 5.5 MA (maximum per supply) shall be supplied to maintain the specified collection efficiency. Integrally mounted door operated electrical safety interlock switch shall be provided to prevent access to the high voltage components without first interrupting the primary input power. The power supply shall operate over a temperature range of -35 to 65 degrees C, be self-protecting and accommodate an LED light indicating the performance status of the ionizing/collecting cell.

High voltage output leads shall be sealed and a bleed resistor incorporated to remove stored electrical charge when the power supply is de-energized.

All power supply components shall be designed for ease of mounting and servicing. High Voltage power supply cable shall be furnished in fifty (50) foot lengths as standard, for connection of the control cabinet power supplies to the ATS Unit collector cells. High voltage power cables shall be of one continuous length, splicing is not acceptable.

#### STATIC PRESSURE DROPS

| <u>ESP Section</u>                            | <u>S.P.D H<sub>2</sub>O</u> |
|-----------------------------------------------|-----------------------------|
| 40% Open perforated prefilter or after filter | 0.14"                       |
| Metal Mesh prefilter or after filter          | 0.15"                       |
| 2" Impinger                                   | 0.10"                       |
|                                               | 1.25"                       |
| <u>Fan Transition</u>                         | 0.11"                       |

The ESP section must have both an internal prefilter and an after filter, select and add for each. External losses for ductwork, exhaust hoods, manufacturing equipment with associated entry losses, kitchen hoods, etc. must be added with the above internal equipment losses to calculate total fan static pressure required.

#### 14.1.10 Electrical:

Factory wiring will be in accordance with the National Electrical Code. Required field wiring shall conform to the national Electrical Code and any local code requirements.

### 15.0 CLOSE CONTROL UNITS

#### 15.1 Cabinet

The cabinets shall be with rounded corners with flush fitting panels and compression locks for effective sealing.

The units shall have a rigid angle frame with hinged / lift off panels to provide access for installation and servicing. Panels are lined with fire resistant, Class 'O' non – eroding insulation material with thermal and acoustic properties.

#### 15.2 Microprocessor Controls

The units shall have easy to use and fully informative designed systems for precise monitoring and control of room conditions, equipment status and alarm warning. The control system shall be also includes a graphical display with real time trend logging.

#### 15.3 Switchgear



The electrical switchgear shall includes the following :- Bottom entry gland

plate with grommited cable entry holes. Interlocked door isolator.

MCBs and contactors for fans, heaters, humidifier and compressors as appropriate. MCBs for remote condensing units.

MCBs plus fan speed controllers for remote condensers. Easy connect customer terminals.

Colour coded and numbered cabling, neatly run in trunking segregating high and low voltages. All wiring complies with IEC60204-1.

#### 15.4 Filters

The filter shall be and disposable and have an efficiency of 30 percent ASHRAE 52/76, Eurovent 4/5 arrestance value of 90 percent (G4). Fitted in the return air stream, these shall be held by quick release retaining clips and are accessible from the front of the unit.

#### 15.5 Cooling Coil

Dual 'V' frame R407 or equivalent optimized coils shall be multi-row constructed from rifled copper tubes with aluminium fins. Sloped stainless steel drain trays ensure efficient removal of condensate via factory fitted transparent traps.

#### 15.6 Heaters

Electric heaters shall be fitted as standard, have incoloy elements with stainless steel finning. Balanced over three phases, they incorporate safety over – heat protection. The heaters shall be thyristor controlled.

#### 15.7 Fans

Low speed fans shall be forward curved impellers and 'sealed for life' bearings are used. Each failure – alarmed fan shall have its own TEFC motor which can be inverter controlled. Pressure sensors monitor airflow shall be provided.

#### 15.8 Humidifier

A 'fail safe' electrode boiler humidifier, located in the air stream shall provides clean sterile steam. The output shall be controlled and the microprocessor controller (Humidifier self – cleaning system) shall be provided.

#### 15.9 Compressor (for DX System)

The compressors shall be scroll type and have the following characteristics :

1. High EER / COP
2. Axial and radial compliance.
3. Liquid tolerance design – the compressor shall ingest liquid refrigerant without damage.
4. High volumetric efficiency – no clearance volume and no valves shall be provided.
5. Low vibration level – continuous compression process (3 rotations) and vibration amplitude shall be less than 0.1mm.

6. Light load start, smooth starting torque. Low motor torque variation.

#### 15.10 Air Cooled Condensers

The condensers shall be copper tube with aluminium fins.

#### 15.11 Controls

The controller shall be stand alone with the following feature. User friendly  
simply easy to use graphic display.

Non – volatile memory.

16 bit surface mounted technology.

Real time trend logging.

Modular software and hardware.

Full networking facility.

32 point alarm annunciation. Main function  
icon display.

### 16.0 SMOKE CONTROL STATION

Fire fighter's smoke control station (FSCS) shall be located in the central command / control room or as indicated on drawings.

The control station shall include the following features:

- a) The fire fighter's smoke control station shall provide full monitoring and manual control capability over all smoke control systems and equipment.
- b) The control station shall have the highest priority control over all smoke control systems and equipment. Where manual controls are also provided at other building locations for control of smoke control systems, the control mode selected from the FSCS should prevail. FSCS control should override or bypass other building controls such as hand off auto and start/stop switches located on fan motor controllers, and controls intended to protect against electrical overloads, provide for personnel safety, and prevent major system damage.
- c) The fire fighters smoke control station shall contain a building diagram that clearly indicates the type and location of all smoke control equipment (fans, dampers, etc.). The building areas affected by the equipment shall also be clearly indicated.
- d) The actual status of the systems and equipment that are activated or are capable of activation for smoke control shall be clearly indicated at the fire fighters smoke control station.
- e) Status indication shall be provided for on and off status of each individual fan used for smoke control. 'ON' status should be sensed by pressure difference as proof of air flow.

### 17.0 CARBON MONOXIDE DETECTION SYSTEM

- 17.1 The system to comprise of remote sensors, exact location to be agreed with the Consultant. Sensors to be wired to multiple channel carbon monoxide monitors.
- 17.2 Control shall be 24 volt, 1 ph, 50 Hz.
- 17.3 Monitor controls to be housed in a lockable NEMA-12 enamel finished cabinet with hinged door and be complete with window.
- 17.4 Cabinet to be fitted with indicating lights to indicate sensor being monitored and level of carbon monoxide.
- 17.5 Carbon monoxide sensors will continuously monitor the concentrations of carbon monoxide (CO) prevailing within the car park.
- 17.6 Under normal circumstances with the CO levels below 25 ppm, the extract fans shall run at low speed to achieve an air change rate of 3.
- 17.7 On sensing 25ppm and above at any CO sensor, a visual and audible alarm shall be initiated at the panel. The extract fans shall be triggered to run at high speed to achieve an air change rate of 5.
- 17.8 Zone indication of CO sensor being monitored shall be indicated on the multiple channel CO monitor.
- 17.9 The CO monitor shall be interfaced with the BMCS.
- 18.0 SUBMISSIONS.
- 18.1 Initial technical submissions are to accompany the bid including full technical details of every item of equipment proposed for the project, with the appropriate figures and details highlighted in marker pen or similar in standard manufacturer's catalogues.
- 18.2 After final checking of sizing after award of contract, resubmission will be made with any changes specifically noted.
- 18.3 Technical submissions for the following are particularly required to be submitted.
  - Air handling units.
  - Fan coil units.
  - Outdoor Units
  - Control Centre.
  - Fans.
  - Close Control Units Controls
- 18.4 In the event that the Consultant requires further information on any of the proposed items the Contractor shall arrange visits to similar units which he has installed. In any event, all equipment proposed for the project shall be currently installed and working satisfactorily locally. The Contractor must be able to provide references of at least three satisfied Owners and Consultant with equipment of the particular size and model proposed.

End of Section

## **SECTION 4**

### **DUCTWORK AND AIRSIDE EQUIPMENT**

## SECTION 4

### DUCTWORK AND AIRSIDE EQUIPMENT

#### INDEX

|      |                                             |    |
|------|---------------------------------------------|----|
| 1.0  | DUCTWORK MATERIAL. ....                     | 3  |
| 2.0  | DUCTWORK SPECIFICATION.....                 | 3  |
| 3.0  | DUCTWORK GENERAL. ....                      | 4  |
| 4.0  | LOW PRESSURE DUCTWORK. ....                 | 5  |
| 5.0  | FLEXIBLE DUCT WORK - LOW PRESSURE.....      | 6  |
| 6.0  | FLEXIBLE CONNECTION. ....                   | 7  |
| 7.0  | ACCESS PANELS .....                         | 7  |
| 8.0  | BALANCING DAMPERS.....                      | 8  |
| 9.0  | BACKDRAFT DAMPERS.....                      | 8  |
| 10.0 | SPLITTER DAMPERS. ....                      | 9  |
| 11.0 | FIRE DAMPERS. ....                          | 9  |
| 12.0 | SILENCERS.....                              | 10 |
| 13.0 | VIBRATION ISOLATION (FOR ALL SECTIONS)..... | 10 |
| 14.0 | DUCT DRAINS AND DRIP PANS. ....             | 12 |
| 15.0 | DIFFUSERS AND REGISTERS.....                | 12 |
| 16.0 | EXTRACT AIR LOUVRES .....                   | 15 |
| 17.0 | SAND TRAP LOUVRE FOR AIR INTAKE.....        | 15 |
| 18.0 | FILTERS .....                               | 16 |
| 19.0 | MOTORIZED SMOKE AND FIRE DAMPERS.....       | 16 |
| 20.0 | SUBMISSIONS .....                           | 17 |

## 1.0 DUCTWORK MATERIAL.

- 1.1 Make all ductwork, unless specifically noted otherwise, of galvanized sheet steel conforming to BS 2989 or ASTM A653A, 653M, G90 (Z275) and having a nominal coating of 1.25 oz./sq. ft. The galvanizing shall be carefully done and the sheets shall be of such quality that they may be bent flat on themselves with no fracture to the coating or the base metal.
- 1.2 Flexible ductwork shall be light weight aluminium laminated duct suitable for low and medium pressure systems. The aluminium laminated construction shall encapsulate a high tensile steel wire helix between two layers of 0.9 micron thick aluminium plus 12 micron thick polyester. Flexible ducts shall be with factory applied thermal insulation and vapour barrier. Insulation shall be 25mm thick, 16 kg/m<sup>3</sup> density.
- 1.3 All supports, stiffeners, flanges etc. used in ductwork construction and installation shall be hot dip galvanized.
- 1.4 Extract ductwork handling corrosive gases (battery rooms) shall be stainless steel 316 construction.
- 1.5 Smoke Extract and stairwell pressurization ductwork shall be fire rated designed, constructed and tested to meet the requirements of BS 5588 Part 9 and BS 476 Part 24. Ductwork complete with supports, accessories and fittings shall to be tested and approved to meet the following requirements with fire inside and outside.

Stability - 240 minutes

Integrity - 240 minutes

Insulation - 120 minutes

## 2.0 DUCTWORK SPECIFICATION.

- 2.1 Ductwork fabrication and hanging methods shall be as described in the "Equipment" Volume of the American Society of Heating, Refrigerating and Air Conditioning Engineers Guide and Data Book, latest edition, and/or current edition of applicable manuals published by the Sheet Metal and Air Conditioning Contractors National Association Inc. (SMACNA), where methods described in these volumes are not at variance with the requirements of any authority having jurisdiction, or do not conflict with methods described hereafter. Ductwork shall not be shop fabricated until work has been job measured and interference situations coordinated.
- 2.2 Any Contractor who is more familiar with British Standards may refer to B.S. CP. 352 and DW. 144. Any discrepancy will be clarified by the Consultant on request.

In any event, the standard of work required is the best available, and the specifications are to be so interpreted.

### 3.0 DUCTWORK GENERAL.

- 3.1. All dimensions shall be checked on site before ductwork manufacture is commenced.
- 3.2. The whole of the ductwork installation shall be carried out by an approved specialist in ductwork manufacture and installation. No ductwork shall touch the building structure or building finishes direct, but shall be isolated with insulating spacer.
- 3.3. The fabrication shall be carried out in a neat and workmanlike manner with all ductwork true in size and cross-section, braced and stiffened as specified and with all internal and external surfaces free from projections and sharp edges.
- 3.4. At each main branch in ductwork and at each fan discharge and suction, provide sufficient number of Pitot tube test holes for balancing systems. Also provide test holes for traverse fan discharge and at all equipment. Test holes shall be located within easy reach of catwalks or ladder. Each test hole shall have 20 mm clear opening, provided with a metal ring plate with a threaded hole in the boss, and matching screwed head plug. Where these plugs are installed in insulated ductwork, provide an extension collar against which the insulation can be finished.
- 3.5. Reinforced holes shall be provided where thermometers, manometers, thermostats, gauges, damper rods etc., occur in ductwork. Extended collars shall be provided for the reinforced holes where these occur on insulated ductwork. Where copper tubing passes through ductwork, or casing, provide a rubber grommet to prevent damage to copper tubing.
- 3.6. Ductwork shall be rigidly suspended or supported from building structure. Expansion type concrete inserts shall be placed so that the fastener is in shear rather than tension. Powder actuated fasteners placed by an explosive charge will not be accepted. Angle type trapeze hangers with rod supports spaced at 2 meters maximum shall be used. "C" type beam clamps will not be accepted. Provide necessary steel angle iron required for bracing of ductwork or equipment, and for supporting ductwork from building structure.  
  
A layer of felt strip 12.5 mm compressed thickness shall be provided between any support member which is designed to clamp or grip the duct (e.g. circular duct band clip) or on which the duct is to rest. All supports shall be hot dip galvanized.
- 3.7. Increase in duct size shall be gradual. Where width or largest dimension of a duct is over 450 mm, duct shall be stiffened by bending in a break across corners in both directions. Ducts shall be self-supporting and complete in themselves. Single thickness partition between ducts will not be accepted. Visible internal portions of duct outlets to grilles and registers shall be painted in dull black.
- 3.8. All necessary allowances and provisions shall be made in the installation of the ducts for the structural framing of the building and when changes or offsets are necessary, the required cross-sectional areas shall be maintained. All of these changes however, shall be approved, and installed as directed at the time.
- 3.9. During installation the open ends of ducts shall be protected with blank, flanged sheet metal baffles, securely attached to prevent debris and dirt from entering.

- 3.10 Where ducts are shown connecting to masonry openings and/ or along the edges of all plenums at floors, walls, etc., provide a continuous 30 x 30 x 4 mm galvanized angle iron which shall be bolted to the structure and made airtight to same by applying caulking compound on the angles before they are drawn down tight. The sheet metal at these locations shall be bolted to the angle iron framing.
- 3.11. All air ducts, casings, plenums etc., shall be constructed of lock forming quality prime galvanized steel sheets, which are free from blisters, slivers, pits, imperfectly coated spots etc. No second quality sheet metal allowed.
- Where damage (or rusting) has occurred on galvanized duct work the affected section shall be made good by painting with two coats of zinc-rich paint and approved finishing paint, or where the damage in the Consultant's opinion cannot be made good, then a new section of ductwork shall be provided at no cost to the contract.
- 3.12. Duct shall be constructed using double or Pittsburgh lock corner seams. All seams shall be hammered down and made airtight by applying sealant before hammering down. For transverse joints, refer to current ASHRAE guide for low pressure ductwork.
- 3.13. Support the vertical ducts installed in the various shafts at each floor level with galvanized supporting irons of approved size. Install these angles across the width of the shaft, with their ends attached to angle irons securely anchored into the masonry walls of the shaft, or attached to the framing of the floor openings. The ducts shall be bolted to these supporting angle irons.
- 3.14 Ensure that all openings required through floors, walls, partitions etc., for the duct system are provided in the exact location required.
- 3.15 Each piece of ductwork shall be wiped inside and out before installation and all open ends shall be capped and sealed to prevent entrance of dirt during construction. Ensure that ductwork systems are clean and free from dirt, dust, grime, debris etc., before initial operation of fans. Fans shall not be operated until the filters are installed and approval from the Consultant has been obtained.
- 3.16 The bottom joint and 150 mm of vertical joint on outside air intake ducts and mixing chamber ducts shall be soldered and made watertight. Provide drain connection and run copper drain pipe to nearest floor drain.
- 3.17 All fixing devices including nuts, bolts, washers etc., used in the construction or support of galvanized ductwork shall be Sherardized, or cadmium-plated.
- 3.18 Final connections to diffusers shall be carried out using flexible ductwork.
- 4.0 LOW PRESSURE DUCTWORK.
- 4.1 This applies to ductwork with mean velocities less than 2,000 fpm (10 M/Sec) and static pressures of 2 inches water gauge, (500 pascals) or less.



- 4.2 Rectangular low pressure ductwork shall be fabricated from prime quality, re-squared, tight coat galvanized steel sheets of the following gauges.

| Size of Duct (Milli metres) | Gauge of Sheet Metal.            |
|-----------------------------|----------------------------------|
| 300 mm less in width/depth  | No.26 US Std. Gauge (0.6 mm) 300 |
| to 750 mm width/depth       | No.24 US Std. Gauge (0.7 mm) 750 |
| to 1350 mm width/depth      | No.22 US Std. Gauge (0.85 mm)    |
| 1350 to 2100 mm width/depth | No.20 US Std. Gauge (1.00 mm)    |
| 2100 mm and over            | No.18 US Std. Gauge (1.3 mm)     |

- 4.3 Reinforcing and joints shall be in accordance with ASHRAE Guide and Data Book, Equipment Volume.

Particular note is to be made to rigidity classification, and joint constructions. Up to 400 mm duct size is to have a "Standing S" joint and size 400 to 1200 mm is to have a "Standing S-angle reinforced" joint.

Size above 1200 mm is to have a "Standing S-angle reinforced" joint but, with intermediate reinforcing angle.

- 4.4 Low pressure suction and discharge plenum chambers shall be fabricated from 1.3 mm (18 gauge) galvanized steel with galvanized angle iron framework and bracing.

- 4.5 In square elbows and in elbows where radius is less than 1-1/2 x width of duct, sheet metal deflector vanes shall be installed the full height of the duct, being securely riveted in place. All vanes shall be double thickness vanes of two gauges heavier than the duct in which they are installed, and shall be factory made, not site fabricated. Vanes shall be tack welded to vane rail. For vane lengths over 1000 mm, tack weld vanes to 10 mm tie-rod at mid-span.

- 4.6 Circular low pressure ductwork shall be fabricated from prime quality as per DW 144 or SMACNA Standards.

- 4.7 Exposed circular low pressure ductwork shall be double wall spiral duct with 30 percent perforated inner shell and with 25mm thick, 24kg/m<sup>3</sup> density fibre glass insulation.

## 5.0 FLEXIBLE DUCT WORK - LOW PRESSURE.

- 5.1 Flexible ductwork shall be manufactured with a two layer aluminized polyester inner core, surrounded by 25 mm thickness of 24 Kg/M3 density fiberglass, all wrapped in a reinforced multilayer aluminium polyester outer jacket.

Ductwork shall meet the standards of NFPA 90A, and be U.L. listed or to meet BS 476 and BS 413. Maximum flame spread rating - 12 (Class 1). Maximum smoke developed rating - 50.

- 5.2 Duct clamps shall be made of an 8mm wide stainless steel band with 'Lifted' edges to avoid damage to the duct. The clamp shall be provided with a 'flip up' and 'quick lock'. Tightening head for ease of firing and speed of application.

- 5.3 Flexible duct installation shall be in accordance with manufacture's instruction. Joints between factory insulated flexible ducts and field insulated ductwork shall be sealed and taped under this section.

## 6.0 FLEXIBLE CONNECTION.

- 6.1 Flexible connectors at inlet and discharge to air handling equipment (AHU / FCU) shall be pre-assembled 24 gauge (0.7 mm) galvanized steel to ASTM A 653A, 653M with minimum of 75 mm wide (exposed) fabric. For higher pressure applications 25 mm of width of fabric shall be used for each 25 mm of static pressure.
- 6.2 Flexible connectors attached to acoustically treated ductwork shall be insulated with 25 mm fiberglass insulation packed between flexible connector and 16 gauge (1.5 mm) galvanized steel housing. Housing shall be fastened to duct with sheet metal screws. At equipment collar caulk between collar and flange on housing with 10 mm thick permanently flexible sealant.
- Care shall be taken to ensure that ducts on both sides of the connection are independently supported and that no "bridging" occurs.
- 6.3 All flexible connection will be by an approved manufacturer, with ratings of fire spread, strength etc., listed by a recognized Testing Authority. All flexible connectors shall be UL certified.
- 6.4 The fabric shall be rated for use up to 200 degrees F (93 degrees C) working temperature with a tensile strength of not less than 100 psi (690 Kilopascals).
- 6.5 The material shall also be impervious to moisture, dimensionally stable, and shall not rot.
- 7.0 ACCESS PANELS
- 7.1. Duct access panels shall be minimum 450 x 300 mm. unless restricted by duct dimensions. All access panels shall be constructed from galvanized steel sheets, 25 mm apart with necessary reinforcing inside for rigidity, with space filled with glass fibre insulation. Panels shall be made airtight with a continuous neoprene rubber gasket.
- 7.2. Openings in ductwork shall be provided with continuous galvanized reinforcing bars, which on insulated ductwork, shall be extended to the face of the insulation. 'Small' panels shall be provided with at least two self tightening camlocks. All panels shall have drawer type handle.
- 7.3. Apparatus access doors shall be minimum 600 x 1500 mm with angle or channel frame. Provide two 75 mm strap hinges with brass pins; and two handles minimum, operable from inside and outside. All access panels shall be constructed from double thickness 18 gauge (1.3 mm) galvanized steel, with 25 mm insulation between sides with necessary reinforcing for rigidity. Doors shall be made airtight with "tadpole" gaskets.
- 7.4 Provide access panels where shown, required and directed and in the following locations:
- Bottom of all duct risers.
  - Next to outside air intakes and outlets.
  - At each fire damper.
  - Into apparatus casings to facilitate maintenance and cleaning of all components.
- 8.0 BALANCING DAMPERS.
- 8.1 Install volume control dampers in accessible locations at all branch connections and wherever necessary to adjust the flow of air to secure correct distribution. Casing and blades shall be made of galvanized sheet metal, and shall be equipped with an approved device for fastening in any desired position. This device shall be such that the damper cannot move or rattle and pointer shall indicate the position of the damper from the outside of the finished duct insulation.

- 8.2 Dampers shall be aerofoil multileaf opposed blade with blade height not more than 180 mm.
- 8.3 All blades shall be operated by a single operating quadrant with gears and links as necessary.
- 8.4 Spindles shall be non-corrodible, passing through bearings made of special plastic temperature resistant to 100 deg. C. Capped bearings shall be provided on non drive side.
- The whole damper assembly shall be mounted in a galvanized frame with flanges.
- 9.0 BACKDRAFT DAMPERS.
- 9.1 Backdraft dampers shall be low leakage with parallel blades and neoprene edge seals.
- 9.2 Dampers frames shall be constructed from galvanized sheet steel with Aluminium blades. Blade stub shafts shall be brass with PVC bearings. Sealing strips on blades shall be polyester foam.
- 9.3 Pressure relief dampers shall be multi-parallel blade with weighted arm closing assist. The frame shall be anodized Aluminium channel sections with formed Aluminium blades. Maximum blade length shall be 100 mm, and polyester foam seating strips shall be incorporated on blade edges. Bearings shall be in PVC with non-corrodible shafts.
- 10.0 SPLITTER DAMPERS.
- 10.1 In each low pressure system take off where opposed blade duct dampers are not specified, splitter dampers shall be provided. The only justification for not providing splitters is in a system which can be demonstrated to be sized by a static regain programme and which is run in high pressure fittings.
- 10.2 Splitters shall consist of hollow blades in a vane rail assembly, made from galvanized steel by a recognized manufacturer.
- 10.3 A lockable quadrant adjustment lever shall be located outside the insulation, and marked clearly to show vane position.
- 11.0 FIRE DAMPERS.
- 11.1 Provide, where required by ordinances or codes, fire dampers made to British standard B.S. 476 or North American Standards or N.F.P.A. 90A, and complete with angle iron frame, 72 deg. C fusible link, pivot rods, and spring catches. Enlarge duct sections around fire dampers, to allow 100% unrestricted duct area while in open position. Provide approved type access doors, with airtight gaskets, for inspection and servicing of fire dampers. Fire dampers shall be 2.0 hours rated and shall be labelled by manufacturer.
- 11.2 Provide fire dampers in all ducts over 125 sq.cm. in area, in the following locations, whether or not specifically required by ordinances and codes:-
- Duct entering and leaving fire shafts (duct shafts). Ducts passing through designated fire walls.
- Ducts through floors, and not encased in fire shaft. Ducts leaving and entering plant area.
- Ducts entering and leaving storage areas.
- 11.3 Where fire dampers are located remote from fire partition, duct between fire damper and partition shall

be encased in double metal lathe and plaster or other fireproofing acceptable to Authorities having jurisdiction.

- 11.4 Fire damper construction shall be of galvanized steel casing, galvanized steel blades, fitted with catch plates, stainless steel closure springs and stainless steel seals. Blades shall be out of the air stream. Fusible links shall be rated for a fusing temperature of 72 deg. C unless otherwise specified.

- 11.5 Damper installation frame shall be supplied by the fire damper manufacturer.

- 11.6 Dampers shall be UL listed or approved by an equal certification authority.

## 12.0 SILENCERS

- 12.1 Silencers shall be installed when required to achieve the specified noise levels.

- 12.2 At shop drawing stage the contractor will undertake calculation for every unit and every area and determine the insertion loss required to meet the stated noise criterion. Supply and return duct noise must be considered as well as duct breakout noise.

- 12.3 Materials of construction shall be galvanized sheet metal and mineral fibre acoustic fill which is inorganic, inert, moisture and vermin resistant. Silencers shall be so constructed as to prevent erosion and pregnability of the acoustic fill.

- 12.4 The silencers shall incorporate a removable panel which provides complete access to all internal surfaces for cleaning and also permits removal and replacement of acoustic fill.

- 12.5 Silencers shall be factory made, and shall have available certified test data concerning insertion loss. This must be available when required, in advance of ordering units. Where a silencer is made under licence, or part assembled locally, then tests must also be made after assembly and witnessed by an independent authority.

- 12.6 The static pressure loss of any required silencer must be considered in air handling equipment original selection.

## 13.0 VIBRATION ISOLATION (FOR ALL SECTIONS)

- 13.1 All plant and equipment shall be isolated from the building structure in such a manner that noise and vibration is not transmitted through the structure.

Anti-vibration equipment will be manufactured by a specialist company acceptable to the Consultant, and with standard ratings information available.

The following are guideline requirements and must be confirmed by the specialist acoustic consultant as being adequate for the particular equipment proposed by the Contractor.

## 13.2 Pumps.

Water and fire pumps are to have open type spring isolators with equal stiffness in horizontal and vertical planes. They shall be supported on 10 mm thick ribbed neoprene Vibropad and will be colour coded to aid identification.

They will also have level adjustment screws and be rated for 75% more than the anticipated design load.

The spring shall be located between the housing pad and the concrete inertia base. The inertia base for each pump will be designed to give a static deflection of 50 mm at rated design load conditions, and shall be fully independent of any adjacent pad.

## 13.3 Air Handling Unit, Floor Mounted.

Floor mounted air handling units shall be mounted on open spring type isolators with 25 mm static deflection, equal stiffness in horizontal and vertical planes and a 10 mm neoprene Vibropad base.

They shall have built in levelling screws and limit stops, and colour coding to aid in identification.

#### 13.4 Air Handling Units, suspended.

These shall have box type spring isolation with level screws and neoprene snubbers. Spring will be colour coded to aid identification.

Springs will be rated for 50% more than the design load and shall have a static deflection at this design load of 25 mm.

#### 13.5 Centrifugal fans, floor mounted.

Centrifugal fans shall have open type spring isolation with equal stiffness in horizontal and vertical planes. They shall be supported on 10 mm thick neoprene Vibropad and will be colour coded to aid identification.

They will have level adjusting screws, and will be rated at 50% more than the design load and shall have a static deflection of 50 mm.

The springs will be located between the fan and the secondary support steel.

#### 13.6 Axial fans, floor mounted.

Axial fans shall have open type spring isolation with equal stiffness in horizontal and vertical planes. They shall be supported on 10 mm thick Vibropad, and will be colour coded to aid identification. They will have level adjusting screws, and will be rated at 50% more than the design load, and shall have a static deflection of 50 mm.

#### 13.7 Piping.

All piping connected to isolated equipment shall be supported with isolation hangers for at least the first three points of support on all sides of the equipment.

The first point shall have a static deflection of 2 x the equipment deflection, except that the deflection of this point shall not exceed 50 mm ( with pipe work full).

The second and third supports shall have deflections of 25 mm each.

For suspended piping, springs will be box type spring hangers with snubber, level adjustments, and colour coding.

For low level piping mounted on roller systems, the isolators will be open type springs with equal stiffness in horizontal and vertical planes.

#### 13.8 In addition, Flexible connection shall be included at inlet and outlet to all items of equipment.

Flexible connections will be of full line size and will be flanged with gaskets and mating flanges.

Connectors shall be twin sphere manufactured with multiple plies of nylon tire cord fabric and neoprene in hydraulic rubber presses and vulcanized.

#### 13.9 Ductwork.

(See Section 6 for flexible connection.)

#### 14.0 DUCT DRAINS AND DRIP PANS.

##### 14.1 Provide drip pans under cooling coils mounted in ducts in built up plenums.

Drip pan shall be constructed of not less than 18 gauge steel (1.2 mm) hot dip galvanized after fabrication, with all joints welded or soldered. Sides of pans shall be not less than 100 mm upstream and 300 mm downstream of coil.

Pan width shall extend 50 mm outside coil headers and/or return bends to collect any condensation released from these surfaces.

Pan shall be equipped with one or more 25 mm drain which must be run to drainage system as indicated by the Consultant.

##### 14.2 Drain pans for fan coil units shall be factory manufactured and supplied with the fan coil unit. Drain pan must be factory insulated on the sides in addition to on the bottom, with at least 20 mm rigid insulation.

##### 14.3 Drain lines will have traps and falls as specified in the plumbing section.

#### 15.0 DIFFUSERS AND REGISTERS

##### 15.1 Diffusers, registers and grilles shall be arranged for flush mounting in lay-in type ceilings and over lap mounting in plaster, mineral tile and similar ceilings, with concealed fixings unless otherwise directed.

##### 15.2 Grilles, register and diffuser locations shall be adjusted to suit reflected ceiling drawings, or Consultant's site instructions. All grilles, registers, diffusers, louvres shall be from one manufacturer.

##### 15.3 Provide plaster frame for grilles, and diffusers installed in plaster ceilings.

##### 15.4 All diffusers, grilles and registers shall be supplied completely factory powder coated. Finish colour shall be to the approval of the Architect. The interior of all grilles and diffusers is to be factory painted matt black.

##### 15.5 All supply grilles and diffusers will have opposed blade balancing dampers. All will have foam rubber sealing band around the edge to seal to the structure. All pivots will be round section, not of formed sheet, and not relying on a spring steel locking wire.

##### 15.6 Unless otherwise specified basic grilles and diffuser materials shall be Aluminium extruded sections. Sections in the air-stream shall be carefully selected to minimize turbulence.

##### 15.7 All grilles and diffusers supplied on this project shall be tested and rated in accordance with ASHRAE Standard 70-72, ADC Test Code 1062-GRD and ISO 3741.

##### 15.8 Linear bar grilles shall be fabricated from aluminium, with 6.4mm wide bars on 12.5mm centres pressed into a notched steel retaining bar. The core can be either welded into the outer frame, or, where the grille is used in a sill application, held in the outer frame by spring clips fixed to the core retaining bar.

The outer frame shall be 35mm deep and shall have a visible flange 25.4 mm wide. Mitered end caps shall be welded to give a near invisible joint.

The grill shall be complete with an opposed blade damper painted matt black, and shall be fixed with universal mounting brackets. Both the damper and the fixing brackets shall be accessible through the face of the grille.

Continuous grilles shall be provided with positive alignment strips, which fit into special keyways extruded into the frame of the grille to ensure clean unbroken lines.

##### 15.9 Ceiling Diffusers shall be multicone giving 4 way horizontal discharge.

The three centre cones of the diffuser shall be manufactured from pressed aluminium, with the remaining cones and the outer frame fabricated from extruded aluminium welded at the corners to give near invisible joints.

One, two, and three way pattern cores shall be used as indicated on schedules. All cores shall be interchangeable.

The core shall be removable without the use of special tools, but for safety, shall be fixed to the outer frame by a small length of chain.

The diffuser shall be complete with an opposed blade damper painted matt black.

- 15.10 Wall registers shall be double deflection fabricated from aluminium, the front vanes being horizontal, the rear vanes vertical. This grille shall be complete with an opposed blade damper painted matt black and adjustable from the face of the diffuser.

Both sets of vanes shall be fully adjustable without the use of special tools.

- 15.11 Eggcrate return or extract grille shall be provided with a steel lattice core of 12.7mm x 12.7mm openings, giving a free area of 90%.

The core shall be fixed into an extruded aluminium frame, with welded corners and a 25mm face flange.

The grille is complete with an opposed blade damper painted matt black and adjustable through the face of the diffuser.

- 15.12 Circular ceiling diffuser shall be of aluminium construction with two concentric inner spinnings.

The diffuser core shall be fully adjustable for vertical or horizontal air discharge, and shall be removable without the use of special tools.

A flap damper shall be provided in the neck of the diffuser which is adjustable from the diffuser face.

- 15.13 Vision proof door transfer grilles shall consist of a steel core with inverted-vee type blades, and an extruded aluminium frame with matching rear flange.  
The frame shall be adjustable from 28mm to 60mm to suit the door width.

Transfer grilles for 'LIGHT TIGHT' applications such as dark rooms shall be with two vision proof cores back to back.

Transfer grilles on fire rated doors shall include a block of intumescent material as a fire stop.

- 15.14 Linear Slot diffusers shall provide an unobtrusive continuous air diffusion with a pleasing aesthetic appearance. Hairline butt joints shall ensure clean unbroken linear runs for active and dummy sections.

The diffusers shall be complete with pattern control blades, fully adjustable from face of diffuser through 180 degrees and shall be fitted with end caps at each end.

The diffuser members shall be constructed from high quality aluminium extrusions to BS 1474 while the pattern control blades shall be of black rigid PVC.

- 15.15 Wall grilles with terminal filters shall be same as the diffuser described in clause 18.16 except for the wall grilles instead of the ceiling diffusers.

- 15.16 Exhaust valves shall be manufactured from high quality sheet steel spinings protected by a stove enamelled or powder coated paint finish. Flanges shall be fitted with sealing gaskets.

The valves shall be installed with the aid of a mounting ring and air flow adjusted by rotating the central

disc. Finish of valves shall be to Architect's choice.

- 15.17 Circular swirl diffuser shall be similar to Trox model VDL or equal, with adjustable blades complete with outer discharge nozzle, suitable for vertical air discharge controlled by blade position. Suitable for mounting heights of  $\geq 3.80\text{M}$  and variations in supply temperature differentials between  $-10\text{ K}$  to  $+15\text{K}$ . Comprising of diffuser face with rotating blades adjustable manually with top or side entry plenum box.

Diffuser face shall be in galvanized sheet steel to DIN 17162, outer discharge nozzle in aluminium. The face of the diffuser, discharge nozzle and connecting section shall be phosphate treated and white stove enamel finish (RAL 9002 to Architect approval) using electro-dipcoat process, resistant to saturated environment for a minimum of 100 hours without deterioration (DIN 50017). The plenum box shall be in galvanized sheet steel to DIN 17162.

- 15.18 Jet diffusers shall be manufactured from high quality aluminium extrusions and powder coated with colour to Architect choice. Jet diffuser shall come with round nozzle and long throw. The diffusion angle shall be adjustable to  $\pm 180$  degree. The jet diffuser shall have a plenum box behind. The jet diffuser shall be selected based on the required flow, throw, noise level and pressure drop. All these data to be provided in the submission of jet diffusers for Consultant approval.

#### 16.0 EXTRACT AIR LOUVRES

- 16.1 Louvres shall be extruded Aluminium frame with Aluminium blades of not less than 2 mm thickness, and shall be firmly fixed so as not to vibrate. Unsupported blade width shall not exceed 1800 mm
- 16.2 Behind each louvre shall be an insect mesh screen 76 x 6 mm made from 2 mm diameter wire. The screen will be clamped by a 20 mm frame and will be firmly fixed to the outer edges of the louvre. The screen and frame shall be hot dip galvanized after fabrication.
- 16.3 The connection to the louvre shall be flexible and shall ensure no duct load is transmitted to the louvre.
- 16.4 Louvres shall be provided with powder coated finish to the approval of the Architect.

#### 17.0 SAND TRAP LOUVRE FOR AIR INTAKE

- 17.1 Sand trap louvres shall have a double deflection inlet passage to separate sand from incoming air by means of centrifugal forces.
- 17.2 Separation efficiency particle size 350-700 shall not be less than 90% at a face velocity of 1M/Sec and not less than 70% at a face velocity of 2M/sec.
- 17.3 Sand trap louvre shall be of aluminium construction, self cleaning and maintenance free. The base of the louvre shall have self-emptying sand holes.
- 17.4 Pressure drop at 2 M/Sec average face velocity shall not exceed 120 pascals.
- 17.5 Insect mesh shall be included.



- 17.6 Sand louvres shall be provided with powder coated finish to the approval of the Architect.

## 18.0 FILTERS

### 18.1 Pre filters

Filter Media shall be of the non woven cotton fabric type. The filter media shall have an average efficiency of 25-30% on ASHRAE Test Standard 52-76. It shall have an average arrestance of 90-92% in accordance with that test standard.

Media support Grid shall be a welded wire grid with an effective open area of not less than 96%. The welded wire grid shall be bonded to the filter media to eliminate the possibility of media oscillation and media pull away.

The media support grid shall be formed in such a manner that it affects a radial pleat design, allowing total use of filter media.

Enclosing frame shall be constructed of a rigid, heavy duty, high wet strength beverage board, with diagonal support members bonded to the air entering and air exiting side of each pleat, to ensure pleat stability. The inside periphery of the enclosing frame shall be bonded to the filter pack, thus, eliminating the possibility of air bypass.

### 18.2 Bag Filters

Bag filters shall be high performance extended area disposable type filters. Each filter shall consist of high density glass microfibre media with a chemically bonded backer, individual pockets and a corrosion resistant galvanized steel enclosing frame. The filter unit shall be completely factory assembled and shall be classified by underwriter's laboratories.

Pockets shall consists of glass micro fibers chemically bonded to a reinforced backing. They shall be sewn with a tapered stitch which forms a 'V' or wedge shaped pocket resulting in uniform velocity in the passages of the air entering and air exiting sides of the filter. The pockets shall be equipped with a minimum of 40 support points per square foot of the filter media.

All stitching points shall be completely sealed with foam seal or equivalent. The pockets shall be chemically adhered around the periphery of the galvanized steel retainers. Retainers shall have rolled edges to reduce possible cuts to media, or lacerations to installers.

Enclosing frame shall be constructed of a 'J' return channel of galvanized steel.

## 19.0 MOTORIZED SMOKE AND FIRE DAMPERS

- 19.1 The Contractor shall furnish and install, motorized smoke and fire shut off dampers as required for the proper functioning of the system.

- 19.2 All control dampers shall be opposed blade.

- 19.3 Casing shall be in galvanized sheet steel to BS 2989 Z2 G275 M.

- 19.4 Blades shall be of natural anodized extruded aluminium to BS 1475.

- 19.5 Blade tip seals silicon rubber, temperature resistant upto 300 deg. C.

- 19.6 Side seals shall be stainless steel to BS 1449 Grade 302 – 525.

- 19.7 External blade linkage shall be stainless steel to BS 970 304-S 15.
- 19.8 Blade spindles shall be in stainless steel to BS 970 304 S11 supported in plain brass bearings.
- 19.9 Motorized shut off dampers shall have a leakage rate not exceeding 200 M3 / hr / M2 at 1000 Pa. Confirming to UL 5555. The actuator shall be fixed in the factory and certified for proper operation.

#### **19.10 NOISE ATTENUATION**

- 19.11 The contractor shall supply and fix acoustic insulation and noise attenuator units where necessary to reduce the air borne noise transmission through the distribution duct system, so, that the specified noise criteria levels are satisfied.
- 19.12
- 19.13 Internal Acoustic Lining of Duct:
- 19.14
- 19.15 Furnish and install duct lining complete as specified hereinafter and to the approval of engineer.
- 19.16 Where it is possible to achieve the required attenuator by lining the interior of duct runs, this may be accomplished by applying 1" thick lining to the inside of the ducts.
- 19.17 Lining for rectangular duct shall be semi rigid fiber glasses blanket and for round duct shall be flexible fiber glasses blanket. Both with special surface treatment and for approach casings, large plenum and machinery air shaft shall be rigid coated fiber glass board.
- 19.18 Acoustic insulation shall be suited to the atmosphere in which it is to operate and shall be incombustible or self-extinguishing non-flame spread grade, vermin proof and adequately fixed to ducting or mechanical room walls. Liner density shall be 1.5 Lbs. per cu.ft. (24 kg/m3), 25 mm thick and shall comply with NFPA requirements.
- 19.19 Duct liner shall be applied with both mechanical fasteners and an adhesive. Adhesive must be both water and fire resistant. Mechanical fasteners could be either of the welded pin or impact pin types. The link of each pin shall be enough to accommodate the specified liner thickness. Washer on top of the pins shall retain the liner.
- 19.20 Fiber glasses duct lining should be treated with a biocide and must pass a test showing that it will not support the growth of fungi even at relative humidity in excess of 90%.
- 19.21 Duct liner shall be applied for supply and return air ducts with a minimum of 3 m length away from each air handling unit or fan coil unit.

#### **20.0 SUBMISSIONS**

##### **20.1 Technical Submissions**

For any minor revisions in architectural layouts, contractor shall be responsible to carry out part redesign for such area and submit necessary calculations for Engineer's approval before proceeding with the shop drawings.

- 20.2 The Contractor will prepare full shop drawings, including sections, of distribution systems and equipment.

##### **20.3 Hardware Submissions.**

The Contractor will submit catalogue information for all distribution equipment including, but not limited to:

Duct work - Low pressure. Flexible duct work - Low pressure Flexible connections.

Access panels.  
Balancing dampers.  
Motorized dampers.  
Backdraft dampers.  
Splitter dampers.  
Turning vanes.  
Fire dampers.  
Silencers.  
Vibration isolators.

Diffusers,  
grilles and  
registers.  
Louvres, Sand  
louvres.  
Filters.

- 20.4 The Contractor will submit samples for any or all of the above as requested by the Consultant after receipt of the catalogues.
- 20.5 Samples will definitely be required of these items which are exposed, such as diffusers, grilles, louvres. The samples must be of size, specification and finish to be relevant to the project.
- 20.6 Where a Country of Origin is given, this refers to the Head Office in the case of International Corporations. However, for each product not manufactured in that Country, separate approval must be obtained from the Consultant.

End of Section

**SECTION 5**  
**PLUMBING WORKS**

## SECTION 5

### PLUMBING WORKS INDEX

|          |                                                                    |
|----------|--------------------------------------------------------------------|
| Contents |                                                                    |
| 1.0      | SCOPE OF WORK..... 82                                              |
| 2.0      | SERVICES DESCRIPTION..... 82                                       |
| 3.0      | PIPEWORK INSTALLATION: GENERAL ..... 82                            |
| 4.0      | STORAGE AND PROTECTION..... 83                                     |
| 5.0      | GRADIENT AND DRAINAGE OF PIPELINES ..... 84                        |
| 6.0      | PIPEWORK SUPPORTS ..... 84                                         |
| 7.0      | PIPE EXPANSION ..... 85                                            |
| 8.0      | AIR VENTING ..... 86                                               |
| 9.0      | ANTI WATER HAMMER CUSHIONS & ARRESTORS ..... 86                    |
| 10.0     | PUDDLE FLANGES ..... 86                                            |
| 11.0     | PIPEWORK MATERIAL ..... 86                                         |
| 12.0     | PEX PIPES..... 87                                                  |
| 13.0     | PIPE INSULATION..... 88                                            |
| 14.0     | JOINTING OF PIPES ..... 88                                         |
| 15.0     | VALVES ..... 88                                                    |
| 16.0     | STRAINERS..... 91                                                  |
| 17.0     | SUMP PUMPS ..... 91                                                |
| 18.0     | ELECTRICAL HOT WATER HEATERS ..... 92                              |
| 19.0     | PACKAGED WATER PUMPSETS - CONSTANT SPEED ..... 92                  |
| 20.0     | PACKAGED WATER PUMPSET – VARIABLE SPEED ..... 94                   |
| 21.0     | G.R.P. STORAGE TANKS (GENERAL REQUIREMENTS) ..... 99               |
| 22.0     | HYDROSTATIC TESTING OF WATER PIPING SYSTEMS..... 101               |
| 23.0     | TESTING OF EQUIPMENT..... 102                                      |
| 24.0     | REJECTION OF EQUIPMENT Equipment shall be rejected if: - ..... 102 |
| 25.0     | TESTING AND ADJUSTMENT OF CONTROLS..... 102                        |
| 26.0     | GOVERNMENT AND LOCAL AUTHORITIES..... 102                          |
| 27.0     | DOMESTIC WATER PIPING STERILIZATION AFTER COMMISSIONING ..... 102  |

## **1.0 SCOPE OF WORK**

- 1.1 The work shall include design check, supply, installation, testing, commissioning, adjustment and setting to work of the water services installation.
- 1.2 By virtue of carrying out a check on the complete design, the contractor is to assume full responsibility for the correct functioning of the system and to carry any liability or guarantee as may be necessary to protect all parties in this regard.
- 1.3 The plumbing services subcontractor shall be responsible for coordination with other trades and services and shall provide all materials, labor and supervision, equipment, tools, appliance, services, etc. for carrying out the following items of works:
- Domestic hot and cold-water installations.
  - Thermal and protective insulation of pipework, plant and equipment installed within the plumbing and drainage contract.
  - Noise suppression measures for plant and equipment installed within this section of the contract.
  - Protective painting of all materials installed within this section of the contract.
  - Identification of all services.
  - Electrical power and control panels for power supply to equipment including cabling from panels to respective equipment. Power supply terminating in an isolating switch terminating within 3 meters of each equipment shall be provided by the electrical subcontractor.
  - All builder's work including breaking and making good of walls, floors, etc., equipment foundation pads, sleeves, etc.

## **2.0 SERVICES DESCRIPTION**

### **2.1 System Design**

The plumbing system shall be designed in accordance with the recommendations of the Chartered Institution of Building Services Engineers and to the requirements of the Local Authorities. The system shall be installed as detailed in the drawings.

## **3.0 PIPEWORK INSTALLATION: GENERAL**

- 3.1 Pipe materials shall be as specified elsewhere herein. The Contractor shall install unions or pairs of flanges in the pipework system at the following locations:-
- (a) At intervals not greater than 15 m.
  - (b) To enable the disconnection of all items of plant, equipment, pumps and valves for maintenance purposes.
  - (c) To enable sections of pipework to be subsequently dismantled and refaxed without disturbing other pipework, plant, equipment and building fabric.
  - (d) Where indicated on the drawings.
- 3.2 Pipework shall be run neatly, parallel to walls or grid lines on plan and shall, where necessary, set around columns and projections. All vertical pipe shall be plumb. Direction changes in pipe groups shall be set out from a common center point.
- 3.3 All screwed and flanged joints shall be easily accessible.
- 3.4 Where pipes are fitted around columns or piers, the direction change shall be made using short sets and shall be supported from the floor on the center line of the column.

- 3.5 It shall be the responsibility of the Contractor to ensure that the pipework systems are at least 150 mm from any electrical conduit, cable or wiring.
- 3.6 Pipes entering or leaving trenches or ducts, shall be square to them. All pipework shall be fixed in such a manner as to be readily accessible for tools, welding operations or removable without dismantling adjacent pipework.
- 3.7 Where pipework passes through walls, etc., the Contractor shall ensure that these pipes are not subsequently bedded-in by making good by other trades.
- 3.8 Pipework passing through brick or concrete shall be enclosed in a steel pipe sleeve of such diameter that the pipe may be removed without cutting or deforming the pipework. In particular where plastic pipes pass through a fire rated wall, fire arresting sleeves shall be used with a packing of intumescent strip between the pipe and sleeve.
- 3.9 Pipework passing through internal or external dry sheet construction walls shall pass through formed holes of such size that the pipework may be removed without cutting or deforming the pipe.
- 3.10 All holes through such walls shall be closed after the erection and testing of the pipework by means of removable split closing plates on both sides of the wall. All closing plates shall be manufactured from galvanized sheet steel of not less than 16g thickness in the case of brick or concrete walls. All closing plates shall be painted or finished in accordance with the requirements of this specification.

#### 4.0 STORAGE AND PROTECTION

- 4.1 The Contractor shall ensure that all pipework and fittings delivered to the site for inclusion in the Works are stored in an approved manner to avoid deterioration due to accidental damage and atmospheric conditions.
- 4.2 The uPVC pipework components should, at all times, be stored in a location offering protection from direct sunlight, i.e. from ultra violet light and the associated heating effect, it being remembered that materials close to the ground will be heated by reflection in sandy locations.

a) The storage site should be flat, level and free from sharp stones, flints etc.

b) Pipes should not be stacked to heights exceeding the following:

| <u>Pipe size inch/MM</u>  | <u>Max. Stacking Height</u> |
|---------------------------|-----------------------------|
| Up to 3 in or 90 mm       | 10 x pipe size              |
| 4 in-6 in or 110-160 mm   | 6 x pipe size               |
| 8 in or 200 mm or greater | 4 x pipe size               |

N.B: Smaller pipes should only be nested inside larger pipes which are adjacent to the ground.

- c) uPVC pipes should be protected from direct sunlight and heat of the sun by providing shaded areas.
- 4.3 Any part of the pipework installation which, in the opinion of the Engineer, is sub-standard, unduly distorted, marked by tools or in a deteriorated condition shall be removed and replaced at the expense of the Contractor.
- 4.4 Before final erection, all pipework, shall be free of scale, rust, oil or other forms of corrosion. Such defects shall be removed and the pipework made good in a manner approved by the Engineer. Any stoppage or restriction occurring after the pipework

systems have been put into operation which is attributable to neglect by the Contractor shall be rectified by the Contractor at his own expense.

- 4.5 The open ends of all pipes prepared for welding, shall be protected by plastic plugs. Plugs of paper, cardboard, cotton waste, hessian or similar material will not be permitted.
- 4.6 Open ends of all pipes intended for threading shall be provided with either a protective cap or screwed socket to protect the threaded ends of pipe in addition to the plastic plug.
- 4.7 uPVC Pipes shall be protected from direct sunlight. Where pipes are exposed to sunlight, it should be painted by a Titanium Dioxide based industrial quality white paint and NOT of the cellulose type. Pipes shall be lightly abraded, cleaned, wiped with white spirit and painted with white under coat and followed by white gloss paint.

## **5.0 GRADIENT AND DRAINAGE OF PIPELINES**

- 5.1 All pipework shall be erected in such a manner that air may be vented and liquids drained from the whole or isolated sections of the system. Except where specified elsewhere herein, 32mm diameter key-operated cocks with hose unions shall be fitted at all low points of water systems to ensure complete drainage.
- 5.2 Pipelines shall be graded to fall in the direction of flow unless indicated otherwise on the drawings or elsewhere herein.
- 5.3 At low points in water pipework systems, all vessels and on the dead side of each pair of isolating valves in the distribution mains, drain points are to be provided in accessible positions. These shall be fitted with drain cocks of a type specified elsewhere herein.

## **6.0 PIPEWORK SUPPORTS**

- 6.1 All pipework shall be adequately supported. All support installations shall be in accordance with relevant British Standard Specifications and Codes of Practice, except where modified or extended by the contract documents.
- 6.2 Supports shall allow free movement for expansion or contraction of pipework and shall be located to ensure that pipework branches or fittings are not fouled by the support during expansion or contraction of the pipework service.
- 6.3 Double banking of pipework from a single support position will be permitted, provided the normal operating temperature of the fluids in the two pipes do not differ by more than 30 degrees C., but only where space restrictions prohibit individual support. Triple banking will not be permitted.
- 6.4 Where double banking is necessary, the larger of the two pipes shall be uppermost, and where pipes are the same size but manufactured from different materials, then the pipe having the material with the lowest coefficient of expansion shall be uppermost. Support intervals for double-banked pipework of different sizes shall relate to the smaller size.
- 6.5 Vertical rising pipes shall be supported at the base and the support shall withstand the total weight of the pipe and fluid contained.
- 6.6 Supports shall not be permitted which clamp the pipe so that it is in contact with building fabric or structure.
- 6.7 All supports shall be specifically designed for the outside diameter of the pipe concerned (including specified packings). Oversized brackets will be rejected.



- 6.8 Where non-ferrous pipework is to be supported using ferrous pipe clamps or rings, an approved plastic coating applied to the clamp or ring shall be used to prevent contact between the ferrous and non-ferrous surfaces. Where cast iron rollers are to be used in contact with copper pipework, these shall be copper-plated.
- 6.9 Pipework shall be supported individually by hangers consisting of malleable split rings with malleable iron sockets or steel clevis type hangers.
- 6.10 Pipework shall be supported as recommended by the manufacturer as required in the specification and in particular on both sides of all changes in direction. Pipework shall be supported such that no load is imposed upon plant machinery. Pipework shall be supported at intervals not exceeding the following: -

- 6.11 Cast Iron, Steel or heavy gauge copper pipework.

(15mm) - (1.8 m) (20mm) - (2.5 m)  
(25mm) - (2.75m) (32mm) - (2.75m) (40mm) - (3.0 m)  
(50mm) - (3.0 m) (65mm) - (3.6 m)  
(80mm) - (3.6 m)  
(100mm) - (4.5 m)

150mm and above at intervals not exceeding the recommendations of BS. 3974 Part 1: 1974.

- 6.12 Steel, light gauge copper      Plastic water pipework

|              |   |         |            |   |          |
|--------------|---|---------|------------|---|----------|
| (15mm)       | - | (1.2 m) | (15 to 50) | - | (1.05 m) |
| (20mm)       | - | (1.5 m) | (65)       | - | (1.02 m) |
| (25mm)       | - | (1.8 m) | (80)       | - | (1.35 m) |
| (32mm)       | - | (2.2 m) | (100)      | - | (1.75 m) |
| (40mm)       | - | (2.5 m) | (150)      | - | (2.00 m) |
| (50mm)       | - | (2.7 m) | (200)      | - | (2.04 m) |
| (80mm)       | - | (2.7 m) |            |   |          |
| (100 & over) | - | (3.0 m) |            |   |          |

All pipe supports shall be manufactured in accordance with BS. 3974 Part 1: 1974.

## 7.0 PIPE EXPANSION

- 7.1 Pipework shall be supported in such a manner as to permit free movement due to expansion and contraction. Pipework supports shall be arranged as near as possible to joints and changes in direction and each support shall take its share of the load. The spacing of the supports shall not exceed the intervals given elsewhere herein. Where there are two or more pipes the support spacing may be based on the centers required by the smallest bore pipework.
- 7.2 Vertical rising pipework shall be supported at the base or as indicated to carry the total weight of the riser. Branches from risers shall not be used as a means of support for the riser.
- 7.3 Provision for movement due to expansion and contraction shall be made either by loops, special expansion joints or by changes in direction of the pipework. Supports at such points shall be arranged to ensure that all movement is taken up by the loop, joint or change in direction of the pipework.

## **8.0 AIR VENTING**

- 8.1 Automatic air eliminators shall be provided at all high points in water pipework system. They shall be installed at the highest points of the sections they are intended to vent.

Insect screen shall be provided on all open vents and overflows.

- 8.2 Air vents on water systems shall be of float type, air eliminator of approved manufacture, having bottom inlet and top outlet with screwed connections. They shall have a stainless steel float valve and valve seat, incorporate a stainless steel check valve and shall be so designed as to allow the internal parts to be removed for maintenance or inspection without disturbing pipework. Air vents shall be suitable for the system pressures and temperatures. An air release pipe shall be connected to the air outlet and taken to the nearest suitable discharge point.

## **9.0 ANTI WATER HAMMER CUSHIONS & ARRESTORS**

- 9.1 In all open circuit water systems there shall be a series of water hammer cushions in each branch connection and in particular each vertical branch or pipe run.
- 9.2 These cushions shall comprise a branch in the pipe with a vertical connection not less than 300mm high above the Centre line of the branch. This connection shall be sealed at the top, thus trapping a pocket of air to act as a cushion.
- 9.3 For pressurized system end of every branch, provide automatic water hammer arrestor.
- 9.4 The arrestor shall comply with ANSI A112-26-1M Standard, and shall have a sufficient volume of gas (air) to control the amount of water hammer in the system.
- 9.5 The Arrestor shall keep the system surge pressure under 150 psi (10 bars) with a factory air charge.

## **10.0 PUDDLE FLANGES**

- 10.1 Where pipework passes through the external walls of the buildings or trenches below ground level, the Contractor shall supply and cast or build puddle flanges into the structure.
- 10.2 Puddle flanges are to be manufactured from the same material as the pipework of which they form a part.
- 10.3 Each puddle flange shall comprise a length of pipe, flanged or screwed at each according to diameter with an undrilled slip-on flange welded on the outside at a point where it will be located mid-way in the thickness of the wall. The puddle flange is to be painted externally with two coats of bituminous paint before being built into the structure.

## **11.0 PIPEWORK MATERIAL**

- 11.1 All low pressure cold and hot water piping (after pressure reducing station "PRV") except those embedded in builder's work shall be in Polypropylene pipes.
- 11.2 Polypropylene pipes shall be manufactured according to DIN 8077, PN – 20 and fittings to DIN 19692, PN -20.

Pipes and Fittings shall be joined by the socket poly – fusion method, using a fusion welding machine as per pipe manufacturer's recommendation, as suitable for their piping system.

PPR Pipes & Fittings are sensitive to Ultra violet rays and therefore must not be stored exposed to direct sunlight.

11.3 All pipes above 100mm dia shall be of uPVC pipes.

11.4 Water transfer pipes shall be welded hard copper to ASTM B88 type – K. The fittings shall be copper to ASTM B16.15.

11.5 All the concealed vertical drops of cold and hot water pipes in wall chase shall be in PEX pipes to DIN 16892 & 16893 with DVGW approval as specified in Clause 12.0 below.

## **12.0 PEX PIPES**

### **12.1 General**

Cold and hot water distribution pipework embedded within builder's work inside toilets, bathrooms, kitchen shall be composed of:

- Crosslinked polyethylene pipe according to DIN 16892 and 16893 pulled in a corrugated conduit of ordinary polyethylene.
- Fittings and valves of bronze / DZR construction.

### **12.2 Pipes**

12.2.1 The inner pipe shall be a cross link polyethylene pipe withstanding upto 95 deg. C at a max. pressure of 10 bar without deformation or damage. Short time temperature loading up to 110-degree C.

12.2.2 The pipe shall be resistant to all natural constituents of drinking water to DIN 2000, disinfectants and cleaning agents to DVWG – W291 and DIN 2000, anti-corrosion agents to DIN 1988 T4 (E) and common building materials, such as mortar, cement and plaster.

12.2.3 The pipe manufacturer shall offer minimum of 50 years guarantee for the pipes and 10 years guarantee for the complete system.

12.2.4 All pipes shall be stamped on equal intervals showing clearly the name of the manufacturers along with the pressure and temperature rating of these pipes.

12.2.5 All pipes shall be laid so that the 16mm PEX water pipe can be replaced, if necessary.

### **12.3 Fittings & Accessories**

12.3.1 Pipes shall be connected to the different fittings via a wall box of reinforced plastic fitted with bronze elbow suitable for 15mm or 20mm threaded connection. The box should be suitably designed in order to enable the replacement of existing pipes in the event of their damage.

12.3.2 All valves, box elbows, tees, bends shall be of bronze. Connection shall be of a cone grip unions type allowing: full flow capacity, minimum pressure loss, easily detachable with torque clearly defined for a fool proof installation. Contractor shall use proper tools for assembly as recommended by manufacturer, i.e. assembly pliers, Ratchet torque wrench, cutters, etc.

### **13.0 PIPE INSULATION**

- 13.1 All hot water supply pipework shall be insulated as detailed in Section - 2.

### **14.0 JOINTING OF PIPES**

- 14.1 uPVC pipe jointing sequence shall be carried out as per the written recommendations of the manufacturer and it must be stated that any omission or variation from the jointing procedure will result in a marked reduction in the efficiency of the joint. It is essential in the jointing process that only cleaner and solvent cement recommended by the manufacturer is used to make joints.
- 14.2 When making a joint in high ambient temperature conditions the following should be observed:-
- (a) Store the cement in a cool shady place.
  - (b) When possible protect the pipe which is about to be installed from direct sunlight.
- 14.3 Copper Pipe Work shall be joined by capillary or soldered connections. Joints to sanitary fittings, valves etc., shall be made with brass compression unions. Soldering shall be carried out using lead free solder. Dielectric joint with neoprene or similar insulator shall be fitted at junctions between copper and galvanized mild steel. Direct joints between copper and steel will not be permitted.
- 14.4 Ferrous pipes shall be screwed or welded. All welds shall be in accordance with the relevant Piping Design Code given elsewhere herein. Methods, other than those covered by the standards, maybe used provided that their suitability is demonstrated and that their use is the subject of agreement with the Engineer.

### **15.0 VALVES**

- 15.1 Valves shall be provided to isolate each item of plant and each main branch.
- 15.2 The Contractor shall make all reasonable attempts to ensure the manufacture of valves is consistent throughout the whole of the installation in order to minimize stock holding of spares.
- 15.3 Valves and cocks of any tank or vessel not within a secure building shall be lockable.
- 15.4 Any valves and cocks exposed to public view, shall be of the easy clean pattern and have a polished finish.
- 15.5 Valves in plant room, tank rooms etc., shall be of the hand wheel pattern.
- 15.6 All valves shall be located so as to provide easy access for operation and servicing. Control valves shall normally be installed in horizontal lines with stems vertical.
- 15.7 Each such regulating valve shall be fitted with a regulation indicator calibrated in tenths of one revolution of the hand wheel.
- 15.8 Ball, Globe and oblique valves shall be used on service pipelines where regulation is required, and shall be supplied and fitted in positions indicated on the Contract Drawings. Valves shall be of the same nominal bore as the pipeline in which they are indicated.
- 15.9 Globe and oblique valves used for circuit regulation shall have characterized plug discs.

- 15.10 Hand wheels shall be designed for operation at a temperature suitable for safe handling. They shall be of the open wheel type manufactured from malleable iron or die cast aluminium alloy. Handles shall have 'open/shut' direction indication.
- 15.11 Provide all supply lines to bidets with check valves.
- 15.12 Provide cold water supply to water heaters with check valves.
- 15.13 Provide angle valves for incoming water supply to all fittings.
- 15.14 Provide back flow preventer at connections to all laundry and kitchen equipment, water coolers and as required by BS 6700.
- 15.15 For specification of valves up to 16 bar pressure rating (Gate valves, Balancing valves, Butterfly valves) refer Section-3 Clause 8.0. Valves of pressure rating exceeding 16 bar shall be ductile iron or cast steel construction.
- 15.16 Valves shall always be installed in accessible locations to permit easy operation and maintenance.
- 15.17 Special care must be exercised in the installation of screwed valves to avoid straining their bodies, and preventing the gate or seat from closing tight. The wrench shall always be applied to the side being attached to the valve. When attaching pipe to a valve already in place, a second wrench shall be used to hold the valve while the pipe is being tightened. A pipe shall not be screwed so far into a valve as to damage the seat.
- 15.18 Pressure Reducing Valves.
- Pressure reducing valves shall be pilot controlled, hydraulically operated, diaphragm type with a low by-pass capability. The low-flow by-pass capability shall be achieved by using a balanced direct acting PRV as an integral part of the main valve. At very low flows when the main valve is almost completely closed, to prevent the possibility of cavitation the direct acting valve shall by-pass the main valve and maintain flow.
- PRVs shall be bronze construction including the trim. The pressure reducing valves shall be suitable for maximum working pressure that exist within the system and downstream pressure should be site adjustable between 1.5 and 3 bar. Refer to Schematic drawings for the minimum locations at which PRV's shall be required.
- 15.19 Relief Valves.
- On hot water storage tanks provide an ASME rated thermostatic, self-closing, temperature and pressure relief valve, located in the relief valve openings of tanks. Valve shall have a minimum thermal discharge capacity equal to the input capacity of the heater standard pressure setting of 600 kPa and standard temperature setting of 100 degrees C. Relief valve pipe to discharge to floor drain.
- 15.20 Ball Valves.
- 15.20.1 General
- Valves shall be non-blowout stem design. Quarter turn of handle shall fully open or close valve. Handle position shall indicate whether valve is open or closed. Handle stops shall be a permanent, integral part of the body.
- 15.20.2 Size 50 mm and Smaller.
- Valves shall be standard port, 2-piece construction with screwed ends. Valves shall be designed for minimum PN23.

Valves shall have bronze or brass body, chromium plated or stainless steel ball, steel handle with vinyl grip, and replaceable Teflon seats.

15.20.3 Size 65 mm and Larger.

Valves shall be standard port, BS 5159 with flanged ends.

Valves shall have steel body, chrome or nickel plated steel or stainless steel ball, replaceable Teflon seats, and steel stem and handle.

Where required for installation in the fire protection system, the ball valves shall be FM approved.

#### 15.21 Float Valves

15.21.1 Float valves shall be installed as indicated in the drawings to provide consistent level control in reserve supply water storage tanks. The valve shall meet the requirements of the Water Byelaws for air gaps and shall be constructed throughout in approved materials and shall prevent back siphoning. Inlet flow shall be co-axial with the piston movement, to ensure high discharge capacity and smooth, quiet operation. The valve shall have 'O' ring piston seals, resilient seated disk, Dezincification resistant bronze fabricated approved solder and shall be hydraulically tested to ensure buoyancy and constructed in accordance with BS1968.

15.21.2 Bronze equilibrium float valves 80 and above shall be flanged end, flat faced and drilled.

15.21.3 Bronze equilibrium float valves up to 50 shall be screwed end BS2779 parallel and shall be provided complete with back nut.

15.21.4 Floats for valve sizes 80mm and above shall be of copper.

#### 15.22 Solenoid Valves

Electrically operated solenoid valves shall be single phase 220V and shall be rated for the system pressure.

#### 15.23 Bib Taps

Bib-cocks shall be in accordance with BS 1010: 1973. They shall be provided with hose union nosepiece and hand wheel operated.

#### 15.24 Automatic Air Valves

15.22.1 Automatic air valves shall have a bronze body with bolted cover and a 9mm top outlet.

15.22.2 They shall each incorporate a suitable float mounted on a stainless steel spindle terminating in a needle valve for closure against a stainless steel seating at the outlet. For low pressure systems the valve shall be provided with a test cock and a brazed float, but for medium and high pressure systems the float shall be of stainless steel.

15.22.3 Automatic air valves for low pressure systems shall, incorporate an internal ball check valve at inlet to prevent air entry to the system. Automatic air valves for cold water services shall be to the approval of the local water authority.

15.25 Unless otherwise required by BS Standards and Codes, backflow preventers shall be of the double check valve type incorporating resilient elastic and positively tight seals designed to permit water to flow in one direction only. Backflow preventers shall be suitable for installation in horizontal or vertical position.

Construction shall be:

Body- Bronze Check valve - Brass Drain valve - Brass  
Seals/Membrane - Nitrile: NBR Rubber

Springs/bolts/Nuts - Stainless steel

- 15.26 All valves shall be pressure rated for at least 1.3 times the system working pressure with a minimum rating of 16 bar.

## **16.0 STRAINERS**

- 16.1 The Contractor shall install strainers in the pipeline before the following items of equipment: -

- a Pressure reducing valves.
- b Solenoid valves.
- c automatically controlled valves.
- d Pumps.

- 16.2 Strainers shall be suitable for withstanding the maximum operating temperature and pressure of the pipeline in which they are installed.

- 16.3 Install strainers in pipeline before all equipment as recommended by equipment manufacturers.

- 16.4 Strainers

Strainers size up to 50mm dia shall be bronze body with threaded ends and open mesh model metal screen.

The strainers size 65mm dia and larger shall be of the cast iron straight Y- type. The strainers shall be complete with cast iron body and cap steel screen rods and standard perforated stainless steel screens.

## **17.0 SUMP PUMPS**

- 17.1 Sump Pumps - Small (up to 4.5 L/S flow)

17.1.1 Sump pumps shall be vertical with fully floodable submersible motor pumps in close coupled design, single stage with level control. Motor shall be jacket cooled single phase, IP67 with integrated temperature switch, cable and shock proof plug.

17.1.2 Motor casing shall be chrome nickel steel absolutely tight and corrosion proof. Motor bearings shall be maintenance free, sealed for life.

17.1.3 Pump casing shall be polypropylene-corrosion, impact and abrasion resistant. Impeller shall be polypropylene. Pump shaft shall be chrome steel.

17.1.4 Pump shall be supplied with control panel complete with integrated fuse elements, hand-off-automatic switch, motor contactor, duty pump selector switch, run/fail lamps for individual pumps etc. Pumps shall be controlled by a unit mounted float control.

17.1.5 Pumps shall be monitored & controlled at the BMS as indicated on data point schedule.

- 17.2 Sump Pumps – Large (Above 3l/s flow)

- 17.2.1 Pump casing shall be cast iron with poly urethane lined wear resistant diffusers.
- 17.2.2 Impeller shall be wear resistant, semi shrouded, multi vane with adjustable diffusers.
- 17.2.3 Motor shall be squirrel cage induction type, class F insulation capable of a maximum of 15 starts / hour.
- 17.2.4 Motor casings shall be with integral cooling ribs for maximum heat dissipation.
- 17.2.5 Shaft sealing shall be with two independent mechanical seals assembled in tandem to provide a reliable and durable sealing performance and maximum resistance to abrasion and thermal shock.
- 17.2.6 An oil filled housing for lubricating and cooling mechanical seal units shall provide an additional leakage barrier.
- 17.2.7 Shaft shall be mounted on pregreased rolling element bearings.
- 17.2.8 Cable junction chamber shall be a sealed terminal chamber with separate gland assemblies incorporating strain relief clamps.
- 17.2.9 Pumps shall be of suitable for a maximum depth of submersion of 20M with the flanged outlet directly vertical upwards.
- 17.2.10 Pumps shall be selected to match the duties indicated on drawings. Pumps shall be controlled and monitored at the BMS as indicated on data point schedules.

## **18.0 ELECTRICAL HOT WATER HEATERS**

Hot water heaters shall be horizontal / vertical type. Heaters shall be fitted with immersion heaters to give the maximum output KW. Vessel shall be constructed for 125 PSi working pressure. Vessel shall be GLASS-LINED with magnesium anode protection. Entire Vessel and electrical controls shall be encased in a rectangular or cylindrical sheet metal enclosure with baked enamel finish. Tank shall be insulated with fibre glass or polyurethane injected insulation. Enclosure shall have hinged locking doors over electric controls. Heating elements shall be complete with prewired terminal leads, control with circuits, master pilot switch, etc. A thermostatic control of contacts shall be provided to balance the water heating input to the actual demand. (Five Year Full Guarantee to be provided).

## **19.0 PACKAGED WATER PUMPSETS - CONSTANT SPEED**

- 19.1 The pump set shall be a completely packaged factory assembled and tested unit complete with all major components, instruments, safety devices, stop-valves and pre-wired control panels.
- 19.2 Pumps shall be of the End suction or Vertical Multistage type as scheduled and to match the capacity indicated on the drawings. A Pressure vessel shall also be provided where scheduled.
- 19.3 The unit shall provide water, within a range of pressures specified or required. The pumps operation shall be controlled by pressure switch mounted on discharge line.
- 19.4 For dry run prevention of the pump, following shall be provided.
  - 19.4.1 Plastic float with micro switches to stop the pump when water level reaches preset low level in the storage tank.



19.4.2 A flow sensor in the suction line to stop the pump if 'no-flow' status is sensed.

- 19.5 The pressure vessel / receiver shall be constructed in accordance with BS. 487 Part 1, and shall be coated internally with epoxy resin. The vessel shall contain a butyl rubber diaphragm which shall be removable for inspection for replacement.
- 19.6 The control panel shall incorporate all starters, pilot lights, fuses and circuit breakers, relays and automatic change over equipment to automatically start the standby pump on failure of the selected duty pump. A selector switch to allow pump to operate as duty pumps shall be fitted. An interlocked handle shall isolate the panel when the door is opened. Cyclic changeover facility shall also be provided.

#### Centrifugal Pumps

- 19.7 The flow characteristic of each pump or combination of pumps should match the design requirements of the system to ensure stability throughout the operating range
- 19.8 The power characteristics of the pump shall provide protection against overloading, e.g. in the event of a burst main or excessive demand. Power absorbed shall not be excessive if the delivery head drops.
- 19.9 The basic design requirements of the pumps are detailed on plans or schedules.

#### Construction of Pump Casing

- 19.10 The pump casing shall be manufactured from cast iron. The internal and external material surface shall be smooth and free from irregularities and casting defects. A drain cock shall be provided at the bottom of the casing and a manual air release valve shall be fitted to the highest point on the casing.
- 19.11 Where the pump is supplied with flanged connections, mating flanges shall be supplied bolted on the suction and discharge flanges with a blank gasket between them to prevent the ingress of dirt.
- 19.12 Where the pump is supplied with screwed connections, unions shall be fitted to the suction and discharge connections. A cap shall be provided on each connection to prevent the ingress of dirt.
- 19.13 Pumps constructed with split casings shall incorporate a positive means of aligning the two sections of the casing, such as dowel pins or lugs.
- 19.14 Unit construction close coupled pumps and all horizontal spindle pumps shall be constructed so that the impeller can be withdrawn for inspection without disconnecting the service pipework.
- 19.15 Lifting rings shall be incorporated in the casing, adequately sized to permit lifting of the complete pump unit, including motor and bed plate.
- 19.16 Pump casings shall be wire brushed or sand blasted, to remove rust and scale. The inner surface shall then be greased and outer surface primed with a paint suitable for the casing material.

#### Pump Impeller

- 19.17 The pump impeller shall be manufactured from bronze or gun metal. The impeller shall be machined to a smooth and accurate profile. All waterways shall be smooth and free from obstruction. Ample guide surfaces shall be provided to ensure alignment.

#### Impeller Shaft

- 19.18 The pump shaft shall be constructed from stainless steel. The impeller shall be keyed to a taper on the shaft and retained by a Hexagonal Nut. Drive couplings and pulleys

shall be keyed to the shaft or retained by proprietary clamping devices designed to minimize stress and vibration.

- 19.19 The impeller and shaft assembly are to be dynamically balanced as a complete unit. All rotating elements shall be in static, dynamic and hydraulic balance.
- 19.20 The critical speed of the shaft assembly shall be substantially higher than the operating speed of the shaft. Changes in shaft diameter shall incorporate rounded shoulders to reduce stress concentration at these points.

#### Pump Bearings

- 19.21 The pump bearings shall be designed to absorb any axial thrust forces produced by the impeller to ensure that axial forces are not transmitted to the drive couplings.
- 19.22 Bearings shall be either bush type with oiling ring and reservoir, bush type with oil impregnated material, or ball or roller bearings with oil or grease lubrication. Oil reservoirs shall be fitted with content indicator and drain plug. Greased bearings shall have grease nipples, or screw down auto lubricators provided in accessible positions.

#### Pump Seals

- 19.23 Mechanical seals between the stationary and moving parts of the pump shall be selected to suit the pump operating temperature, pressure and rotational speed, and shall be chemically compatible with the liquid and materials in the system.
- 19.24 Where packed glands are provided, the stuffing boxes shall be of split construction to facilitate easy removal. The studs holding the stuffing boxes in position shall be fitted with self-locking nuts.
- 19.25 Liquid passing through packed glands is to be drained from the pump to an open tundish near the pump base.

#### Drive Couplings

- 19.26 An approved pattern of resilient drive couplings shall be installed between the prime mover and the pump shaft. The couplings shall be designed to absorb stresses caused by minor misalignment of the motor and pump shafts and to minimize the transfer of vibration.

The couplings shall be capable of being disconnected easily to facilitate dismantling for pump or motor overhaul.

### **20.0 PACKAGED WATER PUMPSET – VARIABLE SPEED**

#### 20.1 General

20.1.1 The pumping unit shall be a variable speed system. The pumping system shall include, as a minimum, the programmable pump logic controller, adjustable frequency drive, remote sensor / transmitters, motors and pumps.

The pump manufacturer shall be responsible for the successful operation of the system supplied by him.

20.1.2 The manufacturer of the packaged pumping system shall provide a minimum of one day's training at the job site for the owner's personnel the operation and maintenance of the packaged pumping system.

20.1.3 The variable speed pumping system shall be as manufactured by the pump manufacturer. A pump manufacturer is defined as a company who is regularly engaged in the design and manufacturer of the pumps specified, herein.

The pump manufacturer shall have been in the business of manufacturing complete pumping systems for a minimum of ten (10) years and shall catalog variable speed pumping systems.

An assembler of pumping systems, not engaged in the design and manufacturing of the specified pumps or a representative of a pump manufacturer assembling systems., shall not qualify as a manufacturer of pumping systems.

20.1.4 The system consists of variable speed pump (s) sized to meet system duty condition as indicated on the pump schedule, adjustable frequency drive(s), bypasses, a pump logic controller, and sensors / transmitters all as described herein.

20.1.5 At the time of submittal for approval, the pump manufacturer shall provide a complete system profile analysis which shall include as a minimum: the pumps performance (variable speed pump curves) and the operating characteristics in the system (system curve).

20.1.6 This system profile analysis shall include pump motor and adjustable frequency drive efficiencies, staging points, horsepower, and kilowatt / hour draw.

20.1.7 A spring loaded non slam triple duty check valve with soft seat shall be installed on the discharge of each pump. Valve shall be iron body with a pressure rating equal to ANSI 125 lb. Valve trim shall be bronze with stainless steel spring. It shall be equipped with brass readout valves for more accurate system balance.

20.1.8 Pump shall be bronze fitted centrifugal, single stage, end suction type or vertical multi- stage type. Pump shall be rated for a minimum of 175 PSIG working pressure. Casings shall have gauge ports, vent and drain ports at top and bottom of casing.

20.1.9 Headers shall be copper – type L.

20.1.10 Motors shall be TEFC, Class 'F' insulation, 2900 RPM suitable for 415V/3PH/50HZ supply.

## 20.2 Pump Logic Controller

20.2.1 The pump logic controller shall be provided and the operator interface flush mounted in the door of the packaged pumping system control cubicle. The pump logic controller assembly shall be listed by and bear the label of Underwriters' Laboratory, Inc. (U.L). All components within the pump logic controller panel shall be U.L. listed or recognized components. The pump logic controller shall meet FCC Part 15, for Class A computing devices.

The pump manufacturer shall supply and be responsible for the operation of a programmable solid stage Variable speed pump logic controller. The controller shall function to a proven program that safeguards against damaging hydraulic conditions including:

- a) Motor overload
- b) NPSHR above NPSHA
- c) Hunting

20.2.2 The adjustable frequency drives shall be controlled by a direct digital pump logic controller with discrete input signal selection. Use of a multiplexer for multiple sensor inputs is not acceptable. All individual sensor / transmitters shall be individually wired to the pump logic controller for continuous scan and comparison function. The pump logic controller shall be microcomputer based and hold its software in EPROM. On line field modified data entries, such as set point, shall be stored in RAM memory.

EPROM memory storage shall prevent accidental loss of data due to voltage surge or spike. RAM memory shall be backed up by EPROM. In the event of a complete power

outage, all factory preset values remain stored and shall be able to be recalled by the operator.

20.2.3 The pump logic controller shall be capable of receiving up to two process variable analog input signals. It will then select the analog signal that has deviated the greatest amount from its set point. This selected signal will be used as the command feedback input for a closed loop hydraulic stabilization function to minimize hunting. Each input signal may be programmed for a different set point and set points shall be able to be changed in the field by keypad entry through password security.

20.2.4 The pump logic controller shall be capable of displaying an additional analog input corresponding to flow. The analog input signal shall be the feedback and the set point shall be the command. The closed loop hydraulic stabilization function shall have the capability of performing the traditional proportional, integral and derivative modes. The proportional function shall have range and be infinitely programmable with that range.

Rates shall be in percent gain. The integral function shall also be infinitely programmable and compatible with the adjustable frequency drive (s) AFD(s) and the hydraulic system. It shall be entered as repeats / minute. Derivative shall be entered in units of seconds and be fully programmable within the range allowing infinite entries with that range.

20.2.5 The scan and compare rate that selects the command set point and process variable signal shall be continuous and automatically set for optimum operation. Each sensor shall be scanned every 20 milliseconds.

20.2.6 The pump controller shall have off line and on line diagnostic software. Off line diagnostics shall consist of CPU, EPROM, EEPROM and RAM memory test. It shall have self-calibration of all analog inputs with diagnostics, display test, program test, and all I/O tested and status of above displayed. The controller shall include a watchdog circuit to provide system protection / operation in the event of a controller failure. In the event of a controller failure, the system shall be able to be operated manually across the line operation.

On line diagnostics include off line diagnostics and operational diagnostics including but not limited to individual pump fault (s), AFD fault (s), overloads and sensor / transmitter fault(s).

20.2.7 Further fault information may be accessed by interrogating the pump logic controller through the front mounted keypad. In case of failure, the controller shall fail in a predetermined safe operating mode. Reset after failure shall not result in a no flow condition unless all equipment fails.

20.2.8 The pump logic controller and sensor / transmitters shall be self-powered. The pump logic controller shall be powered by 220 / 1Ph / 50 Hz. The incoming power and I/O circuitry shall reject electromagnetic (EMI) and radio frequency (RFI) interference. All digital outputs shall be electrically isolated.

20.2.9 All external transmitters furnished by the pump set manufacturer shall be powered by the pump logic controller through a quality integral 24 volt power supply. Overvoltage and short circuit protection shall be on board.

20.2.10 The pump logic controller shall operate the pump (s) in a predetermined manner as indicated in the sequence of operation section. Failure of any pump, motor or drive in the system shall cause the controller to not only display the fault conditions, but alternate or transfer the operation to another operating unit as indicated in the sequence of operation section. The pump logic controller shall be capable of utilizing individual differential pressure switches to determine pump failure.

20.2.11 Pump, AFD and overload trip faults shall not clear automatically. They must be manually reset after correction of the fault. All above faults shall be displayed

before starting. First faults shall always be displayed. Faults shall not trigger erroneous fault messages.

20.2.12 The pump logic controller shall be self-prompting. It shall display all messages in plain English. The messages shall include, but not be limited to, how to obtain default values, when to turn it on, and if it is in the active remote control mode.

20.2.13 The operator interface shall have the following:

- (a) Multi fault memory and recall.
- (b) On screen help functions.
- (c) LEDs (pilot lights) and switches.
- (d) Light and switch shall also be displayed in plain English messages scrolled continuously.

20.2.14 The operator interface shall be capable of remotely being started / stopped.

20.2.15 The operator interface and process controller units shall be modular. They shall be able to be interchanged with locally available parts without further programming required.

20.2.16 The readout shall be bright LCD super twist backlit capable of displaying any set point value, I input or output signal (s), and alarm function in engineering units. There shall be forty 6mm characters in a two line display.

20.2.17 the pump logic controller shall be capable of operating in automatic, manual or off line diagnostic modes.

20.2.18 Three levels of password and software security shall be provided for protection of field programs and use accessible features.

### 20.3 Variable Frequency Drive (VFD)

20.3.1 Drive shall be of a microprocessor based PWM design, employing a diode bridge rectifier to convert 3 phase AC to a fixed DC voltage. No input transformer may be used for either power supply or DC bus power within the drive. A constant displacement power factor of .95 shall be at all speeds and rated loads.

20.3.2 The DC bus shall consist of a choke and a capacitor bank to eliminate line disturbances. A switching power supply operating off the DC bus will provide excellent under voltage and ride through capability.

Control and inverter logic shall be isolated from the power section.

20.3.4 Bipolar power transistors shall be used in the inverter section converting a fixed DC voltage to a 3 phase adjustable frequency output with a minimum 97% efficiency at rated load.

20.3.5 For maintenance purpose, the drive shall be capable of starting, stopping and running with stable operation with the motor completely disconnected (no load).

20.3.6 Drive must protect itself against all phase to phase and phase to ground faults, removal of load, single phasing, power outages, and prevent starting with wrong phase rotation.

20.3.7 Protection is defined as normal shut down with no component damage.

20.3.8 The drive shall employ adjustable torque limit control which will override the speed command and decrease the frequency while maintaining the correct volt/hertz ratio whenever the load level surpasses the drive design level.

Torque limit shall be settable from 50 to 115% of full drive rating. When torque limit engages, the output frequency is steadily reduced until the load reduces to the design capacity. At that point the speed will stabilize. If the load reduces further the drive will reaccelerate to the preset speed.

Electronic motor overload provides electronic equivalent to thermal motor overload and shall be adjustable from 60 to 100% of the drive rating.

The drive shall be capable of being started by a 2 wire control circuit, a 240 Volt AC signal or application of input power. Adjustments shall include:

|                                   |                           |             |
|-----------------------------------|---------------------------|-------------|
| Current Limit                     | Min / Max Speed           | Motor Boost |
| Accel Time                        | Decal Time                | V/Hz Ratio  |
| Speed Reference signals shall be: |                           |             |
| Potentiometer input               | 10,000 ohms Voltage input | 0 to 10 VDC |
| Current input                     | 0 to 20 or 4 to 20 mAmp.  |             |

Analog output shall be a non isolated 0 to 5 VDC, selectable for frequency or amperage.

The drive shall provide as standard 3 relays to indicate drive run, drive fault, and current limit condition.

Full RS 232 communications shall be provided as optional equipment.

The variable frequency drive including all factory installed option shall have ETL or UL agency approvals.

#### VFD specifications

For detailed VFD specifications, refer electrical specification volume. Input voltage: 220V / 1PH/50Hz

#### 415V/3PH/50HZ Input Range

|                       |                              |                                       |
|-----------------------|------------------------------|---------------------------------------|
| :                     | +/- 10%                      |                                       |
| Speed Range           | :                            | 20:1 Maximum or as specified. Ambient |
| Temperature           | :                            | 0 – 40 C                              |
| Humidity              | :                            | 0 – 95 % Non-Condensing Altitude      |
| :                     | to 3300 feet above sea level | Efficiency                            |
| greater at rated load | Power Factor                 | :                                     |
| load.                 |                              | 0.95 regardless of speed or           |

Adjustments : Min. / Max. Speed, accel / decal  
independently adjustable to min. 120 sec., current limit, 12 T.

Drive Protection : Over temperature, under / over voltage, phase  
to phase short circuit, phase to ground short circuit, overcurrent, current limit, decal limit.

Status Indication : Drive enabled, power fault indication for under  
/ over voltage, over current, motor trip, accel / decal limit.

#### Sequence of Operation

With the pump set in the "Automatic" mode and the duty pump selector on "pump No. 1, the duty pump 1 shall operate through the VFD which is being controlled by a signal generated by the pump logic controller. Pump 2 will start if duty is not met and pressure cannot be maintained. Pump 2 shall generally remain as standby.

In the event of a system differential pressure failure, due to duty pump or overload fault, the pump logic controller will automatically initiate a timed sequence of operation to start the backup pump in the variable speed mode. A solid red indicator light shall signal this condition and a message shall be displayed on the operator interface of the pump logic controller indicating the failure.

In the event of a system differential pressure failure, due to an VFD fault, the pump logic controller shall automatically initiate a timed sequence of operation to start the selected pump across the line. A solid red indicator light shall signal this condition and message shall be displayed on the operator interface of the pump logic controller indicating the failure.

A subsequent overload trip or duty pump failure shall automatically start a timed sequence of operation to start the standby pump across the line.

When in the "by pass" mode and the pump selector on "Pump No.1 ", Pump No. 1 shall operate across the line at constant speed with motor overload and ground fault protection. A solid red indicator light shall signal this condition. The standby pump shall be locked out.

20.3.9 In the event of an overload fault while in the "by pass" mode, the backup pump shall be locked out.

In the event of the failure of a zone sensor / transmitter, its process variable signal shall be removed from the scan / compare program. Alternative zone sensor / transmitters, if available, shall remain in the scan / compare program for control.

20.3.10 The zone number corresponding to the failed sensor / transmitter shall be displayed on the operator interface of the pump logic controller.

20.3.11 In the event of failure to receive all zone process variable signals, the VFD shall maintain 100% speed, reset shall be automatic upon correction of the zone failure.

20.3.12 Pump, VFD, overload fault shall be continuously scrolled through the display on the operator interface of pump logic controller until the fault has been corrected and the pump logic controller has been manually reset.

## **21.0 G.R.P. STORAGE TANKS (GENERAL REQUIREMENTS)**

- 21.1 This section covers for the supply, delivery, installation, testing and commissioning and leaving in complete working order water storage tanks and cisterns, as indicated on the Contract Drawings.
- 21.2 The tanks shall be insulated glass reinforced plastic construction.
- 21.3 Each tank shall be provided with a heavy duty cover formed from standard panels and complete with manholes and vent.
- 21.4 The termination of vent pipes into cisterns and tanks shall permit easy removal of the covers.
- 21.5 Tanks shall be of the type and capacity shown in the schedules.
- 21.6 Jointing materials shall be suitable for the liquid to be stored. In the case of drinking water, jointing materials shall be of a non-toxic nature.
- 21.7 All facilities for pipe connections shall be made at the manufacturer's works. Pipe connections for up to 50mm nominal bore pipework shall be BSP screwed sockets. Pipe connections for pipework 65mm nominal bore and above shall be studded pads suitable to receive counter flanges to BS. 10.
- 21.8 Where shown tanks shall be divided into equal compartments using standard panels and components. Divisions shall be stayed to enable one compartment of the tank to be emptied whilst the other compartment is in service.

- 21.9 All tank covers shall be provided with purpose made manholes of a minimum diameter of 458mm with hinged or bolted lids. Where tanks are provided with internal ladders separate manholes shall be provided to the ladder and to the ball float valve or float switch. Tanks with no internal ladder shall be provided with one manhole positioned to allow access both to the tank interior and to the ball float valve.
- 21.10 Tank cover(s) shall also be provided with cowls to adequately ventilate the interior.
- 21.11 Tanks 2000mm deep and over shall be provided with internal GRP and external galvanized steel access ladders suitably protected against corrosion and bolted to the tank walls. In elevated positions external tank ladders shall be fitted with safety hoops.
- 21.12 Each tank shall be fitted with a ball float valve. Ball float valves shall be of the sizes indicated and suited to the pressure available.
- 21.13 Overflow pipes shall be twice the bore of the ball valve fitted 32mm size, whichever is the greater. Overflow pipes shall be run to discharge outside the building via floor gulley to the nearest manhole.
- 21.14 Vent cowls and overflows shall be fitted with a suitable gauze screen to prevent insects and vermin from entering the tank.
- 21.15 The tanks shall be sectional pattern, glass reinforced plastic manufactured by the hot press molded process, constructed with plain diaphragmed panels 1m square.
- 21.16 Tanks to have heavy duty covers with hinged ventilated cover. Jointing of panels to be with flexible sealing strip and stainless steel bolts. External support of tanks in excess of 1.84m deep shall be formed from rolled I section steel members to BS. 4, Part 1, 1972, hot dipped galvanized to BS. 729:1971. The design of the corsetry is to comply with BS. 449, Part 1: 1970.
- 21.17 All tanks shall be assembled on the supporting bases in accordance with the manufacturer's instructions, using torque wrenches to prevent damage to the sections due to over-tightening of nuts and bolts.
- 21.18 Detailed construction drawings of all tanks shall be submitted to the Engineer for approval prior to manufacture.
- 21.19 The Contractor shall provide and install a layer of bituminized felt for insertion between the tank and its bearers.
- 21.20 All metallic parts in continuous contact with water shall be of stainless steel.

#### Commissioning and Testing of Cisterns and Tanks

- 21.21 All tanks operating at atmospheric pressure shall be tested under "tank full" conditions for tightness and structural soundness.
- 21.22 All cold water tanks shall be subject to a standing leak test for twenty four hours, during which time all joints shall be carefully examined. Any defects shall be rectified immediately and the test repeated.

#### **G.R.P. Tanks (Particular Specifications)**

- 21.23 The Contractor shall supply and assemble at Site sectional water tank of hot pressed molded GRP panels (Glass Reinforced Plastic) of grey color provide with lockable manhole cover air vent with overflow and all piping connections. Insulated GRP panel tank to be provided with partition. The panels shall comply with BS 7491: Part 3: 1994 and shall have convex shape to assist a complete draining of water of the following characteristics:-



- (a) Specific gravity of 1.8
- (b) Tensile strength of not less than 850 kg/cm<sup>2</sup>
- (c) Flexural strength of 1800 kg/cm<sup>2</sup> to ISO 178.
- (d) Impact strength of 80 kg/cm<sup>2</sup> to ISO 179
- (e) Thermal expansion of  $2.1 \times 10^{-5}$  deg. C to ATM D 696.
- (f) Thermal conductivity of 0.6 W/m<sup>2</sup> C to BS 7491: Part 3.
- (g) Water absorption less than 0.2% to ISO 62.
- (h) Glass content more than 30%
- (i) No light transmission
- (j) Gasket of synthetic rubber
- (k) Bolts and nuts of stainless steel 316 / A4.
- (l) Frame angles and frame plates of hot dipped galvanized steel.
- (m) External ladder of hot dipped galvanized steel.
- (n) Internal ladder of GRP or S.S. according to Manufacturer's standards.
- (o) Tie rods of stainless steel
- (p) G.R.P. panel thick not less than 8mm
- (q) Resin cover of 2mm
- (r) Glass tube level indicator for all compartments.

## **22.0 HYDROSTATIC TESTING OF WATER PIPING SYSTEMS**

- 22.1 All water piping systems shall be hydrostatically tested for at least 1.5 the working pressure for ensuring complete tightness under the test pressure and for the duration of time as specified under the respective system concerned.
- 22.2 Systems may be tested as a whole or in sections to facilitate the progress of the work.
- 22.3 No part of any piping system shall be tested to pressure less than the specified test pressure measured at the highest point of the system.
- 22.4 Care shall be taken not to subject any equipment, apparatus or device to a pressure exceeding its prescribed test pressure as obtained from its name plate data or from manufacturer's published data. Pressure tests shall be applied before connecting piping to equipment. Relief valves instruments, automatic air vents and all devices that might be damaged by the test pressure shall be removed, disconnected or blanked off.
- 22.5 No pressure shall be applied against the closed gate of gate valves. All valves shall be in the open position but not completely back seated during testing. End valves shall be capped.
- 22.6 In testing flanged piping, temporary blank flanges shall be installed and firmly anchored to accommodate all developed end thrust.
- 22.7 All piping that can be damaged by end thrust developing from hydrostatic testing shall be properly anchored during testing especially at changes of direction.
- 22.8 The piping system to be tested shall be closed by plugging and blanking all openings in the system in an approved manner and filled slowly with water making sure to vent all entrapped air. Plugs shall be released temporarily to ensure that water has reached all parts of the system.

Pressure shall be applied to the system by means of a hand pump drawing from a water container. The pump discharge shall be connected to the system through a globe valve, check valve and recently calibrated pressure gauge of suitable range to have the test pressure read in the middle of the range.

- 22.9 After the test pressure is reached, the pump shall be blocked off by closing the globe valve and the variations of pressure in the system monitored on the pressure gauge.

- 22.10 The test pressure shall be maintained on the system for 24 hours during which time the system shall have no noticeable drop in pressure.
- 22.11 While the system is under pressure, a careful inspection shall be made of all pipes and joints and if any leaks in joints or evidence of defective pipe or fitting is disclosed the defective work shall be corrected immediately by replacing defective parts with new joints and materials. No make shift repairs or application of any repair compound will be permitted.
- 22.12 After the correction is made the pressure test shall be repeated until a completely tight system is ensured.
- 22.13 The test pressure shall be released slowly so as not to produce shocks and sudden contractions that might damage the piping.
- 22.14 All piping shall be tested to at least 1 1/2 times the system working pressure.

### **23.0 TESTING OF EQUIPMENT**

- 23.1 After installation, all equipment shall be tested for performance and proper functioning and for demonstrating to the full satisfaction of the Engineer that the equipment meets the design requirements as shown on the Drawings and as specified.
- 23.2 Should any equipment prove defective, unsatisfactory or does not comply with the Specification, the Contractor shall correct, repair or replace such equipment or his own account as directed by the Engineer. After the correction is made, subsequent tests shall be performed until the equipment is proven to fulfil the requirements to the entire satisfaction of the Engineer.

### **24.0 REJECTION OF EQUIPMENT**

Equipment shall be rejected if: -

- (a) It has a capacity less than required.
- (b) Its electric motor is overloaded.
- (c) It causes objectionable noise or vibration.

### **25.0 TESTING AND ADJUSTMENT OF CONTROLS**

All controls and instruments shall be operated simultaneously with the equipment or system that they control and tested for proper, accurate and dependable functioning and operation.

All controls and instruments shall be calibrated and adjusted to the satisfaction of the Engineer.

### **26.0 GOVERNMENT AND LOCAL AUTHORITIES**

Perform at appropriate times during the currency of site work, all tests required by Government and Local Authorities who may from time to time have jurisdiction over the work, and contractor to obtain the necessary certificates of approval. Local regulations shall be strictly followed.

### **27.0 DOMESTIC WATER PIPING STERILIZATION AFTER COMMISSIONING**

On completion of the potable cold water supply services system, including any parts of the existing service they shall be sterilized by the application of chlorine. The minimum requirements for chlorination shall be to flush the pipework thoroughly to remove dirt. Chlorine shall then be added to the storage tanks and water drawn from all draw-off points until chlorine is detected at each outlet point. The mains are then to be allowed to stand for a period of 24 hours, after which the outlets are to be tested for chlorine. If chlorine is not found to be present, all the above shall be repeated until residual chlorine is detected. After this, the system is to be completely flushed with clean water and precautions taken to avoid subsequent contamination of the installation.

**End of Section**

## **SECTION 6**

### **SOIL, WASTE, VENT AND STORM WATER SYSTEMS**

## SECTION 6

### SOIL, WASTE, VENT AND STORM WATER SYSTEMS INDEX

#### Contents

|      |                                                                |     |
|------|----------------------------------------------------------------|-----|
| 1.0  | SCOPE OF WORK.....                                             | 106 |
| 2.0  | SYSTEM DESCRIPTION .....                                       | 106 |
| 3.0  | INSTALLATION.....                                              | 107 |
| 4.0  | CLEANOUTS ON SOIL WASTE VENT AND STORMWATER INSTALLATION ..... | 108 |
| 5.0  | MATERIAL .....                                                 | 108 |
| 6.0  | FLOOR / ROOF / BALCONY DRAINS .....                            | 108 |
| 7.0  | GENERAL REQUIREMENTS .....                                     | 109 |
| 8.0  | INSTALLATION.....                                              | 109 |
| 9.0  | MANHOLES .....                                                 | 111 |
| 10.0 | GULLY GRATES .....                                             | 111 |
| 11.0 | TESTING OF DRAINAGE AND VENT PIPE SYSTEMS. ....                | 111 |

## **1.0 SCOPE OF WORK**

- 1.1 The work shall include design check, supply, installation, testing, commissioning, adjustment and setting to work of the drainage installation.
- 1.2 By virtue of carrying out a check on the complete design, the contractor is to assume full responsibility for the correct functioning of the system and to carry any liability or guarantee as may be necessary to protect all parties in this regard.
- 1.3 The drainage services subcontractor shall be responsible for coordination with other trades and services and shall provide all materials, labor and supervision, equipment, tools, appliance, services, etc. for carrying out the following items of works:
- Soil and waste drainage systems.
  - Storm water drainage systems.
  - Noise suppression measures for plant and equipment installed within this section of the contract.
  - Protective painting of all materials installed within this section of the contract.
  - Identification of all services.
  - Electrical power and control panels for power supply to equipment including cabling from panels to respective equipment. Power supply terminating in an isolating switch terminating within 3 meters of each equipment shall be provided by the electrical subcontractor.
  - All builder's work including breaking and making good of walls, floors, etc., equipment foundation pads, sleeves, etc.

## **2.0 SYSTEM DESCRIPTION**

### **2.1 System Design**

This shall be a gravity drainage system designed in accordance with the recommendations of the Chartered Institution of Building Services Engineers and to the requirements of the Drainage Department.

### **2.2 System Description**

#### **Sewer Drainage Network**

2.2.1 The system shall be DOUBLE stack together with vent installation. All work shall be installed and tested in accordance with BS EN 12056 code of practice for sanitary works.

2.2.2 All internal drainage pipes of building shall be connected by gravity to external drainage network.

#### **2.2.3 Surface water drainage network**

Rain water from the roof shall be piped down via separate pipe work to connect as shown on the drawings.

2.2.4 Rain water from the balconies shall be piped down and connected as shown on drawings.

2.2.5 Condensate drain from Air-conditioning equipment shall be connected to nearest waste stack through a water less trap. Contractor to coordinate with A/C layouts for exact locations.

## **INTERNAL DRAINAGE**

### **3.0 INSTALLATION**

- 3.1 Vent pipes shall extend through roof and terminate 600mm above it with vent cowl. Vent pipes passage through the roof shall be made watertight by proper flashing.
- 3.2 All changes of direction shall be gradual and not abrupt, 45 degree fittings shall be used wherever possible, and 90 degree fittings shall be of the long sweep type. All unnecessary turns and off sets shall be carefully avoided, and run as directly as possible from the sanitary fixtures to the vertical stacks.
- 3.3 Concealed pipes shall be installed in such a manner as to permit easy accessibility for maintenance this applies particularly to valve locations.
- 3.4 All pipes shall be fixed in neatly arranged lines, and adequately pitched horizontal lines to allow the system to be properly vented and drained. Air pockets, traps and sags shall be carefully avoided.
- 3.5 Supports, clamps and hangers shall be made of galvanized steel with rubber internal rubber seal, fixed with drilled plugs. Cutting and pinning of fixings will not be permitted.
- 3.6 Run building drains at a minimum grade of 1% (1:100) pitch unless otherwise noted, downward in the direction of flow. Pitch branch connections to stacks from fixtures at 2% (1:50) where possible.
- 3.7 Provide all the required appurtenances to make the drainage system complete in compliance with code requirements including traps, pipe fittings, hangers, and the like.
- 3.8 Wherever possible, vent stack offsets shall be made with 45 degree fittings.
- 3.9 Take special care in setting roof drains to ensure that they are set at an elevation which will prelude formation of puddles.
- 3.10 Install connections to roof drains in conjunction with the roofing specified under civil works, so that the building is adequately protected during construction from damage by storm water.
- 3.11 No short radius bends to be used. Use short "Tee-Wye" fittings in vertical piping only.
- 3.12 Any piping passing through roofs shall be so arranged to be a minimum of 300mm (1 ft.) from walls or other obstructions so as to permit proper flashings which are provided by another trade.
- 3.13 Where drainage pipework crosses fire rated partitions, walls and floors, provide proprietary fire rated intumescent sleeves with a fire rating equal to or greater than the fire rating of the respective wall or floor.
- 3.14 horizontally long running soil & waste shall be vented from the top end, weather shown or not on layouts.
- 3.15 Velocity breaker shall be provided for the vertical riser stacks as shown in the drawings and necessary accessories shall be fitted for the same

#### **4.0 CLEANOUTS ON SOIL WASTE VENT AND STORMWATER INSTALLATION**

- 4.1 Cleanouts shall be installed at each change of direction of drainage pipes, greater than 45 degrees, inside the building, and where indicated on the drawings. Cleanouts shall be not more than 10m apart in horizontal lines. A cleanout shall be provided at or near the foot of each vertical waste or soil stack.
- 4.2 Cleanouts on concealed piping shall be extended through and terminate flush with finished wall or floor. Pits or chases may be left in the wall or floor, provided they are of sufficient size to permit removal of the cleanout plug and proper cleaning of the system.
- 4.3 Where it is necessary to conceal a cleanout plug, a heavy duty covering plate shall be provided, which will permit ready access to the plug.
- 4.4 Cleanout plugs shall be of heavy duty with seal and lock. Final finish shall be to the approval of the Architect.
- 4.5 Cleanouts shall be of the same nominal size as the pipes up to 100mm pipe diameter and not less than 100mm for larger piping.
- 4.6 Cleanouts shall be so installed that there is a clearance of not less than 45cm for the purpose of rodding and cleaning.
- 4.7 Provide cleanouts at foot of all stacks, changes of directions, at the ends of branch runs, in straight runs as required and where indicated. Terminate as specified under "Cleanouts".

#### **5.0 MATERIAL**

- 5.1 Unless otherwise indicated drainage, vent and rain water pipes and fittings inside the building shall be of uPVC to BS 4514 for pipe sizes 80mm dia and above and MuPVC/ABS to BS 1455-1 for pipe sizes 50mm dia and smaller. Fittings shall be push fit type.
- 5.2 Acoustic lagging shall be provided for horizontal drainage pipes, as detailed in drawings.
- 5.3 Drainage pipes buried under ground up to manhole shall be uPVC to BS 4660.
- 5.4 Sewage pipe between manholes shall be uPVC to BS 4660 protected with concrete encasement.
- 5.5 All pipes under building and those subject to traffic specially under roads shall be protected with reinforced concrete.

#### **6.0 FLOOR / ROOF / BALCONY DRAINS**

- 6.1 Floor drains unless otherwise indicated shall have uPVC traps of a minimum water seal of 7cm, and shall be provided with adjustable and removable strainers. The open area of strainer shall be at least two thirds of the cross-sectional area of the drain line to which it connects. Floor drains shall have heavy duty stainless steel cover with a removable strainer and cover plate. Floor drain shall have built in rodding eye.

All floor drains / cleanouts must be coordinated with floor tiling layout.

- 6.2 Roof drains shall be all epoxy coated, cast iron body drain with membrane, flashing clamp, seepage flange, and convertible 45° and 90° outlets, gravel guard to suit insulation and ballast or pavers thickness, cast metal, oven cured epoxy coated dome.



- 6.3 Balcony drain outlet shall be of stainless steel with non-trapped sump.

## **EXTERNAL DRAINAGE**

### **7.0 GENERAL REQUIREMENTS**

- 7.1 Pipe connections to manholes, collection tanks and percolating pits shall be made in a completely watertight and approved manner.
- 7.2 Pipes shall be kept clean until final acceptance of the work. Exposed ends of all incomplete lines shall be closed with wooden plugs and adequately secured at all times when pipe laying is not actually in progress.
- 7.3 Pipes shall be installed on a good foundation and adequate means taken to prevent settlement. Pipes laid in trenches shall be provided with a solid uniform bearing throughout their entire length.
- 7.4 Pipes shall not be buried at less than 60 cm below finished grade for protection against mechanical damage. Pipes shall not be run closer than 1m to building bearing walls and footings for protection against building settlement.
- 7.5 All pipes shall be laid to a uniform slope. Slopes of sewer drain shall be limited to 3% maximum. The free vertical drop of a sewer pipe into a manhole shall be limited to 45cm between the invert level of the pipe opening and the bottom of the manhole. Where conditions necessitate that the drop would exceed 45cm at the maximum slope of 3% a drop manhole shall be used.
- 7.6 Trenches shall be kept free of water by pumping, use of well points, underdrains or other approved means during pipe laying operations so that all pipe joints are made in the dry.
- 7.7 Precautions shall be taken to protect incomplete work from floating due to storms or from any other cause. All pipe lines or structures not stable against uplift during construction shall be well braced or otherwise protected.
- 7.8 All completed underground lines shall be subject to the inspection and approval of the Engineer. All pipes shall be true to line and grade. The full circle of the pipe shall be visible at the manholes.
- 7.9 Pool backwash line water and cleaning water shall be discharged separately through collection tank as per the drawings

### **8.0 INSTALLATION**

- 8.1 The drains shall be laid truly straight in line and to an even gradient.
- 8.2 Excavations shall be made true and even to falls. The bottoms being trimmed to correct level and well rammed. Remove mud, rock projections, boulders and hard spots and replace with approved fill material well consolidated.
- 8.3 Minimum width of trench shall be 300mm greater than external diameter of pipe.
- 8.4 Any trenches excavated in error to a greater depth than required shall be in-filled back to the required level with concrete mix, at the Contractor's expense.
- 8.5 Before laying, all pipes and components shall be checked for defects and joint spaces cleared of dirt.
- 8.6 Socketed pipes to be laid with sockets uphill.
- 8.7 Flexible joints shall be made in strict accordance with manufacturer's instructions.

- 8.8 Where lubrication of the joint is required the pipe manufacturer's recommended lubricant shall only be used.
- 8.9 Spigot and socket cement joints where required shall be made with ring of tarred gasket well caulked in socket and remaining socket space filled with 1:3 cement mortar, finished in 45 deg. fillet.
- 8.10 Lay and compact bed of granular material to provide 100mm thickness over the full width of the trench. Scoop out locally at pipe sockets where socketed pipes are used. Adjust pipes to line and level and ensure that pipe barrels rest uniformly on the bedding.
- 8.11 Add granular side fill uniformly up each side of pipes compacting by hand.
- 8.12 Any trench sheeting should be lifted before the fill is compacted.
- 8.13 The granular material shall be compacted in 100mm layers by hand up to 100mm minimum distance above top of pipes.
- 8.14 Granular material shall be locally available crushed rock graded in relation to pipe size as follows:-
- |                             |                             |
|-----------------------------|-----------------------------|
| 10mm nominal size aggregate | - 100mm dia. pipes          |
| 10 or 14 mm. ditto.         | - 150mm dia. pipes          |
| 10, 14 or 20mm. ditto.      | - 200mm dia. pipes & above. |
- 8.15 During bad weather, or in wet fine-grained soils such as clays, silts or sands, it is important to prevent the trench bottom being churned up by mean working in the trench. In such cases a blanket of granular material 75mm thick laid over the trench bottom immediately after excavation, or alternatively a sealing layer of weak concrete 50mm thick, is required.
- 8.16 Provide concrete bed Grade A (CP 301 Table 1 Specified Works Cube Strength at 21 N/mm<sup>2</sup> at 28 days) to drain pipes and concrete infill (Concrete Mix 1:15) where drain track is below level of bottom of foundation base and within 900mm horizontal distance from foundation base. The concrete infill to be carried up to the level of the bottom of the foundation base.
- 8.17 Drain tracks beyond 900mm of adjacent foundations and where the bottom of the trench is lower than a depth beneath the foundation equal to the horizontal distance between the nearside of the trench and the foundation less 150mm, shall be in filled with concrete up to that depth.
- 8.18 Concrete bed and surround shall incorporate movement joints at each drain pipe joint.
- 8.19 Each joint shall form a plane surface in the concrete above and below the drain and vertical to the drain and Centre line, and shall separate the lengths of concrete with 13mm thick resilient fibre board or expanded polystyrene pre-cut to pipe diameter and concrete cross-section height and width and left in situ.
- 8.20 Drains below roadways, car parks, and any area subject to vehicular traffic where less than 900mm of cover shall be bedded and surrounded with concrete Grade 'A' (CP 301 Table Specified Works Cube Strength at 21 N/mm<sup>2</sup> at 28days) 150mm minimum thickness all round with provision for movement joints.
- 8.21 Granular fill shall be laid in 100mm layers and had compacted to a level 300mm minimum above top of pipe followed by main backfill material placed and compacted in 300mm layers any trench sheeting being withdrawn as the work proceeds. Heavy

mechanical compactors shall not be used until there is at least 300mm cover over the pipes.

- 8.22 Construction vehicles shall not be allowed to cross drain trenches until the final surface is placed except where timber sleepers or steel plates are positioned to bridge the trench.

## **9.0 MANHOLES**

- 9.1 All manholes shall be constructed to the requirements of the Drainage Department.
- 9.2 Internal manhole covers shall be bolt down airtight double seal, recessed top.
- 9.3 Covers are to be inset concrete filled, with surface finish to match surrounding floor. All covers are to be heavy duty, locking and sealed, with a clear opening of 600mm x 600mm.
- 9.4 Covers for external manholes subject to vehicular traffic shall be heavy duty ductile iron single seal, non-rock heavy duty with square push fit seal plate. Covers to be complete with lifting key-holes, and generally constructed in accordance with BS. 497:76 and shall have a clear openings of 600x600mm unless otherwise indicated. Double seal covers shall be provided where indicated on drawings.
- 9.5 All covers shall be of ductile iron and shall be grey - epoxy coated.

## **10.0 GULLY GRATES**

- 10.1 Where indicated on drawings provide kerb gully unit of ductile iron hinged grate, road retaining bar with clear opening size 532x405mm. Unit to be black bitumen coated and shall be designed to deter ingress and help prevent blockage. A GRP grid shall be provided as standard.
- 10.2 Storm water gratings shall be kitemarked, heavy duty ductile iron, non-rock clear openings size 600x600mm, grey - epoxy coated and shall comply with BS 497.

## **11.0 TESTING OF DRAINAGE AND VENT PIPE SYSTEMS.**

- 11.1 Before the sanitary fixtures are installed, drainage and vent pipe systems shall be subjected to a water pressure test to ensure and prove their tightness and to a flow test to ensure their freedom from obstructions.
- 11.2 The water pressure test shall be applied to the system in its entirety or in sections. All openings in the piping shall be tightly closed with special cast iron pipe plugs or other suitable means and the system filled with water to the point of overflow from the highest opening. The plugs shall be temporarily opened to make sure that all air has been vented and that water has reached all parts of the system.
- 11.3 No section shall be tested to less than a 3-metre head of water. In testing successive sections at least the upper 3 meters of the next preceding section shall be tested so that no joint or pipe, except the uppermost 3 meters of the whole system, shall have been subjected to a test of less than a 3-metre head of water.
- 11.4 The water shall be kept in the system or in the portion under test for at least 4 hours before inspection starts. While the system is under pressure, a careful inspection shall be made of all pipes and joints and if any leaks in joints or evidence of defective pipe or fitting is disclosed the defective work shall be corrected immediately by replacing defective parts with new joints and materials. No makeshift repairs or application of any repair compound will be permitted.

- 11.5 After the correction is made the pressure test shall be repeated until the system is proved tight.
- 11.6 Underground drainage pipes shall be tested by plugging the end of the pipe and filling with water to a minimum head of 3 meters. The test pressure shall be maintained for 24 hours. Pipes and joints shall be inspected and approved before backfilling the trench.
- 11.7 All drainage systems shall be tested for proper flow to ensure their freedom from any obstruction. The Contractor shall disassemble, clear, repair and re-assemble obstructed piping at his own expense. After re-assembly the piping shall again be subjected to the pressure test.

**End of Section**

**SECTION 7**  
**FIRE FIGHTING SYSTEM**

## SECTION 7

### FIRE FIGHTING SYSTEM INDEX

|     |                                                    |     |
|-----|----------------------------------------------------|-----|
| 1.0 | PIPING FOR FIRE SERVICES .....                     | 115 |
| 2.0 | FIRE FIGHTING ACCESSORIES .....                    | 115 |
| 3.0 | IDENTIFICATION SIGNS .....                         | 117 |
| 4.0 | AUTOMATIC SPRINKLER PROTECTION SYSTEM .....        | 117 |
| 5.0 | COMBINED FIRE PUMPSET .....                        | 120 |
| 6.0 | TESTING .....                                      | 126 |
| 7.0 | OPERATING INSTRUCTIONS .....                       | 126 |
| 8.0 | FIRE STOP SYSTEM.....                              | 126 |
| 9.0 | FM 200 TOTAL FLOODING FIRE SUPPRESSION SYSTEM..... | 129 |

The contract works shall include fire protection pipework and equipment as specified herein and as shown on the drawings. The complete system shall be installed by a Specialist Contractor. All the fire protection installation shall conform to and comply with all applicable NFPA standards (13, 14, 20, 2001, etc.) and latest Sharjah Civil Defense Regulation.

## **1.0 PIPING FOR FIRE SERVICES**

The piping system should be as straight as possible. Necessary changes of direction should be made with standard bends, springs or long turn fittings; elbows should NOT be used.

The piping system should be of (ERW) welded galvanized steel piping to ASTM A53 SCH 40, screwed and socketed. The fittings should be of malleable iron or wrought iron galvanized and of steam quality. Grooved type fitting (UL/FM approved are also acceptable.

THESE MAINS SHOULD BE ELECTRICALLY EARTHED TO AN EARTH ROD INDEPENDENT OF THE MAIN ELECTRICITY EARTHING SYSTEM.

## **2.0 FIRE FIGHTING ACCESSORIES**

### **2.1 Landing Valves**

Hydrant outlets should be of a type acceptable to the public fire brigade and should comprise a valve 65mm bore constructed in gun metal, or aluminium bronze flanged fittings for attachment to the riser and fitted with a 65mm instantaneous female coupling and a blank cap secured by a suitable length of chain. The valve gate should lift clear of the waterway and the valve cover should be securely fitted to the valve body so that it does not unscrew when operated. The valve spindle should not be less than 22mm diameter and fitted with a gun metal handwheel about 150mm in diameter which should be marked with OPEN and SHUT directions; openings should be anti-clockwise. The whole fitting should be of sound construction and shall be suitable for a working pressure of 24 bar.

The valve must have an in-built integral pressure reducing device for adjusting the water outlet pressure, at lower floors to 6.9 bars.

Landing valves should be installed at a height above floor level between 750mm and 1100mm. The hydrant outlet should be installed to give the following clearances: not less than 150mm on both sides and below the valve and not less than 200mm clearance above the handwheel.

### **2.2 Fire Hose**

65mm dia hose mtr length, red color fire hose with efficient density pvc/rubber lining, sealed pin holes, reinforced with synthetic nylon jacket as integral part with minimum working pressure of 14bar.

Robust alloy coupling should be bound and tested in factory. Binding should be with corrosion resistant material. Hose to comply with BS 6391 type 3 or DIN standards. Nozzle should be made to aluminium alloy to BS 1490 and fitted with instantaneous adaptor as per BS 336 and should have settings as diffuser/jet/spray/on/off.

### **2.3 Inlet Breeching Piece**

The riser shall be fitted with a four way inlet brass or bronze breaching piece for fire brigade use in addition to the fire pumping station, as required by the local Fire Authority. The inlet shall be protected by a non-return (check) valve. The outlet shall be 100mm to BS 4504.

The breeching piece shall comprise 2 inlets consisting of a 65mm instantaneous male coupling and a non return valve and shall be protected by a cap secured by a suitable length of chain. The coupling shall comply with BS 336.

A 25mm drain valve should be fitted to the riser in the inlet box. The valve should be kept strapped and padlocked closed.

#### 2.4 Inlet Box

The breeching piece and drain valve shall be housed in an inlet box with a glass fronted door.

The door should be as follows:

- i) Glazed with wired glass and conspicuously indicated by the words: "FIRE BRIGADE - WET MAIN INLET" or "WET RISER INLET" in block letters on the inner face of the glass, both in English and Arabic.
- ii) Fastened only by means of a spring lock which can also be operated from inside without the aid of a key after the glass has been broken.
- iii) Made large enough for hose to be connected to inlets even if the door cannot be opened and the only means of access is by breaking the glass.

The box should be constructed in accordance with BS 3980 with minimum dimensions 610mm x 400mm x 305mm.

The box shall be built into the structure with a fall of one in 12 from the rear of the base to the front of the base. Construction material for the cabinet shall be brushed / polished stainless steel as required by Architect / Engineer.

#### 2.5 Fire Hose Reels

Fire hose reel shall be slim type, recess mounted, swinging type to comply or exceed with BS 5274/EN 671 and shall be housed in cabinets.

The main contractor to Architectural specification shall install installation of doors and identification marks.

All the internal parts used shall be made of material according to BS 1490 LM 16/24. The hose tubing should be 25mm internal diameter rubber reinforced with double braiding and 30mtr long to comply with BS 3169 type 1 class B or EN 694 with the exception of car parking areas where hose reels, with extended coverage tubing to be used.

The hose reel nozzle to comply with EN 671-PN 16/DN 25/6.

The unit shall be complete with lock shield valve and 25mm pressure reducing valve to maintain pressure of 3 bar at outlet.

For testing and maintenance drain line with non-return valve to be used. Brief operating instructions should be clearly displayed on each hose reel.

#### 2.6 Combined Landing Valve, Fire Hose and Fire Hose Reel Cabinet

The back box / cabinet shall be constructed from 16swg Zinter powder coated finish, shall have the knockouts for pipe entry. The cabinet shall be divided into 2 section, top section shall accommodate the 25mm hose reel and lower cabinet shall house the 65mm hose, landing valve, nozzle, fire extinguishers and fire hose rack all as shown in the drawing. Dimension shall be as per drawings.

Architrave shall be 20 swg stainless steel (brushed / polished).



Door shall be 18 swg brushed or polished stainless steel as per the Architect requirement. Door shall be fixed to the cabinet by concealed hinges left or right as required to facilitate the hose reel pull direction.

Identification of the fire hose reel equipment shall be as per the Architect requirement in compliance to NFPA Standards.

Fire equipment shall be provided as specified elsewhere.

## **2.7 Portable fire extinguishers**

2.7.1 Fire extinguishers shall be 5 kg carbon dioxide type and 4.5Kg dry powder type to the approval of the local authorities.

2.7.2 Fire extinguishers shall be provided in each Mechanical, telephone and Electrical Rooms as shown on drawings and shall be furnished with wall bracket and shall be mounted on walls.

2.7.3 The pressure test on the extinguishers shall remain valid for 5 years.

2.7.4 Diesel generator room shall be equipped with 6Kg. Foam extinguisher mounted on wall brackets.

## **3.0 IDENTIFICATION SIGNS**

Drain valves, test valves, control valves, shall be fitted with approved enameled signs indicating their use.

## **4.0 AUTOMATIC SPRINKLER PROTECTION SYSTEM**

### **4.1 General**

The design of the sprinkler and wet riser and wet riser system indicated on the drawings will be fully developed and installed by the contractor strictly in accordance with the requirements of the local fire officer and the rules for automatic sprinkler installation to NFPA – 13,14, 20 & 24.

The firefighting system shall be fed from the underground storage tank located inside the building, which shall be constructed in reinforced concrete.

The system shall be complete with automatic fire pumping station and valves, water motor alarm and gong, pressurized water mains, main distribution pipes, distribution pipes, range pipes, hangers and supporters, sprinkler heads and main drain system. The alarm panel in order that the alarm bells in various locations can be actuated.

The water motor alarm and gong should consist with a simple water turbine having the shaft connected to a rotary ball clapper mounted within a domed gong. The test valve shall be incorporated on a branch pipe from the alarm valve to allow operational conditions to be simulated for test purposes. Auxiliary pressure sensing devices shall be incorporated in the feed to the alarm gong to actuate the main fire alarm panel.

Water from the test valve shall be properly drained to a pit. The alarm system shall be self-winding, adjustable recycling non thermal type and equipped with signal retarding device to prevent false alarms due to surge in the water system.

All isolating valves related to water tank, pump room and main distribution line shall be OS and Y type and shall be monitored by fire alarm system for open / close position.

#### 4.2 Main Control Valves

The system shall be provided with a set of installation control valves, main stop valve, alarm valve and test valve.

The main stop valve should be strapped and double padlocked in the open position. The alarm valve should be closed in the static position with the pressure below and above the valve, at normal level and no water flow. In the event of a pressure drop above the valves due to activation of a sprinkler head, water is admitted to the branch pipe connected to the water motor alarm and the gong.

#### 4.3 Sprinkler Heads

The glass bulb sprinkler heads shall be quick response type and shall be of the upright type and rated for 79 C (174 F) in car park area, concealed type rated for 68°C (154°F) in false ceiling area and concealed side wall extended coverage for other areas. These shall be located such that there is no interference with the discharge pattern and 75-150 mm below ceilings. For other types of sprinkler heads refer description on the sprinkler layout design drawings.

The sprinklers and lines shall not be spaced too close together, in order to prevent an operating sprinkler from wetting and thereby delaying operation of adjacent sprinklers.

In locations where sprinkler heads are liable to be operated or damaged by accidental blows, they should be protected by stout metal guards, care being taken to ensure that the normal operation of the sprinkler head in the event of fire is not thereby impeded.

The contractor shall supply adequate quantity of spare sprinkler heads with necessary tools of the correct temperature rating to the employer for use during an emergency.

Sprinkler heads shall be installed strictly as per the manufacturer instructions and in compliance with NFPA standards.

#### 4.4 Valves

4.4.1 All valves shall be rated for at least 1.3 times the system working pressure.

4.4.2 Ball valves:

50mm and smaller: Bronze, 2-piece, threaded or sweat ends, standard port, blowout proof stem, chrome plated ball, glass reinforced seats, UL approved.

4.4.3 Gate valves:

50mm and smaller: Outside screw and yoke gate valves, bronze body, bronze mounted, screwed bonnet, rising stem, solid wedge, with normally open supervisory/tamper switch with double wire leads.

65mm and larger: Outside screw and yoke gate valves, cast iron body, bronze mounted, bolted bonnet, rising stem, solid wedge, with normally open supervisory/tamper switch with double wire leads and rising stem.

4.4.4 Butterfly valves:

50mm and smaller: Bronze body butterfly valve, geared operator, visible position indicator, normally open supervisory/tamper switch with double wire leads, stainless steel disc and stem.

50mm and larger: Cast or ductile iron body butterfly valve, lug style, geared operator, visible position indicator, normally open supervisory/tamper switch with double wire leads, EPDM resilient seat, EPDM seals, nickel plated ductile iron disc. Valve assembly to be bubble tight with no downstream

flange/pipe attached. Use cap screws for removal of downstream piping while using the valve for system shutoff.

4.4.5 Supervisory/tamper switches:

For O S & Y valve or butterfly valve installations, UL/FM listed/approved, to monitor position of valve, tamper resistant cover screws, single or double SPDT switch contacts, corrosion resistant, for indoor or outdoor use, NEMA 4 & 6P enclosures.

4.4.6 Check valves:

50mm and smaller: Bronze body, threaded end, Y-pattern, regrind able bronze seat, renewable bronze disc, suitable for installation in a horizontal or vertical line with flow upward.

65mm and larger: Cast or ductile iron body, flanged or grooved ends, bronze trim, bolted cap, renewable bronze seat and disc, suitable for installation in a horizontal or vertical line with flow upward.

Provide 12mm automatic drip drain on inlet of fire dept. connection check valve.

4.4.7 Spring loaded check valves:

50 and smaller: Bronze body, threaded ends, bronze trim, stainless steel spring, stainless steel center guide pin, Teflon seat unless only bronze available.

65 and larger: Cast or ductile iron body, wafer or globe type, bronze trim, bronze or EPDM seat, stainless steel spring, stainless steel stem if stem is required.

4.4.8 Pressure-Restricting Valves: The valve shall be brass construction. Include, female NPS inlet and outlet; adjustable setting feature; and straight or 90-degree angle pattern design.

4.4.9 Drain valves:

20mm min. two or three piece bronze body ball valve; threaded ends, chrome plated bronze ball; glass filled Teflon seat; Teflon packing and threaded packing nut; blowout-proof stem; with hose thread outlet and cap.

4.4.10 Double check valves:

Double check backflow preventer with 2 independent spring loaded check valves, 2 isolation ball or gate valves with normally open tamper switch with double wire leads and 4 valve test ports. Construct of bronze or epoxy coated cast iron body with bronze and plastic internal parts, stainless steel springs, silicone rubber valve discs, bronze seats.

4.4.11 Alarm check valves:

Cast or ductile iron body, flanged or grooved ends, bronze grooved seat with O-ring seal, single hitch pin and latch design. Provide trim for bypass, drain, electric sprinkler alarm switch, pressure gages, precision retarding chamber, and drip cup assembly piped to floor or hub drain, fill line attachment with strainer.

#### 4.5 Flow Switches

A vane type water flow switch with metal enclosure with adjustable pneumatic retard and electrical characteristics to be compatible with alarm system. Flow switches shall be UL/FM approved.

#### 4.6 Pressure Switches

Pressure actuated switch with field adjustable settings, metal enclosure and electrical characteristics compatible with alarm system. Pressure switches shall be UL/FM approved.

#### 4.7 Local Alarm Bell

Weatherproof electric alarm bell with red painted metal housing, mounting base and gong; solenoid operator; weatherproofing O-ring seal and electrical characteristics compatible with alarm system. Alarm bells shall be UL/FM approved.

#### 4.8 Pressure Gauges

Cast aluminium, stainless steel or brass case of not less than 9 cm in diameter, double strength glass window, black lettering on a white background, phosphor bronze bourdon tube with bronze bushings, recalibration from the front of the dial, 99% accuracy over the middle half of the scale, 99.5% accuracy over the remainder of the scale. Include bronze 3-way globe valve with plugged outlet for Fire Inspector's test gauge. Pressure gauges shall be UL/FM approved.

#### 4.9 Fire Pump Test Meter

Fire Pump Test Meter, UL/FM approved, incorporating a calibrated venture and attached GPM meter, to be installed on the discharge side of the fire pump, to accurately measure pump performance. Test Meter shall be supplied with grooved ends for installation with grooved end couplings.

#### 4.10 Water Motor Alarm

Factory assembled water motor shall consist of body, impeller, gasket and cover. Body shall be cast from highly corrosion resistant aluminium magnesium alloy.

### 5.0 COMBINED FIRE PUMPSET

#### 5.1 Main Pump – Electric Motor Driven

The fire pump set shall be serving sprinkler system and wet riser system.

The Fire Pump set shall be centrifugal pumps according to NFPA 20. Complete pump set with controllers shall be UL listed and FM approved.

5.2 Pump - Shall be of the horizontal split case, making possible complete servicing without breaking piping or motor connections. Motor pump connection shall be of the flexible coupler type. The pump shall use a mechanical rotating type carbon seal and shall face against a Remite insert. The pump shall be equipped with oil lubricated bronze sleeve bearings for smooth and quiet operation.

5.3 Motor - Shall be drip proof, 2900 RPM and shall be especially selected for quiet operation and shall be so stamped. The horse power of the motor shall be of such a size as to insure non-overloading of the motor through the capacity range of the pump.

5.4 Base - Shall be of a size suitable for the pump, motor and shaft, and shall be constructed of cast iron or welded steel.

5.5 Installation - A concrete base 8 inches higher than the surrounding floor shall be provided. Construct base and install as shown by the detail on the drawings. Provide a minimum of four 1/2" anchor bolts from concrete to pump base.

5.6 Controls - The pump shall be controlled by a combined manual and automatic across-the-line type fire pump control panel. The control panel shall have an externally operated circuit breaker, a pilot light to indicate when power is on, switches or push buttons for manual or automatic operation, a pressure switch control for automatic operation and a remote alarm signal when pump is operating. The entire control panel shall be factory assembled in a steel cabinet.

5.7 Jockey Pump

5.7.1 Type - The jockey pump shall be of the centrifugal base mounted type, especially designed and constructed for quiet operation. Pump shall comply with NFPA – 20.

Pump - The jockey pump shall be bronze fitted, directly coupled to its motor. The pump shall use a mechanical rotating type carbon seal and shall face against a Remite insert. The pump shall be equipped with oil lubricated bronze sleeve bearings for smooth and quiet operation.

5.7.2 Motor - Shall be drip proof, 2900 RPM and shall be especially selected for quiet operation and shall be so stamped. The horse-power of the motor shall be of such a size as to insure non-overloading of the motor throughout the capacity range of the pump.

5.7.3 Base - Shall be of a size suitable for the pump, motor, and shaft, and shall be constructed of cast iron or welded steel.

5.7.4 Installation - A concrete base 3 inches higher than the surrounding floor shall be provided. Construct base and install as shown by the detail on the drawings. Provide a minimum of four 1/2" anchor bolts from concrete to pump base.

5.7.5 Controls - The jockey pump shall have a relief valve set for the proper pressure and a pressure switch for automatic operation. There shall be a magnetic across-the-line starter with "Hand-Off-Automatic" selector switch in cover provided to start and stop the pump.

5.8 Diesel Driven Fire Pump

5.8.1 Fire pump shall constitute a diesel engine and fire pump of the horizontal split case Centrifugal pump. Specification shall be as mentioned above under item 5.2, except for pump speed which shall be according to the choice of the diesel engine speed. (Approx. 2900 RPM)

5.8.2 Rating of diesel engine shall be chosen at the prevailing conditions at Sharjah, i.e. ambient temperature of 120 deg F. and a humidity ration of more than 90%.

5.9 Automatically Controlled Diesel Engine

5.9.1 The diesel engine for driving the fire pump shall be specifically designed for fire pump service. It shall develop sufficient horse power to drive the pump and shall have bare engine brake horse power not less than 20 percent greater than the maximum horsepower required to drive the pump at its rate RPM.

The engine shall be an open type self-contained power unit including the following accessories:

- (a) Stub shaft.
- (b) Fuel pumps and filters.
- (c) Air Cleaner.

- (d) Adjustable governor capable of regulating engine speed within a range of 10 percent between shut-off and maximum load conditions. The governor shall be set to maintain rated pump speed at maximum pump load.
- (e) Lube oil pump.
- (f) Lube oil filter.
- (g) Over speed shut down device arranged to shut down engine at a speed approximately 20 percent above rated engine speed and with manual reset. The overspeed shut down device shall be so supervised that the automatic engine controller will continue to show an overspeed trouble signal until the device is manually reset to normal operating position.
- (h) Electric Starter, generator or alternator and voltage regulator.
- (i) Dual storage battery units. Each unit shall have capacity to maintain cranking speed through a 6 minute cycle (15 seconds cranking and 15 seconds rest, in 12 consecutive cycles). Provide certification that the battery unit complies with this requirement. Batteries shall be furnished in a dry charge condition, with electrolyte liquid in a separate container. Electrolyte shall be added at the time the engine is put in service, and the battery is given a conditioning charge.
- (j) Dual battery chargers, automatically controlled.

Chargers shall be specifically designed for fire pump service. The rectifier shall be a semiconductor type.

The charger for a lead acid battery shall be a type which automatically reduces the charging rate to less than 500 milliamperes when the battery reaches a full charge condition.

The control equipment incorporated in an "off-on" type of charger for a lead acid battery shall start the rectifier hourly. It shall shut off automatically when the battery has been fully charged.

The battery charger at its rated voltage shall be capable of so delivering energy into a full discharged battery in such a manner that it will not damage the battery. It shall restore to the battery 100 percent of the battery's ampere hour rating within 24 hours.

An ammeter with an accuracy of five percent of the normal charging rate shall be furnished to indicate the operation of the charger.

The charger shall be designed so that it will not be damaged or blow fuses during the cranking cycle of the engine when operated by an automatic or manual controller.

A manual charge switch with indicator light shall be provided or, in lieu thereof, the charger shall automatically charge at the maximum rate whenever required by the state of charge of the battery.

When not connected through a control panel it shall be arranged to indicate loss of current output on the load side of the d-c over-current protective device.

Storage batteries shall be substantially supported, secured against displacement, and located where they shall be readily accessible for servicing.

#### (k) Engine cooling system

Cooling water shall be from the discharge of the pump, taken off prior to the pump discharge valve.

Threaded rigid pipe shall be used for this connection. The pipe shall include a manual shut-off valve, flushing type strainer, pressure regulating valve, automatic electric solenoid valve and a second manual shut-off valve. A bypass line with manual shut-off valve shall be installed around the manual shut-off valves, strainer, pressure regulator and solenoid valve.

#### (l) Engine jacket temperature regulating device.

- (m) Engine jacket water pump.
- (n) Instrument panel with water temperature gauge, oil pressure gauge, ammeter, tachometer of the totalizing type or an hour meter.
- (o) Exhaust mufflers and flexible connectors, stainless steel non-sparking.
- (p) The engine shall be arranged for automatic operation with all wiring terminating in a junction box suitable for field hook-up.
- (q) Exhaust piping from flexible connection at engine to muffler and from muffler to point of discharge to atmosphere shall be schedule 40 black steel pipe and fittings, welded.

#### 5.10 Pump and Diesel Engine

Shall be mounted on a common base plate, direct connected through a flexible coupling, and equipped with a coupling guard. Coupling shall be a spacer type to permit pump shaft and impellers to be removed without moving diesel engine or pump bottom casing.

#### 5.11 Vibration Isolation

Fire pumps and diesel engine are to have open type spring isolators with equal stiffness in horizontal and vertical planes. The pump base shall be supported on 10 mm thick ribbed neoprene Vibropad and will be colour coded to aid identification.

They will also have level adjustment screws and be rated for 75% more than the anticipated design load.

The spring shall be located between the housing pad and the concrete base. The base for each pump will be designed to give a static deflection of 25 mm at rated design load conditions, and shall be fully independent of any adjacent pad.

#### 5.12 Automatic Diesel Engine Control Panel

The automatic engine control panel shall be specifically designed for fire pump service. The panel shall be enclosed in a floor mounted duct and moisture resistant housing. The controller shall be the combined manual and automatic type including a "Manual-Off-Automatic selector switch.

- (i) A pressure switch and single phase power failure relay with local visual indication and auxiliary contact for remote indication of 240 volt AC power failure shall be provided which will, upon drop in system pressure or loss of single phase AC power, cause the controller to open or close all circuits necessary to automatically start the engine by performing the following functions:
  - (a) Turn on fuel supply.
  - (b) Open the cooling water solenoid valve.
  - (c) Crank the engine in a series of crank-rest cycles automatically alternating between the dual batteries on each cycle.
  - (d) Disconnect cranking motor upon engine start.
  - (e) Once the engine is started, it shall remain in operation until shut down manually. A signal in the Synoptic Panel shall indicate that the Diesel Engine is operating.
  - (f) If the engine should fail to start after several cranking attempts, the controller shall disconnect all starting circuits and energize an audible and visual alarm to indicate "over crank" in the synoptic panel as well as in the engine control panel.
- (ii) Alarm and Signal Devices on Controller
  - (a) A pilot lamp(s) shall be provided to indicate that the controller is in the "automatic" position. The lamp shall be accessible for replacement.
  - (b) Separate pilot lamps and a common bell shall be provided to indicate trouble caused by:

Low oil pressure in the lubrication system. The controller shall provide means for testing the position of the pressure switch contacts without causing trouble alarms.

Failure of engine to start automatically. Shutdown from overspeed.  
Battery failure. Each battery shall be provided with separate lamps which shall be lighted or extinguished on battery failure on attempt to start.

High engine jacket water temperature.

- (iii) Water Pressure Control: A pressure actuated switch having high and low calibrated adjustments and responsive to water pressure in the fire system shall be provided in the controller circuit. Suitable provision shall be made for relieving pressure to the pressure switch to test the operation of the controller and the pump.
- (iv) Enclosure: The structure or panel shall be securely mounted in an enclosure(s) which will protect the equipment against mechanical injury and falling drops of water striking the enclosure from the downward vertical. All switches required to keep the controller in the "automatic" position shall be within the locked cabinets having to break glass panels.
- (v) A wiring diagram shall be provided and permanently attached to the inside of the enclosure showing exact wiring for the controller, including a legend of identifying numbers of individual components. All wiring terminals shall be plainly marked to correspond with the wiring diagram furnished.
- (vi) Wiring Elements: Wiring elements of the controller shall be designed on a continuous-duty basis.
- (vii) Field Wiring: All wiring leading from the panel to the engine and to the batteries shall have adequate carrying capacity. Such wiring shall be protected against mechanical injury. Controller manufacturer's specifications for distance and wire size shall be followed.
- (viii) Marking: Each operating component of the controller shall be marked to plainly indicate an identifying number referenced to the wiring diagram. The markings shall be located so as to be visible after installation.
- (ix) Instructions: Complete instructions covering the operation of the controller shall be provided and conspicuously mounted on the controller.

5.13 All proposed pump make and model no. shall be UL listed / certified to the approval of Engineer. All control panels shall be UL certified.

5.14 Fire pump set control panel at 15<sup>th</sup> floor shall be equipped with dual motorized changeover breakers (electromechanically interlocked) in order to allow for dual feeder connection arrangement and to Civil Defense requirements and approval.

5.15 Fuel Storage Tank

- (i) The fuel storage tank shall be an integral part of the unit and shall be mounted within the support base of the unit. The tank shall have a capacity of 150 gallons. Provide all piping, fittings and valves required.
- (ii) The unit support base shall be designed to allow space for oil pan clearance.
- (iii) Fuel tank shall be provided with drain and clean out provision.

5.16 Tests



- (i) Prior to shipment, the pump and engine shall be thoroughly shop-tested as a unit by the pump manufacturer. A characteristic curve showing the pump performance based upon the results of the shop test shall be submitted to the Engineer prior to shipment, and shall include the performance pump and engine at the set-governor's speed. Absence of these test reports, indicating that the complete unit has been tested in the factory by the driver, will be ample grounds for the equipment rejection.
- (ii) Each control panel shall be fully assembled and factory tested by the control manufacturer prior to their shipment, and evidence of this test shall be furnished with the controller.
- (iii) Field Acceptance tests
  - (a) The field acceptance test results shall be as good as the manufacturer's certified shop test characteristic curve for the pump being tested within the accuracy limits of the test equipment.
  - (b) Overheating: The pump shall operate at peak load conditions without objectionable heating of the bearings or of the prime mover. The operating pump speed shall be the speed at which the pumping unit would be expected to operate during a fire.
  - (c) The engine shall not show signs of overload or stress and its governor shall properly regulate the speed.

(d) Controllers

The automatic controller shall be put through not less than ten automatic and ten manual operations.

A running interval of at least five minutes at full speed shall be allowed before repeating the starting cycle.

Automatic operation of the controller shall start the pump from all the provided starting features, such as pressure switches, valves, etc.

Length of Test. The pump shall be in operation not less than one hour (total time) during the foregoing tests.

5.17 Unit purchase

The pump, driver, and all controls and necessary attachments, specified herein, shall be purchased under a unit contract. The pump manufacturer shall assume unit responsibility and shall provide the services of a qualified Engineer to supervise the installation and alignment of equipment. Field tests shall be conducted in the field to satisfy Local Authorities having jurisdiction of satisfactory operation of pump, driver and controller. The pump manufacturer's engineer shall be at the job site, supervising the test. The control manufacturer shall have their representative at the job site to train operators in the use of the controls. The pump manufacturer shall see to it that a qualified engine representative is present to train operation and maintenance personnel in the use of the diesel engines.

5.18 Spare Parts and Tools

Spare parts and tools necessary for two years operation, as recommended by the manufacturer, shall be provided as a part of the contract for the engine. A list of these spares shall be included in the offer.

## **6.0 TESTING**

General: All testing shall be performed under the work of this Section. All services required for testing shall be a responsibility of the work of this Section. The contractor shall notify the Engineer of all tests, 48 hours prior to testing.

The Fire Protection Piping Systems shall be tested hydrostatically for not less than 2 hours at 1.5 times the maximum system pressure.

If the systems are tested in sections, the connection to the previously tested section shall be included.

All tests that may be required by the local Fire Authorities shall be performed under the work of this Section in the presence of their representative if so required.

All leaks shall be corrected and the system re-tested until no leaks are found, at no extra cost.

When the various systems are completed, operation tests shall be run on all equipment to demonstrate proper operating conditions. These tests shall be run under the observation of the Engineer.

Cost to repair any damages to the building construction occasioned by pipe leaks or defective materials shall be borne under the works of this Section, at no cost to the Owner. All corrective work shall commence immediately after damage occurs.

## **7.0 OPERATING INSTRUCTIONS**

Upon completion and approval of the systems, this Contractor shall provide an experienced engineer to instruct the Owner's operators in all details of operating and maintaining the system. The Contractor shall provide three (3) sets of type written operation instructions, parts lists and service manuals of all equipment, wiring diagrams, control diagrams and test reports, suitably bound.

## **8.0 FIRE STOP SYSTEM**

### **8.1 Summary**

Applications of firestop systems include:

- Penetrations for the passage of duct, cable, cable tray, conduit, piping, electrical busways and raceways through fire-rated vertical barriers (walls and partitions), horizontal barriers (floor/ceiling assemblies), and vertical service shaft walls and partitions.
- Openings and penetrations in fire-rated partition or walls containing fire doors.
- Multiple electrical boxes located within the same stud cavity.
- All materials shall conform to relevant North American or European Standards.

### **8.2 Definitions**

Firestopping: Material or combination of materials used to retain integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, water and hot gases through penetrations in fire rated wall and floor assemblies.

### 8.3 System Description

Performance Requirements: Provide firestop systems that have been manufactured and installed to maintain performance criteria stated by manufacturer without defects, damage or failure.

### 8.4 Submittals

- General: Submit listed submittals in accordance with Conditions of the Contract and Submittal Procedures Section.
- Product Data: Submit product data including manufacturer's SPEC-DATA product sheet, for specific products.
- Shop Drawings: Submit shop drawings showing layout, profiles and product components.
- Quality Assurance Submittals: Submit the following:

Certificates: Product certificates stating that materials comply with specified performance characteristics and physical requirements.

Manufacturer's Instructions: Manufacturer's installation instructions.

### 8.5 Quality Assurance

- Source Limitations: Obtain firestop through-penetration and joint systems from a single manufacturer.
- A manufacturer's direct representative (not distributor or agent) to be on-site during initial installation of firestop systems to train appropriate contractor personnel in proper selection and installation procedures. This will be done per manufacturer's written recommendations published in their literature and drawing details.
- Proposed fire stop materials and methods shall conform to applicable governing codes having local jurisdiction.
- Firestop Systems do not reestablish the structural integrity of load bearing partitions/assemblies, or support live loads and traffic. Installer shall consult the structural engineer prior to penetrating any load bearing assembly.

### 8.6 Delivery, Storage, and Handling

Deliver materials in manufacturer's original, unopened, undamaged containers with identification labels.

Store materials protected from weather and damage and at temperature conditions in compliance with manufacturer's requirements.

Before handling, read product data sheets and material safety data sheets. Do not use damaged or expired materials.

### 8.7 Firestop Systems

- Penetrations of fire resistant assemblies

"Penetrations of pipes, tubes, conduits, wires, cables, ducts and vents through fire resistant walls and shaft enclosures, and floor, ceiling, and roof elements of fire resistant assemblies shall be protected in accordance with the following:

- Non-combustible penetrations

"None insulated metal ducts and non-combustible pipes shall be protected by an approved through-penetration protection system. The annular gap around the penetrates shall be packed with mineral wool/ rockwool and the ends shall be filled and sealed with Firestop silicone sealant to an appropriate depth.

- Combustible penetrations

Metal ducts with combustible jackets or insulation shall be protected by an approved through-penetration protection system. The annular gap around the duct shall be packed with mineral wool/ rockwool and the ends shall be filled and sealed with Firestop intumescent sealant to an appropriate depth.

- Single cable or bunched cables

The annular gap around cable or bunches of cables shall be packed with rockwool/mineral wool and the ends shall be filled and sealed with intumescent sealant to an appropriate depth.

Cable trays crossings:

Permanent system Fire stop intumescent sealant shall be applied around and between the cables. The opening shall be sealed with towable fire-resistant cement-based mortar.

Temporary system where future changes are expected ready-to-use, intumescent flexible block firmly seated shall be built up inside the opening sealing it completely. Firestop intumescent sealant shall be applied around and between the cables.

Nonmetal pipe crossings less than 50 mm diameter: The annular gap around the penetrates shall be packed with mineral wool/ rockwool and the ends shall be filled and sealed with Firestop intumescent sealant.

Nonmetal pipe crossings of 50 mm diameter and more: The annular gap around the penetrates shall be packed with mineral wool/ rockwool and the ends shall be filled and sealed with Firestop intumescent sealant to an appropriate depth.

The pipes shall be provided with a pre-assembled graphite based intumescent jacket contained in a metal housing, fixed around the pipe with metal fixings.

## 8.8 Execution

### 8.8.1 Manufacturer's Instructions

Comply with manufacturer's product data including product technical bulletins, product catalog installation instructions and product carton instructions.

### 8.8.2 Examination

Verification of Conditions: Examine areas and conditions under which work is to be performed and identify conditions detrimental to proper or timely completion.

### 8.8.3 Preparation

Prepare surface to receive firestop system products in accordance with manufacturer's instructions for surface preparation.

Verify penetrations are properly sized and in suitable condition for application of materials.

Comply with manufacturer's recommendations for temperature and humidity conditions before, during and after installation of firestopping.

#### 8.8.4 Installation

Comply with manufacturer's instructions for installation of through-penetration or construction joint fill materials.

Seal all holes or voids made by penetrations to ensure an air and water resistant seal.

Consult with mechanical engineer, project manager, and damper manufacturer prior to installation of firestop systems that might hamper the performance of fire dampers as it pertains to duct work.

#### 8.8.5 Field Quality Control

Examine sealed penetration areas to ensure proper installation before concealing or enclosing areas.

Keep areas of work accessible until inspection by applicable code authorities.

#### 8.8.6 Protection

Protect materials from damage on surfaces subjected to traffic.

### 9.0 FM 200 TOTAL FLOODING FIRE SUPPRESSION SYSTEM

#### 9.1 Scope

This specification outlines the requirements for a total flooding of clean agent gas fire suppression system suitable for protection of risks involved. Medium must be approved by NFPA and EPA for manned areas and comply with the following specification.

Gas [HFC227EA] should be refilled at the factory of manufacturer and certificate will be required in original with all the shipping documents.

All the hardware and other equipment must be brought exclusively for this project and all the required document must be submitted to confirm that the material is original.

Consultant would reserve a right to visit factory to check the complete lot before dispatch from manufacturer's factory.

#### 9.2 Purpose

This specification is for procurement of a fire suppression system. It shall be used as a standard for equipment, installation, and acceptance testing.

#### 9.3 General Requirements

##### 9.3.1 Application

This specification outlines the requirements for an automatic fixed total flooding fire suppression system.

##### 9.3.2 System.

The automatic fixed fire suppression system shall be dedicated clean agent fire suppression system as manufactured by specified manufacturers. Major components of the suppression system, cylinders, valve and releasing devices shall be supplied by one manufacturer. The design of the fire suppression system shown on the drawings is for guideline and to identify risks and do not restrict the possibility of specialist fire contractors to propose improvement on it within the limits of these specification and acceptance of consultant and other approving authorities.

### 9.3.3 Applicable Publications.

The following publications / codes / standards are equal and form a part of this specification to the extent referenced.

- NFPA 2001 [& the latest] Standards on Clean Agent Fire Extinguishing Systems
- ISO 14520- UNE 23571 [as applicable]
- ANSI B1.20.1, Standard for Pipe Threads, General Purpose
- ASME B31.1, Power Piping Code
- Federal Register, Volume 59, Page 13044, Environmental Protection Agency (EPA) Significant New Alternatives Policy (SNAP) Program August 21, 2003.
- LPCB or UL & FM either of these two approval having all the component listed to be used in the offered system.

### 9.3.4 Units

This specification contains requirements described in U.S. units, with a conversion to metric units in parentheses, where applicable. The U.S. unit is to be considered the specified value, with the metric unit only an approximation.

### 9.3.5 Contractor

#### Requirements.

In order to ensure system effectiveness and owner satisfaction, the Contractor shall engineer, furnish, install, and test all specified equipment and accessories for a complete system as described in the plans.

### 9.3.6 System Drawings.

The Contractor shall specifically prepare plans that are to an indicated scale and easily reproducible. These plans shall show the quantity, location, and marking of all system components. Included shall be a description and routing of all piping. Care shall be taken to locate all agent storage containers in a distance to the protected area to ensure that a system discharge time within the permitted limits of 10 seconds can be achieved esp. If adjoining areas would be fed by a single bank, yet try not to be kept in the same protected room. However, even possible, any system covering more than 2 risk with one bank will not be accepted. System electrical schematics and diagrams shall be provided, including a description of all interlock functions. six sets of all drawings shall be submitted to the Owner and appropriate Authority Having Jurisdiction for approval prior to fabrication.

### 9.3.7 System Checkout, Owner Training, and Acceptance.

Upon completion of installation, a meeting shall be held at the site with the Contractor. Equipment supplier and Owner personnel shall be present. The Contractor shall familiarize Owner's personnel with system components, system functions, and recommended procedures. At this time, a functional test of the system shall be demonstrated. Room integrity tests must be conducted in accordance with NFPA 2001 or ISO standards. The Contractor shall provide the necessary personnel and instruments to conduct this test. The Contractor shall provide the Owner with a complete operation and maintenance manual as well as a written summary of any functional tests conducted. The following shall be included in the Contractor's quotation:

- Room integrity testing
- Duration and location of training (on site if feasible).

#### 9.3.8 Maintenance.

After the award of the project a maintenance agreement shall be submitted to the Owner for consideration valid for 1 years from final handing over date; however, this agreement may or may not be accepted.

#### 9.3.9 Installation

All work shall be completed in a high degree of workmanship and in compliance with current NFPA Standard 2001 and local codes. All openings made during installation shall be sealed at time of system installation to prevent leakage and maintain system integrity with compatible passive fire protection.

#### 9.3.10 Warranty.

System shall be warranted for parts and labour for not less than a period of one (1) year from date of installation. The Contractor shall specify the maintenance to be performed during the warranty period to maintain warranty conditions.

### 9.4 Technical Requirements

#### 9.4.1 Suppression Agent.

The used suppression agent shall meet the component requirements of this specification. Medium should not cause reaction in human metabolism, effect normal circulatory system & or effecting cardiovascular or central nervous system. The proposed medium should have following qualities:

Zero "ODP", environment friendly, low GWP, highly efficient in radiant heat blocking & smoke, having slight cooling effect, approved and recommended for protecting risks as in this project. Discharge noise should be lower than 90 db.

#### 9.4.2 Agent Storage.

The agent shall be stored in steel cylinders meeting the component requirements of this specification. "Bulk" agent storage shall not be preferred. The agent could be stored with the addition of only approved substance, including nitrogen super pressurization but not the inclusion of CO<sub>2</sub>. Due to the risk of explosion and safety the gas cylinders preferably [if space available] should be kept away from the protected risk area. Storage pressure should be 42 bar. The use of explosive devices to actuate agent discharge shall not be permitted.

#### 9.4.3 Suppression System Design.

##### 9.4.3.1 Concept.

The suppression system is dedicated for each risk and shall provide release of clean agent based on the concept of total flooding fire protection for enclosed areas. Irrespective of low pressure or medium pressure, system should be designed maximum as 5 milli bar in the room. In the event of fire followed by discharge, oxygen level should not go below normal comfortable level or cause hypoxia or thermal shock.

##### 9.4.3.2 Sequence of operation

## THE SYSTEM IS AUTO

In the normal conditions with the main power connected to the panel and the battery backup provided only the "main on" led on the panel should illuminate. Alarm signaling and gas release sequence in accordance with NFPA 72 and NFPA 2001. Power supply unit of conventional extinguishing panel to be "UL" listed or equal, subject to acceptability by consultant. In addition as a feature there should be an electronic "LED" visual display in the room [ to be easily seen by person in the room], to show preset count down [ 30 second or as required ] after second stage activation. The panel having built in manual release on front facia are not acceptable.

Any detector in alarm on either circuit of the detection zone will cause the followings:  
Fire led on the Gemini panel to illuminate

The pre alarm indication on the main analogue addressable panel to appear. The alarm bell to operate

Activation of the second detector connected to double knock zone will cause the following events'

Fire led on the panel to indicate/illuminate

The gas release flashing beacon and sounder/strobe to operate.

Actuator output on the panel to operate for gas discharge after the preset countdown time delay elapsed [adjustable between 0 to 100 seconds]

Activate the 2<sup>nd</sup> stage relay to shut down the a/c [to have better fire detection a/c could be shut down on first stage]

The damper and other optional related eqpt to operate.

Activation of the manual release unit provided in the entrance irrespective of whether the system is in auto or manual will become active. Manual controls shall not require a pull of more than 40 lb nor a movement of more than 14 inches to secure operation.

Each remote manual control for activation shall be located not more than 4 feet above the floor.

The same sequence should happen as in case of automatic detection as mentioned above.

### 9.4.3.3 Agent Quantity

#### 9.4.3.3.1 Minimum Design Concentration.

The system shall be designed to provide a uniform minimum concentration of extinguishing agent throughout the protected space(s) at the lowest expected ambient temperature. The Contractor shall survey the expected fuel load of the space(s), and recommend a design concentration, in accordance with the manufacturer's instructions.

#### 9.4.3.3.2 Minimum Design Agent Quantity.

The system shall be designed to deliver at least 95% of the minimum design agent quantity in 10 seconds.

#### 9.4.3.3.3 Actual Agent Quantity.

The actual agent quantity shall include an additional amount to ensure discharge of the minimum design agent quantity in the maximum discharge time per 8.4.3.4. In case of using heat absorbing gases the flooding factor could as per manufacturer guidelines.



#### 9.4.3.3.4 Highest Expected Actual Concentration

The actual concentration, after discharge of all agent per 4.3.2.3, at the highest expected ambient temperature, shall be determined for each protected space. In no case shall the highest expected actual concentration exceed 10% v/v (for Halocarbons).

#### 9.4.3.4 Calculations.

Computer calculations shall be used to determine pipe size, friction loss, and nozzle orifice size. The results shall indicate the discharge time for the system, the agent quantity discharged from each nozzle, and the nozzle pressure. The average nozzle pressure shall in no case more than 280 PSI. Discharge nozzles shall conform to Cl.2.2.5 of NFPA and shall be stainless steel SS 316 or equal.

#### 9.4.3.5 Discharge Time.

The maximum time to deliver no less than 95% of the agent weight needed to achieve the minimum design concentration per 4.3.2.1 of this specification shall be 10 seconds.

#### 9.4.3.6 Concentration Hold Time.

Generally the intended concentration hold time is 10 minutes.

#### 9.4.3.7 Enclosure Pressure Venting.

A calculation of the necessary enclosure vent area required to prevent impact pressure or exceeding the maximum allowable enclosure strength shall be undertaken by the Contractor. The calculation method shall be in accordance with recommendations provided by the system.

#### 9.4.4 Components.

The suppression system shall include all the following components:

- Suppression agent HFC 227 ea.

Irrespective of which system is proposed the storage pressure of cylinders must be below 2150 psi

- Agent storage cylinder(s) with valve(s)
- Storage cylinder strap(s)
- Solenoid releasing device
- Multiple container pressure-releasing components (if necessary)
- Discharge nozzle(s)
- Distribution pipe and fittings
- Drop curtains to be installed where required.

All the component & equipment used in the system must be from European or North American origin.

### 9.5 Component Requirements

#### 9.5.1 Suppression Agent.

Fire suppression agent supplied with the system shall be provided by reputed manufacturer. [Great lakes / DuPont / solvay etc.] Agent shall be filled into the agent storage containers at the system manufacturer's factory for the first fill and original certificate with the project name with complete component list including cylinders, valve, actuators, panels will be provided by manufacturer of both hardware and gas. All the documents should be legalized in manufacturer's country Local refilling as first fill is not acceptable. The complete material would be ordered fresh for the project and all the certification & documents to confirm genuineness of the material to be provided.

## 9.5.2 Agent Storage Cylinders

### 9.5.2.1 General

Agent storage shall be at ambient temperature conditions, within the range of -40°F to +130°F (-40°C to +54°C). Upon discharge of the system through double knock detection, no parts shall require replacement other than gaskets, lubricants, and the gas agent. Systems containing components that have a dated life span and must be frequently replaced shall not be preferred. The containers shall be capable of being filled in 1-kg increments of gas storage cylinders, supports etc.

### 9.5.2.2 Cylinders.

Cylinders shall be steel high-pressure resistant containers with fill capacity of minimum, 93k / L. Cylinders must have a test pressure of 100+ bar. Each cylinder must have embossed serial no.

### 9.5.2.3 Cylinder Valves.

Cylinder valves shall be machined brass forgings. No other material or valve manufacturing method is acceptable. The cylinder valves shall operate on the differential pressure principle, with operation of a pilot check by the release device venting pressure above a piston. Internal container pressure below the piston shall then open the valve. A port shall exist in the piston to allow the pressure above and below the piston to equalize when the valve is closed. The piston port shall be equipped with a ball check to ensure complete opening of the valve when the pilot check is operated. The valve shall be equipped with a safety disc, designed to rupture when an overpressure condition exists within the container. Within the specified fill density limitation, rupture of the disc shall not occur below 130°F (54°C). This rupture disc shall not be the primary means of discharging the agent into the system piping. The safety disc shall be equipped with a diffuser to safely vent the container pressure without dangerous forces acting on the container.

### 9.5.3 Releasing Devices.

Each clean agent system shall use a solenoid-operated releasing device that causes discharge of all agent storage containers in the system. Each releasing device shall be separately series supervised and activated by an output directly from the control panel and as backed by thermal pneumatic sensor. The releasing device shall be easily removable from the container valve without emptying the container. The suppression system shall also be capable of direct mechanical actuation, providing a means of discharge in the event of total electrical malfunction.

### 9.5.4 Discharge Nozzles.

Discharge nozzles shall be specifically designed and tested for use with the gas used. The nozzles shall be with female threads and available in 1/2 inch through 3 inch sizes. Each size shall come in two styles: 180° and 360° dispersion patterns. The nozzle used shall be provided with pipe threads that correspond to the nozzle size. Each nozzle shall be equipped with stamping to indicate the part number of the nozzle. Nozzle must be with female threaded inlet.

### 9.5.5 Distribution Pipe and Fittings.

Distribution pipe and fittings are to be steel, in accordance with NFPA 2001. Pipe junctions 2 inch (50 mm) and smaller shall be threaded. Pipe junctions larger than 2 inch (50 mm) may be welded. Design of the piping system shall be in accordance with ASME B31.1 and NFPA 2001. All pipe threads shall be in accordance with ANSI B1.20.1. The pipes could be Sch 80 and 3000 lbs fittings or as per manufacturer guidelines.

## 9.6 Test and Acceptance

### 9.6.1 General.

The final tests and acceptance shall be conducted in the presence of the Owner's representative and governing authorities. In case of any test failure, the Contractor is responsible for determining required corrective action.

The client / consultant may ask a third party to collect a sample of the product from installed system & to get chemical analysis of gas done by any international reputed laboratory to confirm the genuineness of the product.

### 9.6.2 Inspection.

An inspection shall be made by the SPECIALIST FIRE COMPANY TO ensure that all required dampers, door bottom seals, weather-stripping, caulking, and intumescent sealant / collars / mortar have been installed by civil contractor and that the areas protected shall, in fact, contain the extinguishing medium for the required time.

### 9.6.3 Room Integrity.

Room integrity tests should be conducted. Tests shall be conducted by the Contractor in the presence of the Owner's representative or consultant.

### 9.6.4 Piping.

As mentioned elsewhere in this document.

#### 9.6.4.1 Inspection.

The piping distribution system shall be inspected by the Contractor for the following:

- Compliance with design drawings
- Compliance with design calculations
- Pipe sizes
- Pipe lengths
- Fitting counts
- Means of pipe size reduction
- Orientation of tees
- Pipe fastening
- Nozzles in proper locations with correct aiming
- Nozzle part number

#### 9.6.4.2 Pressure Test.

The system discharge piping shall be pneumatically tested in a closed circuit for a period of 30 minutes at 2 times higher than the working pressure. At the end of 30 minutes, the pressure drop shall not exceed 20 percent of the test pressure. In general, test for mechanical tightness of the piping shall be conducted as per cl. 4- 7.2.2.12 as per NFPA 2001

#### 9.6.4.3 Flow Test.

A flow test (puff test) using nitrogen shall be performed on the piping network to verify that flow is continuous and that the piping and nozzles are unobstructed as per cl. 4-7.2.2.13 and cl. A-4- 7.2.2.13 of NFPA-2001.

#### 9.6.4.4 Training.

As a condition of final acceptance, the Contractor shall provide one operational training session per shift to the Owner's personnel. The training sessions shall include emergency procedures, abort functions, system control panel operation, trouble procedures and safety requirements. Multiple shifts shall receive individual training sessions.

#### 9.6.5 Commissioning.

Upon acceptance by the Owner, the complete system shall be conditioned and the system placed in operation.

### 10.FOAM SYSTEM

#### **General:**

Supply and install, a complete Automatic Foam Water Sprinkler / Spray System for Generator room.

Foam system installation shall comply fully with the requirement of N.F.P.A

#### **Codes & Standard:**

Following codes are identified as applicable code

NFPA 11 Standard for Low Expansion Foam System  
NFPA 13 Standard for Installation of Sprinkler System  
NFPA 16 Standard for Foam Water Sprinkler/Spray System

#### **Design Philosophy:**

The extent of foam system requirements shall be determined from the recommendation in the relevant section of the codes and standards listed above. The criteria for assessing this requirement is that active protection will be provided where it has been demonstrated that its application will prove beneficial in reducing risk of loss of containment of inventory from space, vessel or equipment, where such loss is directly attributed to a design accident event from an adjacent fire area.

#### **Design Parameter:**

Foam extinguishing system shall be employed as the primary form of protection for generator room.

The number of sprinkler heads shall be as per NFPA 13 and design density shall be sufficient to achieve a minimum density of 6.5 l/min/m<sup>2</sup> over the design area of 465m<sup>2</sup>.

1. Application rate and design area shall be as per NFPA.
2. Foam concentrate quantity shall be as per NFPA minimum requirement.
3. Time duration shall be as per NFPA requirement.
4. Foam system shall be Bladder Tank Balance Pressure Proportioning system.

### **Test:**

The system shall be subjected to a hydrostatic pressure test, to the satisfaction and in the presence of the engineer. Pressure shall not be less than 13.8 bar in excess of maximum static pressure when the maximum static pressure is in excess of 10 bar and the test shall be maintained for 2 hours.

### **Sprinkler Control Valve:**

Supply, install control valve as specified in these specifications.

The control valve shall include:

Isolating Valve      Located in such a location as to be always readily visible and accessible to authorized person. It must be secured open by a pad locked or riveted strap.

Alarm Valve          Fixed on the main supply pipe immediately above the isolating valve

Water motor alarm & gong must be installed not higher than 6 meters above the valve. Alarm must be tested at least once a week.

Combined drain and test valve

Retard chamber

Pressure gauges and outlet of alarm check valve

### **Alarm Valves:**

Alarm Valves shall be complete with mechanical alarm gong, all required trimmings, auxiliary valves and drains and with pressure switch for wiring into fire alarm system.

Alarm check valve shall be iron body, bronze trim for 12bar working pressure complete with electric pressure switches, test and alarm, pressure gauges and all other appurtenances for a complete foam sprinkler system.

### **Isolating Valve:**

All control valves shall be gate or butterfly valves with supervisor switches into valve operation.

Valves shall be of one manufacturer with manufacturer's name and pressure rating clearly marked on the outside of the valve body.

### **Isolating Valve:**

All control valves shall be gate or butterfly valves with supervisor switches into valve operation.

Valves shall be of one manufacturer with manufacturer's name and pressure rating clearly marked on the outside of the valve body.

### **Bladder Tank Specification:**

The Bladder tank is an essential part of the Bladder Tank Proportioning System. Used with a mounted ratio controller, this system is designed to inject foam concentrate into a suitable water supply and to automatically proportion foam concentrate over a wide range of flow & pressures.

Foam bladder tanks are welded pressure vessels constructed of carbon steel. A flexible nylon reinforced Buna – N rubber internal bladder holds the foam concentrate and keeps it from coming into contact with water or the inside of the tank shell. The center discharge piping, located within the bladder, insures that the foam concentrate flows to the bottom discharge. During operation, water supplied from the system fills the area outside of the bladder, gradually displacing the foam concentrate inside the bladder causing the bladder to collapse until the supply is exhausted. The bladder tank discharges foam concentrate for the remotely mounted ratio controller at approximately the same pressure as the water supply. Since the bladder tank

system is pressurized during operation, the bladder can be refilled with concentrate only when system is not in use.

**Features:**

- Compatibility shall be considered in chosen the required foam concentrate.
- Requires less floor space than horizontal tanks.
- Bottom foam displacement keeps concentrate feed piping full, which eliminates air in the line and prevents corrosion.

**Applications:**

Frequently used in Aircraft hangars, loading racks, sprinkler systems & offshore platforms.

**Technical Specifications:**

1. Bladder tanks shall be welded pressure vessels constructed of carbon steel with a working pressure of 175psi.
2. Tanks shall be supplied in the vertical configuration and shall be mounted on full skirt, securely welded to tank shell, complete with (4) mounting clips.
3. A flexible nylon reinforced Buna – N rubber internal bladder holds the foam concentrate and keeps it from coming into contact with water or the inside of the tank shell.
4. The tank shall have center discharge piping, located within the bladder, to insure that the foam concentrate flows to the bottom discharge.
5. Bottom discharge connection shall have provision to prevent the internal piping from entering the discharge piping.
6. An internally installed section of 1"dia I.D. rubber hose, between bladder and tank shell, shall extend from the water vent to the water drain connection to prevent the bladder from obstructing these openings.
7. Tank shall include all necessary drain and vent valves, concentrate fill piping and fill cup. In accordance with NFPA requirements for the valve supervision, all tank valves shall be locking type, brass or bronze ball valves with identification labels on the handles.
8. All piping for foam concentrate shall be schedule 40 brass pipe.
9. Water pressurization and foam concentrate supply connections shall be female NPT. External surfaces of tank and piping shall be coated with red high solid epoxy finish.

**Materials of Construction:**

|                  |                                                                |
|------------------|----------------------------------------------------------------|
| Tank             | Carbon steel, ASME code, 175 psi (12 bar) working pressure.    |
| External coating | Red high solid epoxy finish                                    |
| Bladder          | Nylon (0.050"/1.3 mm) reinforced Buna –N (coated both sides)   |
| Internal piping  | Slotted PVC                                                    |
| External piping  | Water side- carbon steel, Schedule 40, Screwed.                |
| Foam Side        | Brass, Schedule 40, Screwed.                                   |
| Vent/Drain       | Ball valve with locking handle, bronze body & brass or chrome. |
| Valves           | Plated Brass Ball                                              |
| Fill Funnel      | Polyethylene, 7-1/2" diameter cup x 1" male NPT spout.         |

**3% AFFF Foam Concentrate:**

Environmentally Responsible AFFF 3% is a superior quality aqueous film forming foam (AFFF), which is used at 3% concentration to extinguish fires in hydrocarbon fuels. This new formulation demonstrates commitment to superior flexibility, firefighting performance, and environmental responsibility. AFFF 3% is suitable for use with most types of proportioning and discharge equipment.

AFFF foam concentrates are designed for rapid-fire knockdown by producing a thin aqueous film, which spreads across the surface of the fuel, separating the fuel from oxygen. This is accomplished by allowing the foam solution to quickly drain from the foam bubble, which in turn, affects long-term seal ability and burn back resistance.

The aqueous film is produced by the fluorocarbon surfactant reducing the surface tension of the foam solution to a point where the solution can be supported by the surface tension of the fuel. The effectiveness of the aqueous film is directly influenced by the surface tension of the fuel. The film tends to be more effective on fuels with higher surface tension such as diesel and jet fuels, and less effective on fuels with lower surface tension such as hexane and gasoline.

#### **Features:**

- Low energy input AFFF - requires minimal agitation.
- Excellent fluidity provides rapid "knockdown".
- Suitable for use with fresh or seawater.
- Compatible with standard proportioning and foam making devices.
- Suitable for use with foam compatible dry powder extinguishing agents.

□

#### **Applications:**

AFFF is used at 3% concentration in fire suppression systems and manually to fight fires involving hydrocarbon fuels such as crude oil, gasoline, and fuel oils. It is not suitable for use on polar solvents or water miscible fuels such as alcohol's, ketones, esters, and ethers. Typical installations include foam water sprinkler systems, aircraft hangars, loading racks, process areas, etc. AFFF 3% is an excellent agent for use in aircraft rescue and fire fighting or other manual fire fighting applications where polar solvent fuels are not encountered. It is also useful as a wetting agent in combating class "A" fires. In general, AFFF foam concentrates may be used with non-aspirating nozzles and sprinklers, however, for best foam expansion and 25% drainage time all foam concentrates should be used with aspirating nozzles and foam making discharge devices.

#### **Typical Physical Properties:**

|                                     |              |
|-------------------------------------|--------------|
| Specific Gravity at 77°F(25°C)..... | 1.03         |
| pH.....                             | 8.0          |
| Viscosity at 77°F(25°C) .....       | 3.0 csk      |
| Freezing Point .....                | -12°F(-24°C) |
| Minimum Usable Temperature .....    | 20°F (-7°C)  |
| Maximum Usable Temperature .....    | 120°F (49°C) |

#### **Foam Water Sprinkler Head:**

Foam water sprinklers are open and air aspirating type. The pattern of coverage is similar to conventional sprinkler head. Foam water sprinklers are designed to operate at a minimum of 2bar pressure and maximum of 4.1bar. The standard coverage per foam water sprinkler is 9.3sq.m.

#### **Technical Data:**

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| Mounting                 | Pendent                                           |
| Maximum Working Pressure | 12.3 kg/Sq.m (175 psi)                            |
| Operating Pressure       | 2.0 bar(30 psi) minimum, 4.2 bar (60 psi) maximum |
| End Connection           | ½" NPT                                            |
| Material                 | Brass or Bronze or Stainless steel                |

#### **Pipes & Fittings:**

Where piping working pressure in the piping is below 20 bar, pipes shall be seamless factory galvanized steel to ASTM A53 grade B schedule 40.

Fitting for piping 50mm. And smaller shall be threaded factory galvanized malleable iron class 150lbs.

Fittings for piping larger than 50m, Shall be joined using factory galvanized mechanical roll grooved fittings and mechanical joints.



Galvanizing shall on the inside and outside of pipe and fittings.

### **Hangers and Supports**

Piping hangers, support shall be fabricated in accordance with NFPA 13.

The hangers, rods support and supporting devices and accessories shall be capable of supporting at least twice the live loads of the product being supported. All hangers, support and accessories shall be approved.

Concrete inserts shall be factory galvanized malleable iron poured in place type, screwed.

Concrete anchors shall be factory made cadmium plated malleable iron steel expansion shield type.

Riser clamps shall be galvanized steel bands shaped to tightly fit O.D of pipe, secured with bolts.

**End of Section**

## **SECTION 8**

### **LIQUEFIED PETROLEUM GAS INSTALLATION**

## SECTION 8

### LIQUEFIED PETROLEUM GAS INSTALLATION INDEX

|      |                                                                                  |     |
|------|----------------------------------------------------------------------------------|-----|
| 1.0  | SCOPE OF WORKS. ....                                                             | 144 |
| 2.0  | G E N E R A L ..... <b>Error! Bookmark not defined.</b>                          |     |
| 3.0  | PIPING, FITTINGS AND VALVES. ....                                                | 149 |
| 4.0  | INSTALLATION OF PIPE, TUBING, PIPE AND TUBING FITTINGS, VALVES<br>AND HOSE. .... | 150 |
| 5.0  | LPG VAPORIZERS .....                                                             | 151 |
| 6.0  | GAS LEAK DETECTION SYSTEM .....                                                  | 152 |
| 7.0  | PERSONNEL .....                                                                  | 153 |
| 8.0  | PAINTING AND IDENTIFICATION .....                                                | 153 |
| 9.0  | TESTING AND COMMISSIONING .....                                                  | 153 |
| 10.0 | LIST OF MATERIAL MANUFACTURERS..... <b>Error! Bookmark not defined.</b>          |     |

## **1.0 SCOPE OF WORKS.**

- 1.1. The work shall include design check, supply, and installation, testing, commissioning adjustment and setting to work of the LPG installation.
- 1.2 By virtue of carrying out a check on the complete design, the contractor is to assume full responsibility for the correct functioning of the system and to carry any liability or other insurance or guarantee as may be necessary to protect all parties in this regard
- 1.3 The subcontractor shall be responsible for coordination with other trades and services and shall provide all materials, labour and supervision, equipment, tools, appliances, services, etc. for carrying out the following items of work:
  - LPG Storage tanks with all supports, valves and accessories as per NFPA 58.
  - LPG distribution pipework with all valves, supports, controls, etc.
  - LPG regulation and metering devices.
  - LPG detection and alarm system.
  - Fire protection installation for LPG tanks like deluge system for above ground tanks and cathodic protection for underground tanks.
  - Protective painting of all materials installed within this section of the contract.
  - Identification of all services.
  - Electrical power and control panels for power supply to equipment including cabling from panels to respective equipment.
  - All builder's work including breaking and making good of walls, floors, etc., equipment foundation pads, sleeves, etc.
  - Hoisting of heavy equipment.
- 1.4 All installation shall be carried out by contractor approved by the Local Civil Defense Authority for this work.
- 1.5 All equipment and materials supplied shall comply with the requirements of the Local Civil Defense Authority.
- 1.6 Contractor to check for the network distributed gas network pipes at the site and coordinate with the authority. Provision for both network and tank type to be provided.
- 1.7 Installation shall comply with the requirements of:
  - The local Civil Defense Authority
  - NFPA 58
  - British Standards Code of practice LPG 1TA.

## 1. GENERAL

1. The system shall be suitable for GAS SYSTEM installation. Central gas system contains the followings:
2. Gas reducing & distribution panel for each building.
3. Gas network within the building.

### 1.1 REFERENCES

- NFPA 54 Fuel Gas Code.
- NFPA 58 Standard for the Storage and handling of Liquefied Petroleum Gases.
- ISO 4437 Buried polyethylene (PE) pipes for the supply of Gaseous fuels.
- ASME Sec IX Welding and Brazing Qualifications

### 1.2 QUALITY ASSURANCE

- 1.2.1. The requirement of the reference standards, specification, rules and regulations of Dubai Civil Defense.
- 1.2.2. The gas distribution system shall be executed only by Contractor and licensed/approved Dubai Civil defense Department.

### 1.3 SUBMITTALS

- 1.3.1. Submit product data and shop drawings in accordance with the followings:
  - a. Manufacturer's printed brochure and catalogue (one original and three copies) with relevant information high lighted (or irrelevant information struck out), along with write up of selection criteria.
  - b. General Piping layout drawings to a scale not smaller than 1:100.
  - c. Equipment installation and piping layout, kitchens, plant room, etc. clearly indicating all fittings, supports, valves etc.
  - d. Typical floor plans indicating shafts, risers locations etc. to a scale of 1:50.
  - e. Installation details
  - f. Electrical wiring diagrams (NTS) and panel installation details
  - g. Calculations/details to substantiate the shop drawings, where required.
  - h. Test and performance certificates of equipment, where required.
- 1.3.2. Product Data: Submit manufacture's product specifications, installation and startup instructions.
- 1.3.3. Shop Drawings: Submit shop drawings indicating tank locations, weight loadings, required clearances and gas system pipe work distribution complete with valves, fittings and controls.
- 1.3.4. Steel Pipes: Statement from manufacturer on his letterhead that the pipe furnished meets the international standards specified.
- 1.3.5. Wiring Diagrams: Submit manufacturer's electrical requirements for power supply wiring. Submit manufacturer's ladder-type wiring diagrams for interlock and control wiring. Clearly differentiate between portions of wiring that are factory – installed and portions to be field – installed.
- 1.3.6. Submit three (3) hard bound copies of Operation and Maintenance Manual", with complete data for each piece of equipment installed, as detailed below:

The manual shall contain:

- a. Table of contents.
- b. Material submittal with the Engineer's and local authority's approval.
- c. Manufacturer installation instructions.
- d. Manufacturer internal wiring diagrams.
- e. Manufacturer assembly details.
- f. Replacement parts, number listing and description.

- 1.4 Delivery, Storage & Handling
- 1.4.1. Handle gas system components carefully to prevent damage, breaking, denting and scoring. Do not install damaged components, replace with new.
- 1.4.2. Store LPG system components in clean dry place. Protect from weather, dirt, fumes, water, construction debris, and physical damage.

## 2. PRODUCTS

### 2.1 PIPING, FITTING & INSTALLATION

#### 2.1.1 MATERIAL

##### SERVICE LOCATION TYPE

Gas distribution main from PRS yard (1st stage regulation station) to building entries (2nd stage Reg. Station)

All Black Steel, Seamless Schedule 40, ASTM A 106 Gr

Gas distribution mains from 2nd stage Reg. Station to appliance connection

All Black Steel, Seamless Schedule 40

Gas vertical droppers from roof header and flats All Black Steel, Seamless Schedule 40

From gas meter to appliance All Copper Pipe

#### 2.1.2 JOINTS

- 1. Black steel, electric arc (A/C) welded joints.
- 2. All joints shall be rated for 350 mbar working pressure.
- 3. Pipe joints in copper pipe shall be brazed.

#### 2.1.3 INSTALLATION

- 1. Route piping in orderly manner, conserving building space and without interference with other systems.
- 2. Install exposed piping parallel or at right angles to the building walls, except where otherwise shown on the Contract drawings.
- 3. Where changes in the pipe sizes occur, install tapered reducer fittings.
- 4. Provide adequate clearance for installation of pipe fittings
- 5. During progress of work, keep ends of all piping closed with factory manufactured plugs or blind flanges, to prevent the entry of foreign matter. Ensure the removal of such plugs or flanges on completion of work.
- 6. All gas droppers or riser shall be installed as per DCD requirements. (Fire rated & dedicated shafts for gas pipes).

#### 2.1.4. PROTECTION

- 1. Exposed black steel piping.

Two (2) coats of red oxide primer.

Two (2) coats of yellow enamel paint, as per color code, if any.

- 2. All copper & steel work in connection with supports for pipes and equipment exposed to the elements shall be painted with two coats of an approved rust preventive paint and finishing coat.

3. All exposed surface of copper & steel pipe work, hangers, brackets, valves, etc., shall be painted with one (1) undercoat and one (1) finishing coat of enamel paint to approved color.

#### 2.1.5 PURGING & TESTING

1. The pipes shall be jointed, supported, plugged and shall have been in position for at least 24 hours, before the tests are carried out. Blow out the complete piping with compressed air/ Nitrogen prior to the start of the test.
2. Test all piping upstream of the 1st stage pressure reducing station isolating valves in the gas storage compound with nitrogen at a pressure of 1.25 of working pressure. Apply soap solution at the joints for checking leaks.
3. The system shall not have any leaks and should hold the pressure for a minimum period of twenty four (24) hours without any realistic drop of pressure.
4. Refer to and comply with further requirements of NFPA-54.

#### 2.1.6. UNDERGROUND PE GAS PIPES

1. Underground gas pipe network: will be HDPE.
2. Underground gas pipes shall be installed at a depth of 1.00 Meter for main, and 0.80 Meter for services.
3. Valves shall be ball valves types with plain ends of PE (100 SDR11) stable for electro fusion for diameters 32 MM up to 120 MM, and suitable to hold pressure of 20 Bars

#### 2.1.7 GAS NETWORK WITHIN THE BUILDING

Central Gas System for each building will be designed as following:

- a) Primary Pressure Reducing & Distribution Panel
- b) Secondary Pressure Reducing & Distribution Panel
- c) Gas Supply Lines
- d) Gas Branches Inside Kitchens
- e) Gas detection & Monitoring Panel

#### 2.1.8 GAS BRANCHES INSIDE KITCHEN

- a) Manual valve
- b) Solenoid valve 220 Volt AC ½"
- c) Safety regulator
- d) Gas meter
- e) Cooker valve
- f) Gas detector connected to the Solenoid valve through emergency shutoff switch.

#### 2.1.9 GAS PRESSURE REGULATING STATION

Located on the on the external wall of the building or inside gas room with louver door, ventilated to open area, this panel will receive the gas from the central network and insure that it will be diverted to the building at maximum pressure of 350 MB.

This Panel contains the following components:

- a) Main Manual Isolating Valve
- b) Solenoid valve
- c) Pressure Gauge
- d) Protective box
- e) Detector Connected to the gas panel

The 1st, 2nd and 3rd stage regulating station are to be supplied and installed in the related areas and shall at least be fitted with the following correctly sized equipment (whichever applicable) as required by local authority and NFPA 54:

- a) Emergency shut-off valve

- b) Solenoid valve
- c) Gas pressure regulating valve
- d) Pressure Gauge
- e) Leak detection equipment

The regulating station shall be sized to deliver the peak gas draw-off from the gas distribution system. The provisional maximum system demand for this installation can be determined from the equipment information provided in the design data information of this document. The final load to be confirmed once the final equipment information is available.

#### 2.1.10 GAS DETECTION AND MONITORING PANEL

1. Located in the main entrance, connected to gas sensor located in shafts or any closed area has gas line passing through, the panel has to interlocked to the main solenoid valve in the First PRDP and shall be interfaced with FAP to shut off the gas completely in case of fire or gas leak.
2. Control Panel
  - a. The central control panel shall be located in the control room located on the site and shall easily be accessible in the event of an emergency.
  - b. The central panel shall have a modular design incorporating input alarm/display module for each gas detector. Each channel shall have indication for low alarm at 15% LEL, high alarm at 30 % LEL and fault at 0-100% LEL indicator.
  - c. Main LPG supply shall be shut off at 30% LEL indication.
  - d. The concentration of the gas shall be displayed on a common indicator with selector switches to display the concentration of the desired channel or zone indicator.
  - e. Below the gas control panel is the system control panel which shall have the following indications:
    - ☐ AC healthy
    - ☐ DC healthy
    - ☐ Sprinkler on
    - ☐ Gas pressure status
    - ☐ Gas Solenoid Valve operated
    - ☐ Common fire signals
    - ☐ Gas leak high
    - ☐ Gas Percentage at Tank

The panel shall have the following switches:

- ☐ Sprinkler manual operate
- ☐ Gas manual shut off
- ☐ System reset

#### 3. Gas Detectors

The gas detectors shall use a pellistor sensor to detect flammable gases and vapors. The sensor shall be housed in a stainless steel, flame proof housing, fitted onto an explosion proof junction box.

#### 4. Cables

MICC or Fire Proof cables of at least 1.5 sq. mm area shall be used to interconnect between each detector and the control panel.

#### 5. Logic of Operation

When low-level gas leak is detected in any zone, local internal alarm and indication shall show on the control panel. When high level gas leak is detected in any zone, the following sequence shall occur:

- a. The solenoid valve in the gas distribution channel shall be closed automatically.
- b. External alarm shall be operated.



Connection to the fire alarm system:

In the event of a fire alarm signal, the following sequence shall occur.

In the event of an emergency, it shall be possible to bypass the automatic system and operate the system manually, this shall include:

- a. Manual shut off/on of the gas through the emergency shut-off push button.
- b. Manual operation of the sprinkler system On/Off
- c. Manual reset.

#### APPROVED SUBCONTRACTORS

All materials proposed for installation of central gas system shall be approved by the Consultant and suit the purpose of use.

#### 3.2 QUALITY ASSURANCE SYSTEM

The Contractor shall institute an approved quality assurance (QA) system that shall be submitted to the Engineer for approval. The records of this QA system shall be kept throughout the duration of the Contract and be submitted to the Engineer at regular intervals as required.

### 4.0 PIPING, FITTINGS AND VALVES.

#### 4.1. General

4.1.1 This section includes basic design provisions and material specifications for pipe, tubing, pipe and tubing fittings, valves used to connect container appurtenances with the balance of the LP Gas system.

#### 4.2 Pipe

4.2.1 LPG Main line shall be seamless steel to ASTM A53 Sch 80.

4.2.2 Distribution pipework shall be seamless steel to ASTM A53, Sch 40.

4.2.3 Final connections to equipment shall be soft copper tubes to BS 2871 Table X. Metal braided pigtails suitable for use with LPG.

#### 4.3 Pipe and Tubing Fittings

4.3.1 Pipe joints in steel, brass or copper pipe shall be welded or brazed.

4.3.2 Fittings used at pressure higher than container pressure, such as on the discharge of liquid transfer pumps, shall be suitable for a working pressure of at least 350 psig (2.4 MPa gauge).

4.3.3 Fittings used with liquid LP-Gas, or with vapor LP-Gas at operating pressures over 125 psig (0.9 MPa gauge) shall be suitable for a working pressure of at least 250 psig (1.7 MPa gauge).

4.3.4 Fittings for use with vapor LP-Gas at pressure not exceeding 125 psig (0.9 MPa gauge) shall be suitable for a working pressure of 125 psig (0.9 MPa gauge).

4.3.5 Soldering or brazing filler material shall have a melting point exceeding 1,000 deg. F (538 deg. C).

#### 4.4 Valves, Other than Container Valves

4.4.1 Pressure containing metal parts of valves (except appliance valves), including manual positive shut off valves, excess flow check valves (either manually or automatically operated), used in piping systems shall be of steel, ductile (nodular) iron, malleable iron or brass. Ductile iron shall meet the requirements of ANSI/ASTM A 395 or equivalent and malleable iron the requirements of ANSI/ASTM A 47 or equivalent. All materials used including valve seat discs, packing, seals and diaphragms shall be resistant to the action of LP Gas under service conditions.

4.4.2 Valves shall be suitable for the appropriate working pressure, as follows:

- a) Valves used at pressures higher than container pressure, such as on the discharge of liquid transfer pumps shall be suitable for a working pressure of at least 350 psig (2.4 MPa gauge). [400 psig (2.8 MPa gauge) WOG valves comply with this provision].
- b) Valves to be used with liquid LP Gas, or with vapor LP-Gas at pressures in excess of 125 psig (0.9 MPa gauge), but not to exceed 250 psig (1.7 MPa gauge), shall be suitable for a working pressure of at least 250 psig (1.7 MPa gauge).
- c) Valves (except appliance valves) to be used with vapor LP Gas at pressures not to exceed 125 psig (0.9 MPa gauge) shall be suitable for a working pressure of at least 125 psig (0.9 MPa gauge).

4.4.3 Manual shut off valves, excess flow check valves and backflow check valves used in piping systems shall comply with the provisions for container valves.

4.4.4 Emergency shut off valves shall be approved and incorporate all of the following means of closing.

- a) Automatic shut off through thermal (fire) actuation. When fusible elements are used they shall have a melting point not exceeding 250 deg.F (121 deg. C).
- b) Manual shut off from a remote location. Manual push button switch for solenoid valves.
- c) Manual shut off at the installed location.

4.4.5 Solenoid valves shall be intrinsically safe and explosion proof construction suitable for 24V DC power supply.

#### 5.0 INSTALLATION OF PIPE, TUBING, PIPE AND TUBING FITTINGS, VALVES AND HOSE.

5.1 When joints are welded or brazed:

- a) The pipe shall be Schedule 40 or heavier (for steel).
- b) The fittings or flanges shall be suitable for the service in which they are to be used.
- c) Brazed joints shall be made with a brazing material having a melting point exceeding 1,000 deg. F (538 deg. C).

- 5.2 Gaskets used to retain LP-Gas in flanged connections in piping shall be made of metal or other suitable material confined in metal having a melting point over 1,500 deg. F (816 deg. C) or shall be protected against fire exposure, except that aluminium O-rings and spiral wound metal gaskets are acceptable. When a flange is opened, the gasket shall be replaced.
- 5.3 Piping in systems shall be run as directly as is practicable from one point to another and with as few restrictions, such as ells and bends.
- 5.4 Provision shall be made in piping including interconnecting of permanently installed containers, to compensate for expansion contraction, jarring and vibration, and for settling. Where necessary, flexible connectors may be used. The use of nonmetallic pipe, tubing or hose for permanently interconnecting such containers is prohibited.
- 5.5 Metallic piping outside buildings may be underground or above ground or both. Above ground piping shall be well supported and protected against physical damage. Where underground piping is beneath drive ways roads or streets, possible damage by vehicles shall be taken into account.
- 5.6 Underground metallic piping shall be protected against corrosion as warranted by soil conditions. Corrosion protection shall comply with the following:
- a) Underground piping shall be protected as needed with a suitable coating to retard the effects of the corrosion conditions existing in the local soil. Coated pipe shall extend at least 6 in (152 mm) above ground on all risers
  - b) When dissimilar metals are joined underground, an insulating fitting shall be installed to electrically isolate them from each other.
  - c) LP Gas piping shall not be used as a grounding electrode.
- 5.7 A hydrostatic relief valve or a device providing pressure relieving protection shall be installed in each section of piping in which liquid LP Gas can be isolated between shut off valves so as to relieve the pressure which could develop from the trapped liquid to a safe atmosphere or product retaining section.
- 5.8 After assembly, piping systems shall be tested and proven free of leaks at not less than the normal operating pressure. Test shall not be made with a flame.

## **6.0 LPG VAPORIZERS**

- 6.1 Vaporizers shall be indirectly electrically heated, FM approved, meeting the requirements of NFPA 58 and shall be suitable for Class 1, Division 2, Group D, hazardous locations as verified per FM approval requirements.
- 6.2 Vaporizers shall be suitable for outdoor installation and shall incorporate suitable automatic means to prevent liquid from passing from the vaporizer to the vapour discharge piping.
- 6.3 Vaporizers shall be equipped, at or near the discharge, with a spring-loaded pressure relief valve providing a relieving capacity as dictated by NFPA 58.
- 6.4 Direct fired vaporizers with an inside diameter of more than 152mm shall be constructed in accordance with the applicable provision of the ASME code for a design pressure of 1,700kPa. Those having an inside diameter of 152mm or less shall be exempted from the ASME code but still be constructed for a minimum design pressure of 1,700 kPa.

## 7.0 GAS LEAK DETECTION SYSTEM

A gas leak detection system shall be installed to monitor for gas leaks at different locations of the gas network.

### 7.1 Control Panel

- a) The central control panel shall be located in the entrance of the building and shall easily be accessible in the event of an emergency.
- b) The central panel shall have a modular design incorporating input alarm/display module for each gas detector. Each channel shall have indication for low alarm at 15% LEL, high alarm at 30% LEL and fault at 0-100% LEL indicator.

Main LPG supply shall be shut off at 30% LEL indication.

- c) The concentration of the gas shall be displayed on a common indicator with selector switches to display the concentration of the desired channel or zone indicator.
- d) Below the gas control panel is the system control panel which shall have the following indications:
  - AC healthy
  - DC healthy
  - Sprinkler on
  - Gas pressure low
  - Gas pressure high
  - Gas solenoid valve operated
  - Common fire signals
  - Gas leak high

The panel shall have the following switches:

- Sprinkler manual operate
- Gas manual shut off
- Test lamp
- System reset

### 7.2 Gas Detectors

The gas detectors shall use a pellistor sensor to detect flammable gases and vapours. The sensor shall be housed in a stainless steel, flame proof housing, fitted onto an explosion proof junction box (Tank Area).

### 7.3 Sounder Flasher

A sounder cum flasher unit shall be installed in the vicinity of the panel located in a visible area.

### 7.4 Cables

MICC cables or FP 200 cables of at least 1.5 sq. mm area shall be used to interconnect between each detector and the control panel.

### 7.5 Logic of operation

When low level gas leak is detected in any zone, local internal alarm and indication shall show on the control panel. When high level gas leak is detected in any zone, the following sequence shall occur:

- a) The solenoid valve in the gas distribution channel shall be closed.
- b) External alarm shall be operated.

Connection to the fire alarm system:

In the event of a fire alarm signal, the following sequence shall occur:

- a) The solenoid valve in the gas distribution channel shall be closed.
  - b) The sprinkler system shall operate.
- In the event of an emergency, it shall be possible to by pass the automatic system and operate the system manually, this shall include:

- a) Manual shut off / on of the gas.
- b) Manual operation of the sprinkler system on / off
- c) Manual reset

## **8.0 PERSONNEL**

The LP gas system shall be installed by specialized personnel supervised by a specialized engineer.

## **9.0 PAINTING AND IDENTIFICATION**

All pipe work, pipe supports and brackets shall be painted with 2 undercoats and 2 finishing coats of approved paint.

LPG piping shall be painted with a specific colour different than other pipes. Orange or yellow colours are recommended. Valves shall also be painted. Direction arrows to indicate the flow of gas in the system shall be painted.

## **10.0 TESTING AND COMMISSIONING**

The system in parts and as a whole shall be tested to ensure that it will operate safely, without leaks and to the applicable specifications.

Commissioning of the system shall be carried out to demonstrate that all the equipment and the whole system are in good working order.

The LPG contractor shall accommodate any additional testing required by authorities such as Civil Defence in order to obtain their acceptance of the system.

The gas network shall be leak tested using gases such as nitrogen or argon. Oxygen or flammable gases will not be used. The leak test shall include all the different pressure stages of the system including:

- a) The filling line operating pressure up to 20 bar.
- b) The tank and high pressure stage operating pressure 16.5 bar
- c) The first stage regulator operating pressure 1.5 bar.
- d) The second stage pressure, operating pressure 37 mbar.

Test pressure shall be at least 1.5 times the operating pressure. The leak test duration shall be 1 hour. The specified test pressure shall be monitored for a minimum of one hour during which the soap water method shall be used to leak test every joint. If no leak is detected and the pressured gauge indicates no drop in the test pressure then the system will be proven to be leak free.

If a leak is found, the leaking joint shall be repaired or the leaking equipment shall be replaced. The testing procedure shall be repeated to prove the complete system as leak free.

A test report shall be submitted to the consultant for approval.

Supply, Installation, Testing, Commissioning, Handing Over & Maintenance of CENTRAL LPG GAS SYSTEM as per below specification in the entire Building wherever required in the entire project to be carried out by the Contractor as per the approved Drawings, Specifications and to the entire approval of Consultant/DM, Dubai Civil Defense & any other concerned department of Government of Dubai. Sub-Contractor/Workshop Drawings & Specifications etc. shall be approved by the Consultant.

The Main Contractor shall carry out the required Civil, Electrical, Plumbing, and Water Supply Provisions towards the above item and to provide complete co-ordination to the LPG System Sub-Contractor to Consultants approval.

**End of Section**

**SECTION 9**  
**IRRIGATION SYSTEM**

**SECTION 9**  
**IRRIGATION SYSTEM**

|      |                                       |     |
|------|---------------------------------------|-----|
| 1.1  | Scope of Work.....                    | 157 |
| 1.2  | Silence of Specification.....         | 157 |
| 1.3  | Shop Drawings.....                    | 157 |
| 1.4  | Approval of Materials .....           | 158 |
| 1.5  | Protection of Existing Services ..... | 158 |
| 1.6  | As-Built Drawings.....                | 158 |
| 1.7  | Manufacturer's Literature .....       | 158 |
| 1.8  | Units of Measurement .....            | 158 |
| 1.9  | Installation and Testing .....        | 158 |
| 1.10 | Warranty.....                         | 159 |
| 2.0  | MECHANICAL.....                       | 159 |
| 2.1  | Scope .....                           | 159 |
| 2.2  | Materials.....                        | 159 |
| 3.0  | PUMP STATION .....                    | 164 |
| 4.0  | ELECTRICAL .....                      | 164 |
| 5.0  | TESTING AND COMMISSIONING .....       | 165 |



## 1.1 Scope of Work

This contract is for construction, completion and statutory maintenance of Automatic Irrigation system as shown on the contract drawings.

This contract shall include for the complete installation, testing and commissioning of the automatic irrigation system summarized below and as per this specification and contract drawings.

- A booster pumping unit with pumps, valves, control system and accessories for the irrigation system. The pumping unit shall be installed inside the pump room located in the basement level.
- An underground pipe network of uPVC mains and submains and polyethylene tubing, with solenoid control valves for the operation of the irrigation system
- A distribution system of sprinklers, sprays and drip emitters with all necessary fittings and accessories.
- Connection of irrigation pumps with water storage tank.
- Connection of irrigation system to the pump room.
- An electrical control system for the irrigation system operation, consisting of satellite controller, low voltage cable and all other necessary accessories.
- All the builders work connected with the installation of irrigation system including core drilling and necessary waterproofing of the core holes.
- Builders work: All required builders work including crossing concrete walls, structures, footpaths, coring through slabs, walls, which deemed to be necessary for the completion of works or as per site requirements shall be the sole responsibility of the contractor.

All works which may be incidental, contingent or necessary for the proper completion of works, but which are not specifically described herein are also considered to be included.

## 1.2 Silence of Specification

1.2.1 The apparent silence of the specification, drawings or other contract documents as to any detail or the apparent omission from them of a detailed description concerning any point, shall be regarded as meaning that only the best general practice is to be used. All interpretations of the specifications will be made by the Engineer on this basis.

1.2.2 The works shall be carried out strictly according to the DM standards.

## 1.3 Shop Drawings

1.3.1 The contractor shall carefully check and verify all dimensions on the tender drawings and shall prepare, for approval by the engineer, shop drawings including but not limited to:

- Detailed layout of main, submain and lateral pipes, valves, irrigation heads and other typical connections, including hydraulic calculations.
- Electric power and control wiring diagrams system for irrigation controller, field wiring and grounding.
- Connection details for the main line connection.
- Irrigation system operation schedule including valves and stations sequencing.
- Shop drawings shall incorporate approved materials and equipment that will be shown to scale.

#### **1.4 Approval of Materials**

1.4.1 All materials shall be as described in the specifications. The contractor shall submit samples of all materials, which are proposed to be used in the system, sufficiently in advance, to the Engineer, for approval, prior to their installation. Material, which are not approved nor to the satisfaction of the Engineer, may be rejected and the contractor shall replace them, at his own cost, with acceptable items.

#### **1.5 Protection of Existing Services**

1.5.1 Particular care is to be taken to avoid damage to underground electricity, telephone and water mains, drains, roads and other underground services during the progress of the works. The Contractor shall be responsible for any damage caused to building works, supplies or services and shall fully indemnify. The owner of the services, or his representatives shall decide the method of execution of repairs.

1.5.2 Prior to starting any work at the site, the contractor has to check existing as built services drawings. Copies of these drawings shall be submitted to the engineer before starting the actual work at site.

#### **1.6 As-Built Drawings**

1.6.1 The Contractor shall maintain one set of contract drawings for the sole purpose of recording "As-Built" conditions of the irrigation system as a whole. All changes, previously approved, and all completed work shall be recorded on these drawings. All valve locations and piping shall be dimensioned and recorded (except cable routes, common to pipe runs, need not be dimensioned).

1.6.2 The As-Built drawings shall be prepared on the latest edition of Auto Cad, and supplied to the Engineer for approval prior to the issue of the Completion Certificate.

1.6.2 On approval of the as-built drawings the Contractor shall forward the following for the Engineer's retention:

- Three complete sets of As-Built drawings properly folded and provided in the plastic folders as part of the Operations and Maintenance manual.
- One set of reproachable mother prints on heavy grade polyester sheet
- One set of computer discs properly labelled and marked.

#### **1.7 Manufacturer's Literature**

1.7.1 At such time as the list of materials has been approved by the Engineer, the Contractor shall provide two (2) sets of manufacturer's technical literature and maintenance literature to the Engineer. Data sheets shall provide sufficient technical information to identify each product and shall include the name and address of the nearest re-supply firm.

#### **1.8 Units of Measurement**

1.8.1 Unless otherwise stated, the units of measurement are metric units.

1.8.2 Where the unit of measurement is the gallon, it shall deem to be the U.S. gallon.

#### **1.9 Installation and Testing**

1.9.1 The contractor shall submit his work method statement for approval prior to commencements of the works. The statement shall detail proposed sequence of work, hold points, testing frequency and document control.

## **1.10 Warranty**

1.10.1 All the works included in this contract shall be warranted by the contractor against all defects and malfunction of materials and workmanship for a period of one year from the date of completion of works and all pumps including all pump room equipments, accessories and Irrigation controllers shall be warranted for the period of five years from the date of completion of works. Should any problem arise with the system during the warranty period all necessary repairs and/or replacements shall be made by the contractor in an expedient manner at no additional cost to the client.

## **2.0 MECHANICAL**

### **2.1 Scope**

2.1.1 The scope of this section includes a description of the materials and methods for the hydraulic element of the system. All equipment, fittings and accessories shall be suitably rated to comply with the system design and operating parameters.

### **2.2 Materials**

#### **2.2.1 UPVC Pipe and fittings**

UPVC piping shall be fabricated from unplasticized polyvinyl chloride according to BS 3505: 1968 and BS 3506: 1969. Pressure rating of pipes shall be class D (12 bar working pressure). Pipe shall be supplied in lengths no smaller than six (6) meters. Each length of pipe shall bear markings to identify type and grade. All piping shall show no evidence of blisters, grooves or other extrusion marks. The Contractor shall provide three (3) samples of each size, on meter in length for approval prior to supply and installation. All pipes 4" and above shall be of the Rubber ring sealing type. The socket shall be an integral part of the pipe and offer hydraulic security under both positive and negative pressure heads.

All joints in pipe shall be made by solvent welding using adhesive solvent and primer as recommended, in writing, by the manufacturer and approved by the Engineer. All cans of solvent and primer shall have labels intact and shall be stamped with date of manufacture. No cans dated over two years old will be permitted. No thinning of solvent or primer will be permitted.

Lubricant and solvent cement used to make joints shall be approved in writing by the pipes manufacturer or his designated representative and the Engineer. Joints up to 2" shall only be threaded. All threaded joints shall be made up using PTFE tape or an approved equal thread sealant. Fittings shall be fabricated from UPVC according to BS 4346. Joints larger than 2" shall be flanged. Flange couplers shall be PN 16 and in accordance with BS 4504. Bolts and nuts shall be 316 stainless steel.

All UPVC pipes and fittings shall be Kitemarked.

#### **2.2.2 Polyethylene Pipe and Fittings**

Polyethylene pipe and fittings for emitter lines shall be manufactured from linear low-density polyethylene incorporating a minimum of 2.8% carbon black and shall have a working pressure of 4kg/cm<sup>2</sup>.

Random samples of drip irrigation pipes shall be subjected to the Teepol stress crack resistance test. Fittings shall be compression type and shall be compatible with the pipe used. Insert barbed type fittings secured by plastic ratchet clips shall not be used.

Pipes for spray head and dripline risers shall be LDPE Class C as per BS 1972-67. Fittings for these pipes shall be compression quick joint type with minimum 12 bar pressure rating.

#### **2.2.3 Steel Pipe and Fittings**

Steel pipes shall conform to API standard schedule 80 or equal and approved. Welded fittings shall be schedule 80 and weld neck. Flanges shall be in accordance with BS 4504 PN16. All bolts nuts and washers used for flanged connections shall be 316 stainless steel.

All pipes and fittings to be coated inside and outside by electrostatic fusion bonded epoxy to average DFT 150 microns or by fusion bonded plastic powder coating to average DFT 500 microns.

#### 2.2.4 Threaded Steel Pipes & Fittings (for 3" and below)

Pipe and fittings shall be heavy duty class (C) hot dip galvanized mild steel, manufactured in accordance with BS 1387/67. Joints shall be threaded and sealed with PTFE tape or some approved method.

Connections to the flanged equipment shall be via threaded flanged adapters. All bolts, nuts and washers used for flanged connections shall be 316 stainless steel.

All pipes and fittings shall be coated inside and outside by electrostatic fusion bonded epoxy to average DFT 150 micron or by fusion bonded plastic powder coating to average DFT 500 micron.

Any galvanized pipe installed in contact with the soil shall be wrapped with at least two layers of an approved wrapping tape of minimum thickness of 1.15mm and applied with 50% overlap.

#### 2.2.5 Gate Valves

Gate valves 50mm (2") in diameter and smaller shall have non-rising stem and be manufactured from bronze according to BS 5154. Inlet and outlet shall be BSP threaded. Gate valves larger than 50mm (2") diameter shall be flanged ductile iron waterworks valves manufactured to BS 5163. All valves shall be coated with solvent free coal tar epoxy DFT 300 micron. Valves installed below ground shall be provided with stemcap for key operation. Nominal pressure rating for the valves shall be 16 bar or more.

#### Solenoid Control Valves

##### 2.2.6 Solenoid Valves

Solenoid control valves shall be normally closed electrically operated, by means of an integrally mounted heavy duty 24-volt AC 50 Hz waterproof solenoid, having a maximum inrush current of 0.41 amps and a maximum 0.23 amp holding current. Valves shall be specifically designed for operation in arid areas with temperatures up to 500 c and 100% humidity. The valve shall be serviceable without removing it from the system and shall have a manually-operated flow adjustment stem for throttling. The body and flow control stem shall be sand cast brass. The diaphragm shall be nylon reinforced nitrile rubber. Valves shall be specifically rated for operation with dirty or effluent water. Control lines shall be protected by self-flushing nylon or grade 316 stainless steel screens that project into the stream flow or by a stainless steel 'scrubber' arrangement designed to clean control passages every time the valve opens. The inlet and outlet shall be B.S.P. threaded.

Valves shall be provided with a pressure regulator, capable of regulating downstream pressures in the range 15 psi to 100 psi + 5 psi. It shall be possible to measure downstream pressure via a schraeder valve, or integral pressure gauge. The pressure regulator shall function when the valve is manually opened and shall regulate successfully for a cross valve head loss of 7.5m (10 psi) or less.

##### 2.2.7 Quick Coupling Valves and Keys

Quick coupling valves shall be solid bronze or brass. Cover shall be rubber covered bronze. The seat disc plunger shall be spring loaded to keep valve closed at zero inlet pressure. Coupler shall be removable from top while valve is under pressure. Valve shall be one piece design with single slot and 1" in size.

The valve shall be operated with plug-in type special coupler key with single lug. The key shall be cast bronze. For connection of hose, swivel elbow to be

used. The keys are to be from same manufacturer to ensure compatibility of the connection.

A key complete with swivel elbow and 25m of 3/4" diameter reinforced rubber garden hose pipe shall be provided for every five valves installed.

#### 2.2.8 Valve Boxes

All solenoid valves, flush valves, quick coupling valves and wire pull boxes shall be installed in lockable access boxes of sufficient size to permit ready removal of the valve and inner assembly. Valve boxes shall be constructed from ABS, GRP or HDPE and shall be supported on block work to prevent shrinkage. The inside of the box and cover shall be labelled with permanent paint or plastic tags with valve identification details. All valve boxes shall be installed in non-traffic areas and preferably in landscape zones.

Valve boxes shall be fabricated from recommended size shall be as follows:

|                                                                  | COVER        | DEPTH  |
|------------------------------------------------------------------|--------------|--------|
| Q.C.V.                                                           | 250 mm (dia) | 230 mm |
| Pull box, Air valve 1" & Flush valve                             | 430 X295 mm  | 300 mm |
| Solenoid valves up to 2"                                         | 590X 349 mm  | 380 mm |
| <b>Solenoid valves above 2"</b><br>and Isolation valves up to 3" | 825 x 495 mm | 450mm  |

Measurements are taken from the top of boxes.

Covers shall be secured by stainless steel 316 bolts.

#### 2.2.9 In-line Drip Tubing

Inline drip tube shall consist of linear low-density Polyethylene tubing with pressure compensating, self-flushing type integral drip emitters. The tubing shall have a minimum outside diameter (O.D.) 16mm and thickness of 1.1 mm + 0.05mm. Pipes for drip lines shall be manufactured from low density polyethylene incorporating a **minimum of 2.8%** carbon black, antioxidants in an amount not exceeding 0.5% and shall have a nominal pressure PN 6.

A random samples of drip pipes shall be subjected to the Teepol crack resistance test. The samples shall be immersed in a bath containing a solution of 10% Teepol and 90% water maintained at 50 degrees C. The pipe should resist the solution without showing any fatigue or crack for a period of 150 hours.

The test should be carried out in an independent laboratory within U.A.E. All fittings for drip line shall be compression type.

EMITTERS:

The emitters must be pressure compensating according to ISO9260 and emission rate of the emitter shall remain constant at varying water pressure from 1bar to 4.5 bar. The flow path in emitter must ensure turbulent flow and emitter must confirm to excellent pressure compensation and self-cleaning/ Flushing mechanism. The emitter must be clogging resisting and the filtration requirement not more than 150 mesh. Coefficient of variation shall be less than 5%. The flow of emitter must remain same /constant at temperatures varying from 0 to 60 deg. Celsius.

The product must have warranty against cracking and performance for 7 Years.

No. of drip line risers shall be as per Engineer's approval. Separate drip line risers shall be for each planter. Each tree in grass areas shall be provided with separate drip line riser. Drip stakes shall be provided at exact point of change in direction of drip line and spacing between drip stakes shall not be more than 10.0 meters

The capacity and spacing of the inbuilt emitters in the drip line shall be as follows:

FOR GROUND COVERS:

Emitter discharge rate shall be 2.3 LPH and spacing between emitters on drip line shall be 50 cm.

FOR SEASONAL FLOWER BEDS:

Emitter discharge rate shall be 2.3 LPH and spacing between emitters on drip line shall be 30 cm. Spacing between drip lines shall be 50 cm.

FOR HEDGES:

Two rows drip lines at 50 cm spacing shall be provided for hedges. Emitter discharge rate shall be 2.3 LPH and spacing between emitters shall be 50 cm.

FOR TREES:

A loop of drip line (around tree) with 3.5 LPH in-line emitter at every 30 cm. Length of drip line in a loop shall be not less than 3.5m.

FOR SHRUBS:

2 no. in-line PC emitters (each emitter 3.5 LPH) per plant shall be used for shrubs if spacing between shrubs is less than or equal to 1.0m.

A loop of drip line with 3 no. in-line PC emitters (each emitter 3.5 LPH) per plant shall be used for shrubs if spacing between shrubs is greater than 1.0m.

2.2.10 Emitter accessories

(i) Drip line stakes

Drip line piping shall be staked at minimum 10m centres or as shown on the drawings. Stakes shall be fabricated from polypropylene and shall be at least 220mm long. Stake head shall feature a serrated surface to grip drip line. Tails of the stakes shall be provided with integral barb type flaps (min 4 No.) to increase retention in the soil.

2.2.11 Pop-up Sprays

The sprinklers shall have a high-level impact strength corrosion resistant body, stem and nozzle. Pop - up height should be 4". The sprinkler shall have a heavy duty stainless steel retraction spring with minimum diameter of 1.5mm. The sprinkler shall incorporate a wiper seal to protect the internal mechanism against ingress of foreign material and prevent pressure blow out and shall have an easily accessible under nozzle screen. The wiper seal should be co-moulded type that ensures flow-by of less than 0.10 US. gpm at low pressure and complete sealing at pressure greater than 10 psi. The sprinkler shall have 12mm (½") threaded inlet.

All sprays should be provided with built-in seal-a-matic check valve and in - stem pressure regulator to maintain constant outlet pressure.

The sprayer shall have the following performance specification at 2 Bar operating pressure:

| Pattern       | Radius (M) | Trajectory | Discharge (L/S) |
|---------------|------------|------------|-----------------|
| Full          | 3.5 - 4.0  | 30 degrees | 0.16            |
| Half          | 3.5 - 4.0  | 30 degrees | 0.08            |
| Three Quarter | 3.5 - 4.0  | 30 degrees | 0.12            |

the line of the trench shall be identified and exposed. In addition, the line of the trench shall be swept using an approved detection device, capable of detecting both live and 'dead' cables.

Trenches shall be made as narrow as practicable, but not less than the pipe diameter plus 300mm, and shall provide sufficient room for laying jointing and testing. The width of the trench at the crown of the pipe shall not exceed the pipe diameter plus 600mm. Sheeting, bracing and shoring shall be used to provide for the safety of all personnel for all trenches.

The trench bottom shall be composed of stable uniform sand with no protuberances that might cause point loading of the pipe. The trench formation shall be trimmed to an even finish which will provide continuous support for the pipe. Additional excavation shall be carried out at the position of the pipe socket to ensure proper joint assembly and pipe support.

If the existing formation is unsuitable for direct laying, pipes shall be laid on a bed of imported sand 100mm thick (minimum). This sand may be sourced from sieved excavated material, but shall not contain any particles larger than 5mm, and shall have good self-compacting properties. Pipes shall be centre loaded leaving joints exposed within three days of laying. Backfill material shall be sand and water compacted once the level of backfill reaches the crown of the pipe. Initial back-fill shall be by hand in 300 mm lifts, and carried out in such a manner as to ensure adequate support under the pipe and distributed evenly to ensure there are no voids under or to the side of the pipe.

(ii) Solvent Cement Welded Joints

The prospective joint should be free of moisture. The jointing surfaces shall firstly be cleaned with a light wiping of PVC cleaner. A light application of cement shall be first applied to the inside of the fitting followed by a slightly heavier application to the exterior of the pipe. The pipe shall be inserted into the fitting, with a 1/8th turn to distribute the solvent cement and the excess on the exterior of the pipe wiped clean. The Contractor shall ensure that excessive quantities of the cement are not permitted to be forced into the interior of the fitting. In addition, care shall be exercised to allow no disturbance of the fittings recently cemented on the line. Cement that becomes heavy or unduly thick shall not be thinned or reused.

If trench conditions so require, the pipes can be jointed into manageable lengths at the trench side and carefully lowered to the trench bottom. If this method is adopted then the assembled line must be carefully inspected in the trench to ensure that individual joints have not been withdrawn during the installation operation.

(iii) Thrust Blocks

Concrete anchor blocks must be provided at all directional changes, branches, tees, valves and end caps. Anchor blocks will be constructed to adequately transmit the thrust forces applied at these points to the solid undisturbed wall of the pipe trench. Anchor block concrete shall not be allowed to totally encase the fitting. Fittings shall be wrapped in a minimum of 3 layers of heavy gauge polyethylene film prior to concreting.

2.3.3 Dripline Pipe

(i) Dripline pipe shall be installed as shown on the detail drawings.

(ii) The planting plans accurately detail the location and density of palm trees, ornamental trees, shrubs and groundcovers. The irrigation pipeline system should take these locations into account, but the pipeline does not necessarily pass through the planted areas. The Contractor shall use his judgement to irrigate the planting by making suitable take-off connections from the dripline lateral pipe. However, the dripline take-off points shall be limited by the following criteria.

1. No dripline shall exceed 50m in length.
2. No dripline shall exceed 100no. emitters along its length.

(iii) Low Head Drainage From Drip Zone Irrigation Laterals

The Contractor shall provide sufficient measures to ensure that low head drainage is prevented. Take-off points shall be supplemented with a ½" check valve where appropriate.

2.3.4 Electric Control Valves

(i) Electric control valves shall be installed as shown on the detail drawings. The valves shall not be installed until all mains have been flushed thoroughly.

(ii) Electric control valves shall be located in planted areas.

#### 2.3.5 Quick Coupling Valves

(i) Quick coupling valves shall be installed as shown on the detail drawings. Installation shall be carried out in a workmanlike manner with all operating devices easily accessible by special key or by hand.

#### 2.3.6 Valve Boxes

(i) Valve boxes shall be installed as shown on the detail drawings, and shall be firm, level and aligned with surrounding final levels.

#### 2.3.7 Painting

(i) All metal work provided for the purpose of piping, supports, etc. or any ancillary parts, exposed to the possibility of corrosion shall be thoroughly degreased, cleaned and then coated with epoxy paint. The paint shall be applied as one (1) coat of primer of a thickness of a minimum of 100 microns followed by two (2) coats of top whose total thickness does not exceed 300 microns.

### 3.0 PUMP STATION

**Main Pump** shall have the following specifications:

- Vertical Multistage Inline Centrifugal Pumps
- Suitable for TSE Water
- Be protected against dry running through a system provided inside pump room and in the storage tank
- Have cast iron body and base
- The power rating of the motors shall be 30% above the rated power shown on the manufacturers catalogue.

### 4.0 ELECTRICAL

4.1 All electrical equipment shall be in accordance with local Electricity Authority regulations and electrical specifications. The enclosures shall be protected in accordance with IP55 for equipment installed above grade and IP58 for equipment installed below grade. Wiring to and from the control panels shall be routed in PVC conduits below the ground. Above grade conduits to be liquid tight flex or galvanized steel.

#### 4.2 Underground Cables

4.2.1 All electric wires for underground use shall be single core solid "Underground Feeder" type insulated with flame retardant thermoplastic compound rated for 600 volts and suitable for direct burial. The contractor shall prepare a wire sizing chart with effective lengths of each station and common wires, which shall be submitted for approval prior to commencement of the works.

4.2.2 Cable markers shall be used for indicating valve numbers for each cable at both ends as well as at splices. Cables shall be laid in PVC conduits in Main Line trenches. Power and signal cables shall be laid in separate conduits. Separate wire shall be laid for each solenoid valve and one spare cable shall be provided for every five cables.

#### 4.3 Wire Connectors

4.3.1 All wire connections at electric control valves and all splices of wire in the field shall be made using wire connectors. The wire connectors shall be specially designed to ensure water-proof underground wire connectors.



The connector shall consist of precision moulded PVC base socket, moulded sealing plug, wire crimping sleeve and shall be easily installed in the field to give a permanent water-proof joint by using sealer for joint make up. All wire connections to be arranged in plastic pull boxes at fixed interval distances of maximum 100 metres and/or at any change of direction.

#### 4.4 Irrigation Controller

The Contractor shall supply; install pedestal mounted field satellite controllers as required for an efficient centralized control & operation of the overall irrigation system. Controllers shall be solid state computer controllers capable of independent timed operation and also capable of incorporation into a centralized computer controlled system, and have the following features:

- Station timing 0 - 99 minutes (in minutes increments).
- 8 start times per station.
- Independent station programming.
- Simultaneous station operation to 4 stations.
- Water budgeting from 25% to 200%.
- Monitor capability.
- Current time maintained during power outage via rechargeable battery.
- Manual test programme.
- Station copy facility
- Up to 4 valves maximum per station excluding pump start.
- 220/240/260 50 Hz transformer.
- Electrical output of controller will be sufficient to operate a total of 8 valves plus pump start relay.
- U.L. listed and tested.
- 12 stations (minimum)

Controller shall be pedestal-mounted in a lockable robust stainless-steel cabinet. Input shall be 240 VAC 50 Hz. Output shall be nominally 24 VAC and shall be sufficient to accommodate the maximum number of valves.

Field wires shall be terminated in screw terminal blocks connected to the controller by a pre-installed cable harness.

The programme memory shall be nonvolatile in the event of a power failure and shall have the capability of remaining effective for a minimum of 10 years.

The real time shall be maintained without re-setting in the event of a power failure lasting a minimum of 24 hours.

Lighting protection shall be provided for all field controllers.

All equipment shall be grounded to a local grounding rod strictly in accordance with the manufacturer's printed instructions.

## 5.0 TESTING AND COMMISSIONING

### 5.1 Pipework and Fittings

All pipes shall be pressure tested within two weeks of installation. Pipes shall be tested after centre loading and generally with joints exposed. Length of pipe tested at any time shall not exceed 500 metres and the rate at which pipelines are successfully tested shall be of the same order of magnitude as the rate of pipe laying. Centre loading shall be sufficient to prevent buckling or deformation due to application of pressure.

5.1.1 All pipework shall be pressure tested for a minimum of 6 hours. The pressure variation over the test period shall not exceed 10%

Test pressures shall be as follows:

Main lines, pipework 10.00 kg/cm<sup>2</sup>

5.1.2 All mainline fittings shall be left exposed for the purpose of this test and shall be inspected for evidence of leakage. Any major break in the mainlines requiring shut-down of the test shall cause the test to be re-run entirely. The Engineer shall govern the re-scheduling of the test.

## 5.2 Electrical tests

5.2.1 Each electrical equipment, cable and complete system shall be thoroughly inspected and tested before finally placing in service under the full responsibility of the Contractor. All tests shall be made in compliance with respective regulations, recommendations and standards.

5.2.2 Any modifications or repairs deemed necessary upon completion of the tests shall be executed at the Contractor's expenses. Further tests shall be carried out on any modified or repaired equipment until it is certified trouble free and acceptable for its intended service. All testing shall be demonstrated to the Engineer in a manner to be agreed later.

## 5.3 Irrigation Systems Tests

5.3.1 After successful pressure test and acceptance thereof by the Engineer, the automatic portion of the system including all electric control valves, quick coupling valves, sprinklers, sprayheads and emitters, shall be activated in a fully automatic manner and run through a complete cycle. Station timings of the irrigation controller will be programmed by the Engineer. Each function shall be demonstrated in accordance with an approved commissioning procedure developed by the contractor. Failure of any components to function in accordance with the design will require a complete new start for a final acceptance test. The Engineer shall re-schedule the time for re-testing.

## 5.4 Operation and Maintenance Manuals

5.4.1 The Contractor shall provide three copies of the operation and maintenance manual for the Engineer's approval prior to issuing of the completion certificate. Manuals shall contain comprehensive operational schedules, recommended spare parts lists, manufacturer's operating data, catalogues and exploded parts diagrams, original guarantees where applicable for the entire irrigation scheme. The material used for the irrigation system shall be listed in the manual.

Draft, manuals shall be submitted for approval prior to commencement of commissioning and shall be revised in accordance with Engineer's instructions and reflect and record the results of the commissioning procedures described above.

One soft copy of the manual on CD shall also be provide.